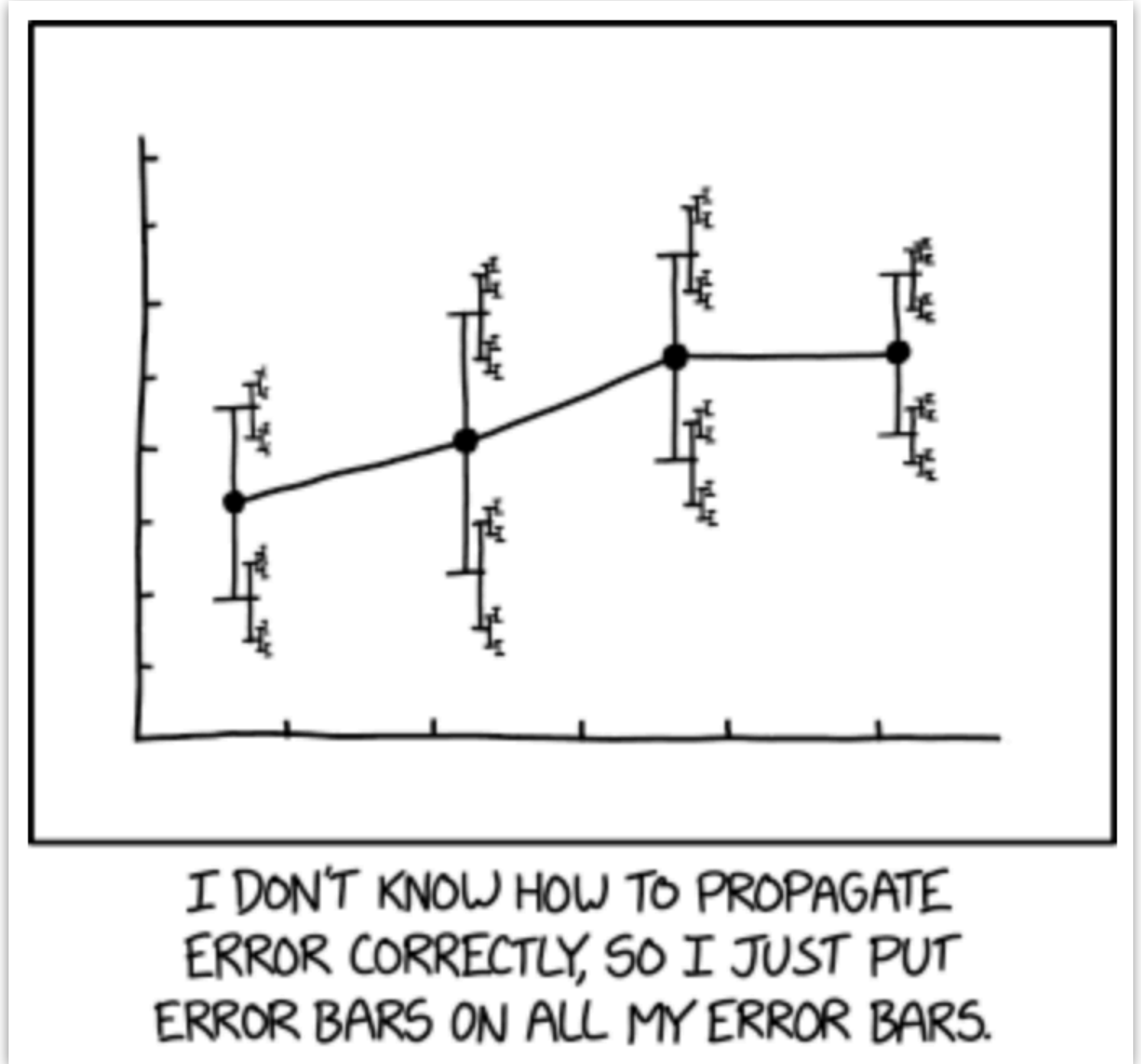
PSYCH 252: Statistical Methods for Behavioral and Social Sciences

Linear mixed effects models 1,2



02/18/2026

Logistics

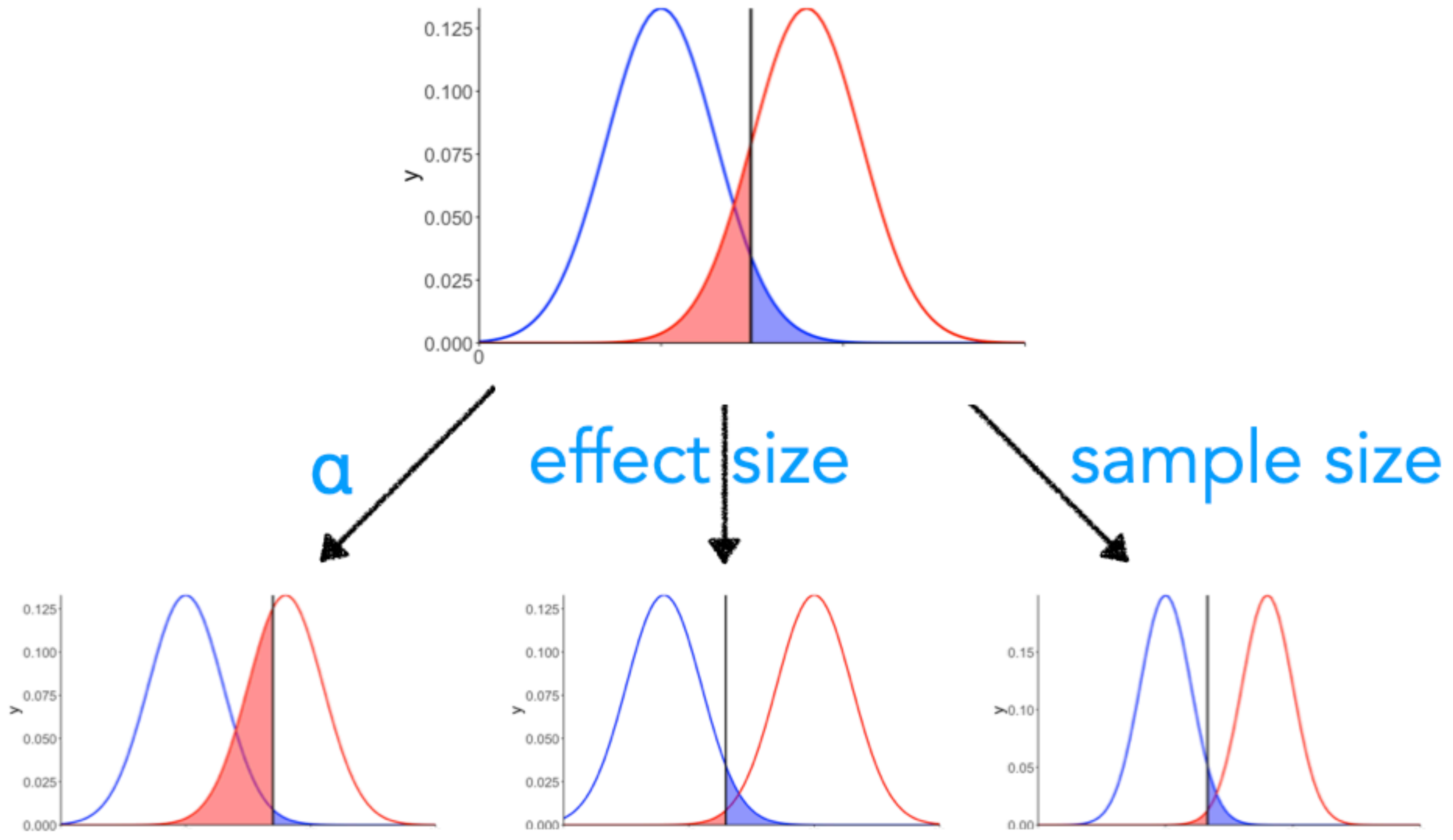
Plan for today

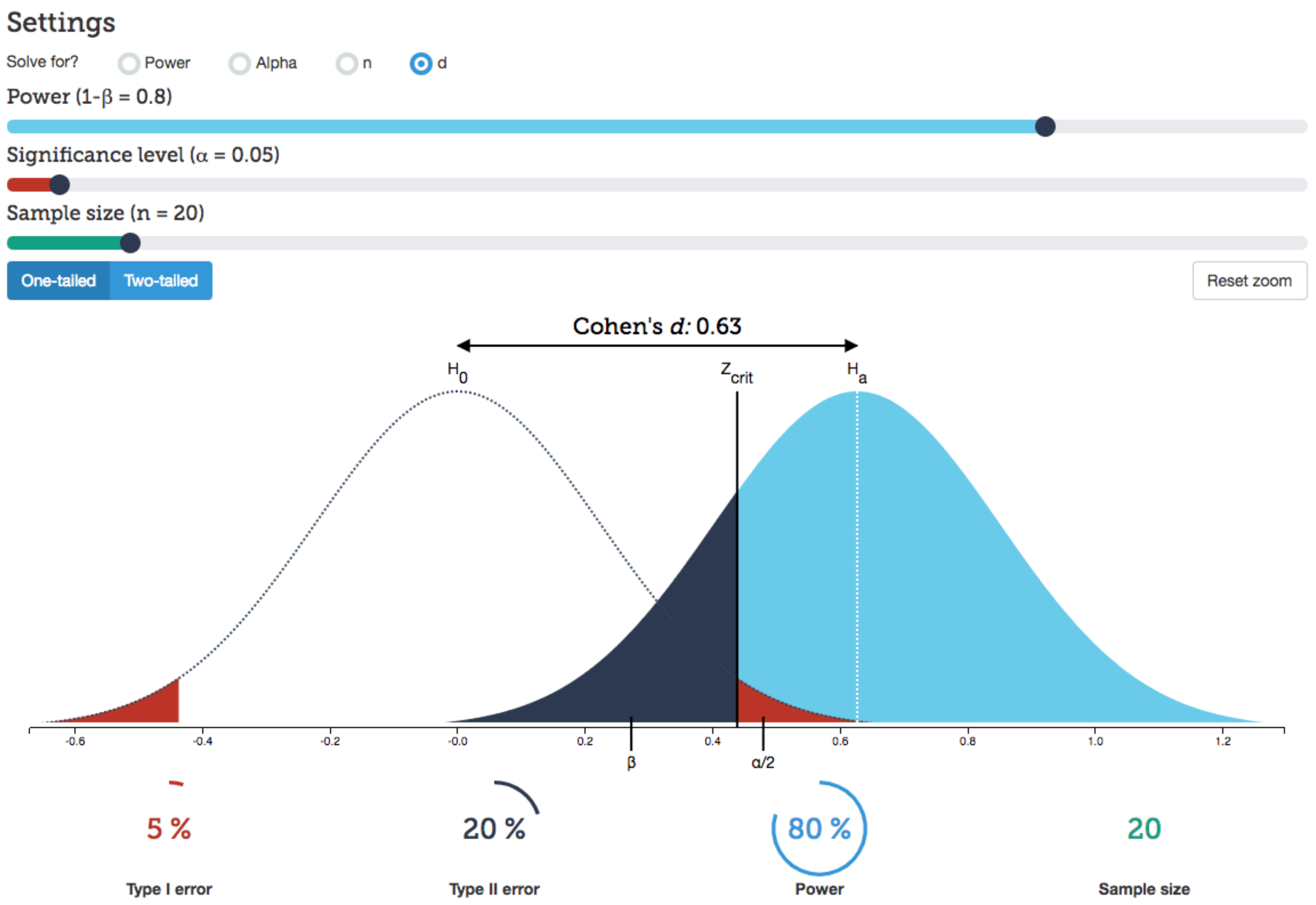
- Quick recap
  - Model Comparison (cross validation, AIC, BIC)
- Linear Mixed Model
  - Accommodating non-idependence in data
  - Understanding lmer() syntax
  - A worked example
  - Reporting results
  - Simpsons Paradox

Quick recap

Quick recap: Power analysis

The knobs we can turn to affect power

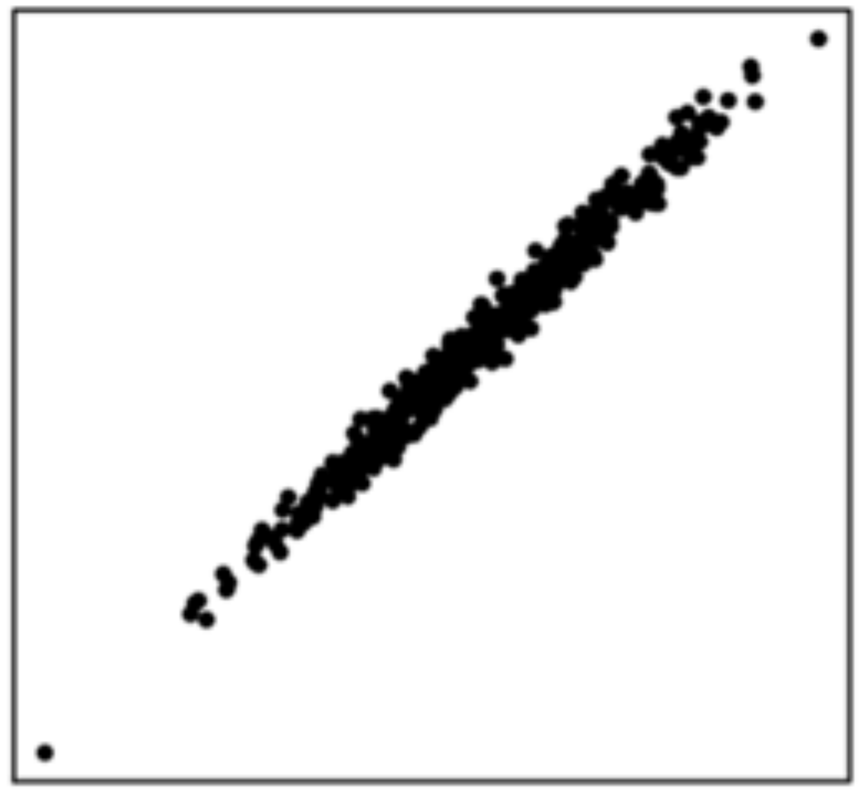




Quick Recap: Effect sizes

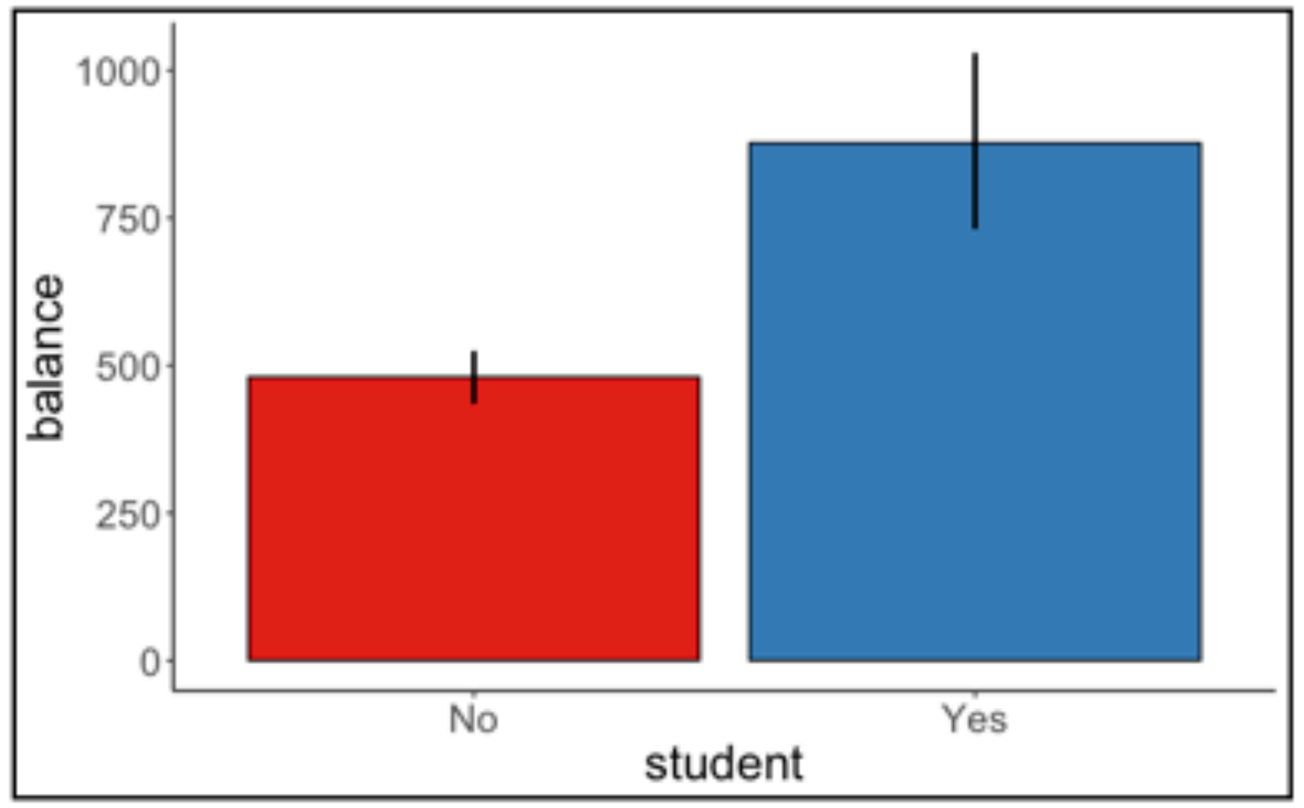
- a p-value tells us whether we can reject the H_0
- **effect sizes is a measure of the strength of the effect**

Relationships between variables



r correlation

Differences between groups



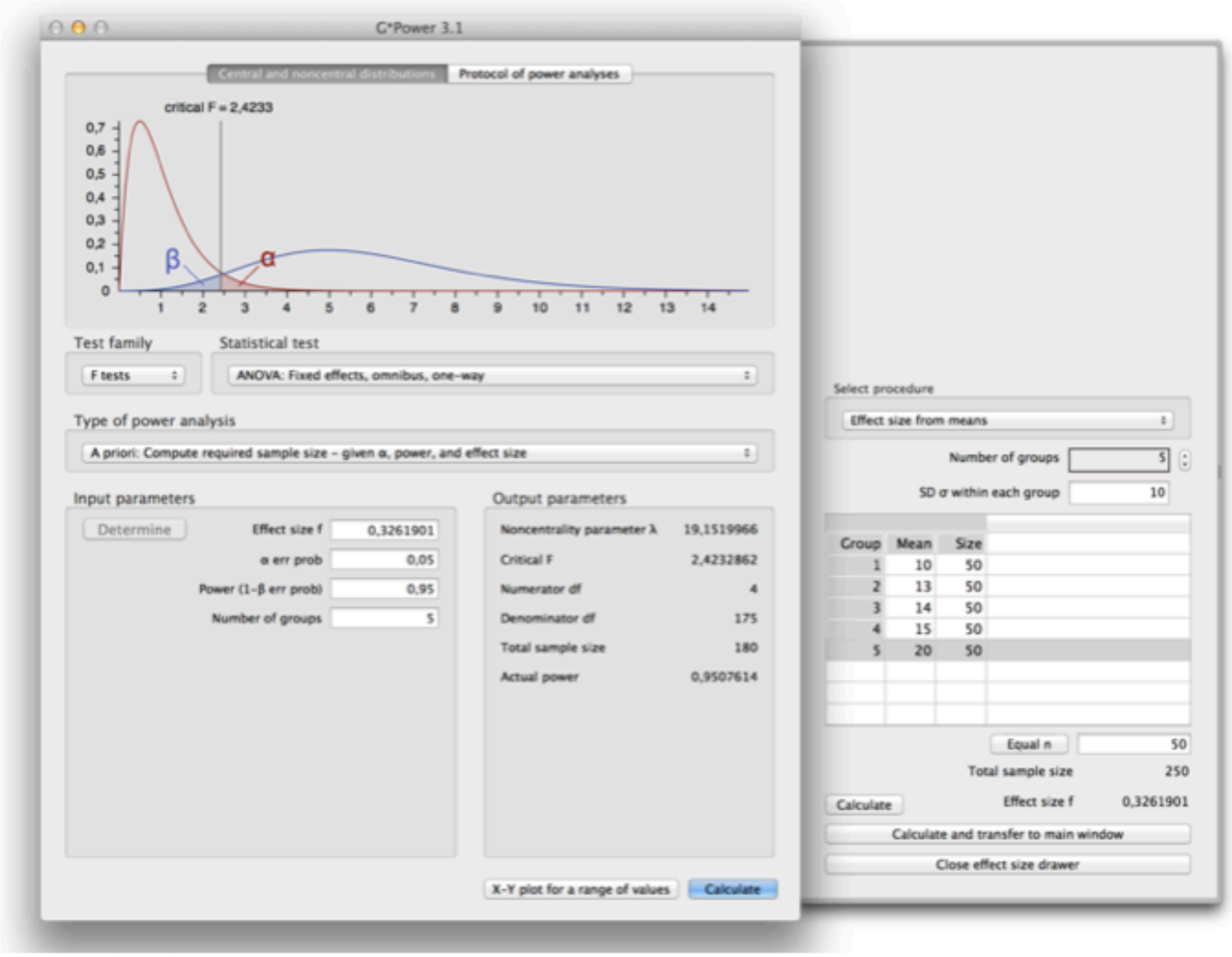
Cohen's d

statistical vs. practical significance

Quick recap: Doing Power analysis (via Algebra)

G*Power 3.1: Alternative software for power calculations

Option 1



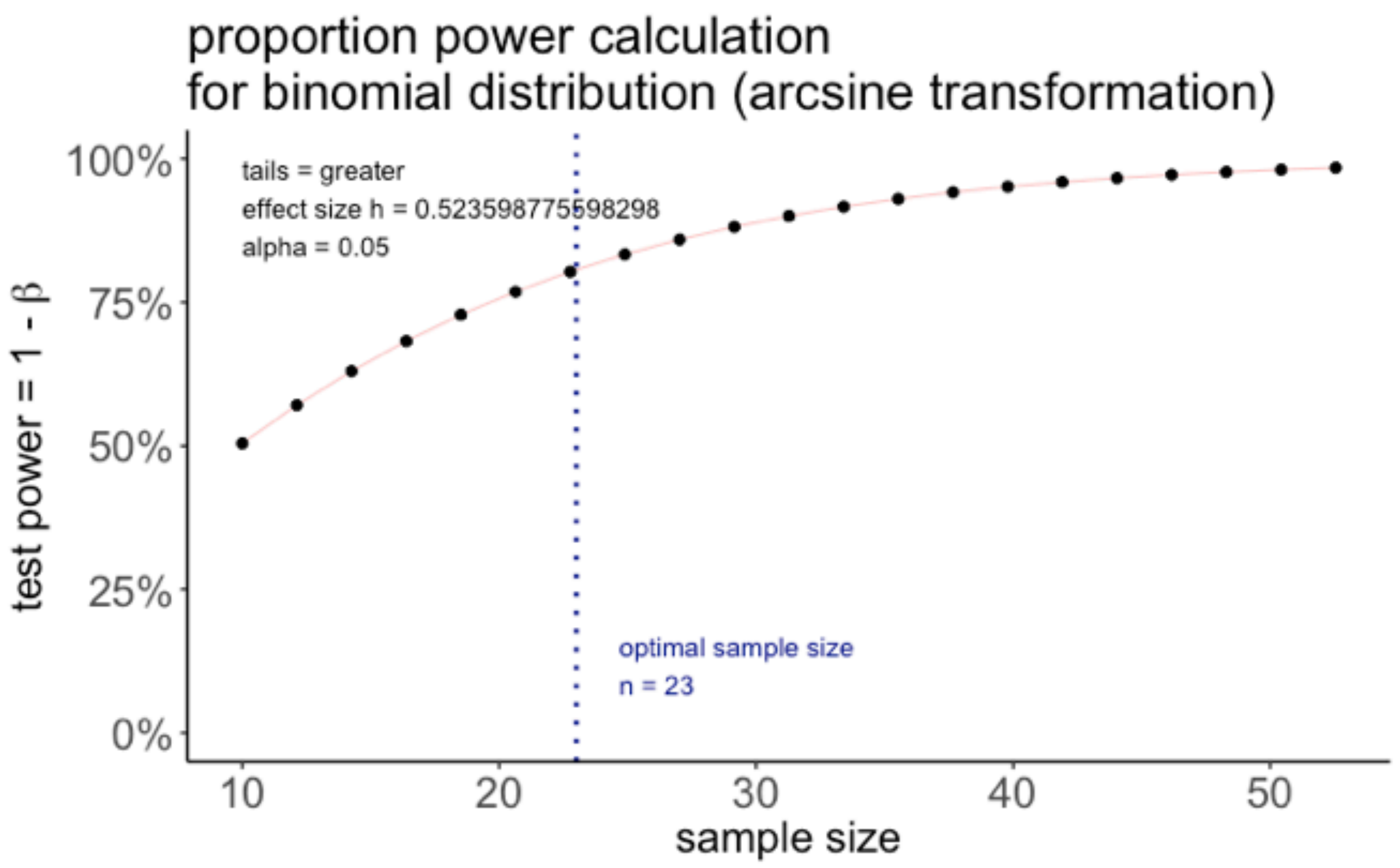
http://www.gpower.hhu.de/

31

"pwr" package in R

Option 2

```r
library("pwr")
pwr.p.test(h = ES.h(p1 = 0.75, p2 = 0.50),
           sig.level = 0.05,
           power = 0.80,
           alternative = "greater") %>%
 plot()
```



32

Quick recap: Doing Power analysis via Simulations

Option 3

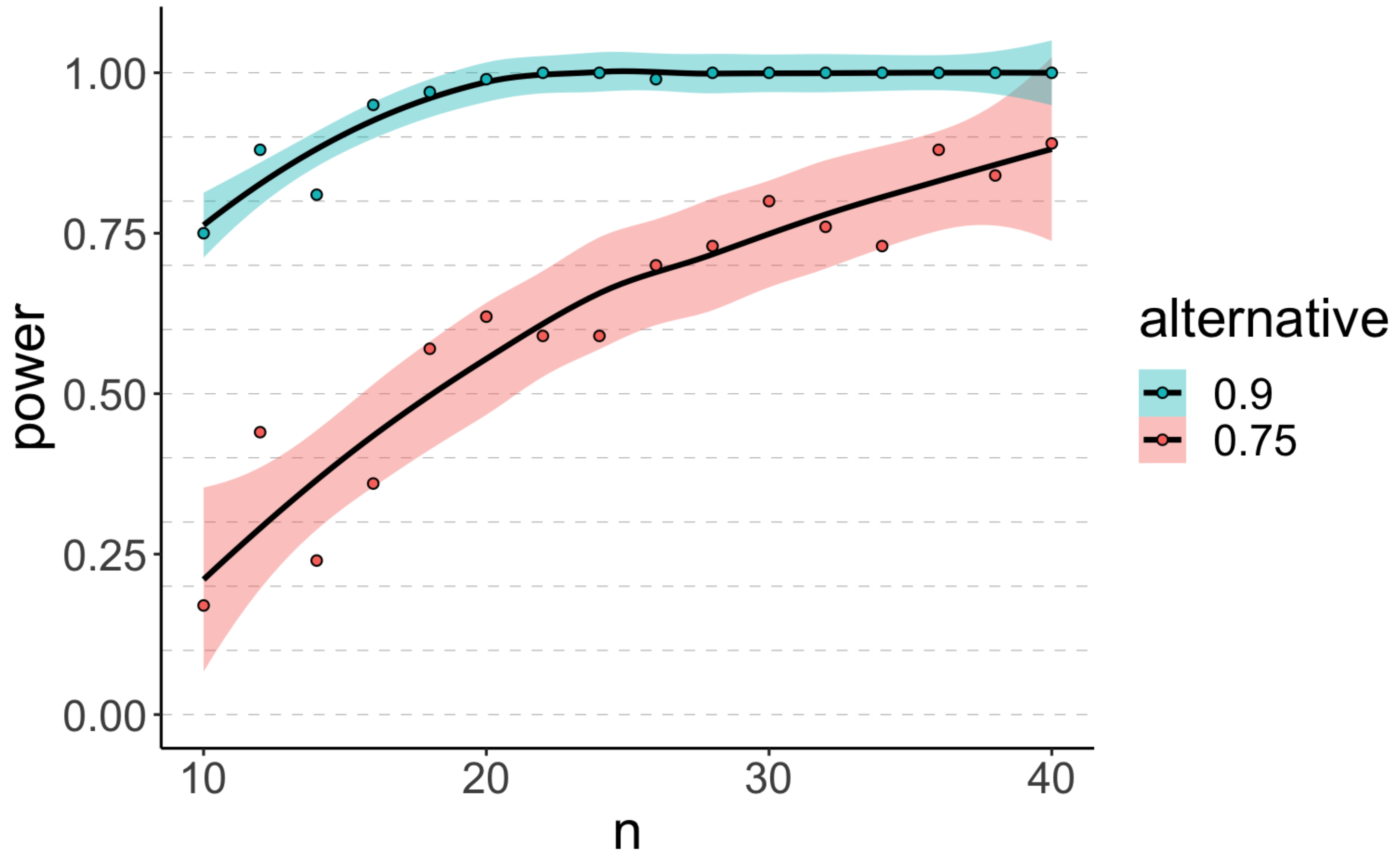
Power simulation recipe

- assume:
  - α, n, effect size
- simulate a large number of data sets of size n with the specified effect size
- for each data set, run a statistical test to calculate the p-value
- determine the probability of rejecting the H_0 (given that H_1 is true)

Let's simulate ...

```r
# make reproducible
set.seed(1)

# number of simulations
n_simulations = 5

# run simulation
expand_grid(n = seq(10, 40, 2),
            simulation = 1:n_simulations,
            p = 0.75) %>%
  mutate(index = 1:n(),
         .before = n) %>%
  group_by(index, n, p, simulation) %>%
  mutate(response = rbinom(n = 1,
                           size = n,
                           prob = p),
         p.value = binom.test(x = response,
                              n = n,
                              p = 0.5,
                              alternative = "two.sided")$p.value) %>%
  group_by(n, p) %>%
  summarize(power = sum(p.value < 0.05) / n())
```

n	p	power
10	0.75	0.2
12	0.75	0.2
14	0.75	0.4
16	0.75	0.2
18	0.75	0.6
20	0.75	0.8
22	0.75	0.6
24	0.75	0.4
26	0.75	0.6
28	0.75	0.8
30	0.75	0.8

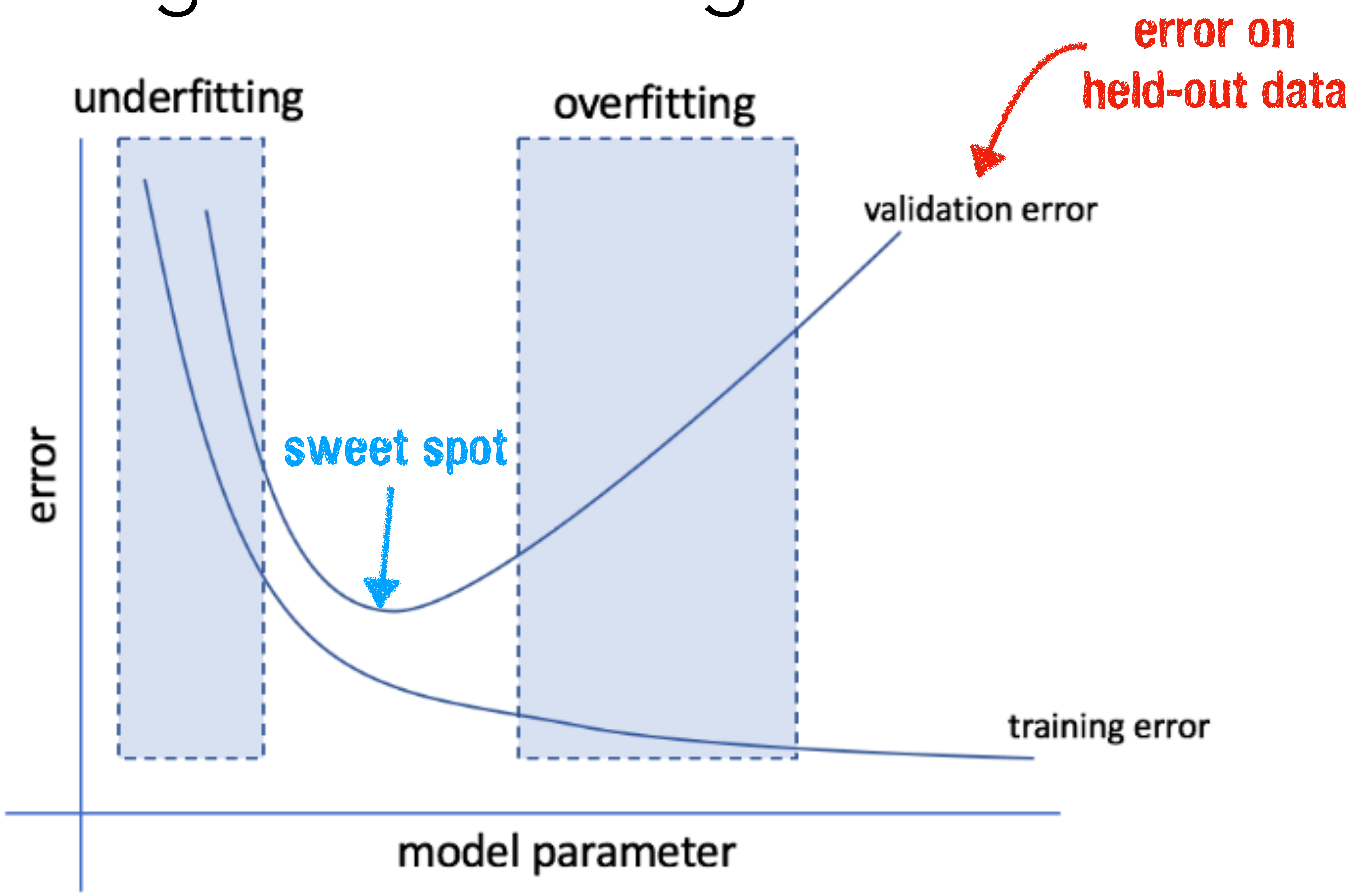


Quick recap: Model comparison

More complex models fit the data better.

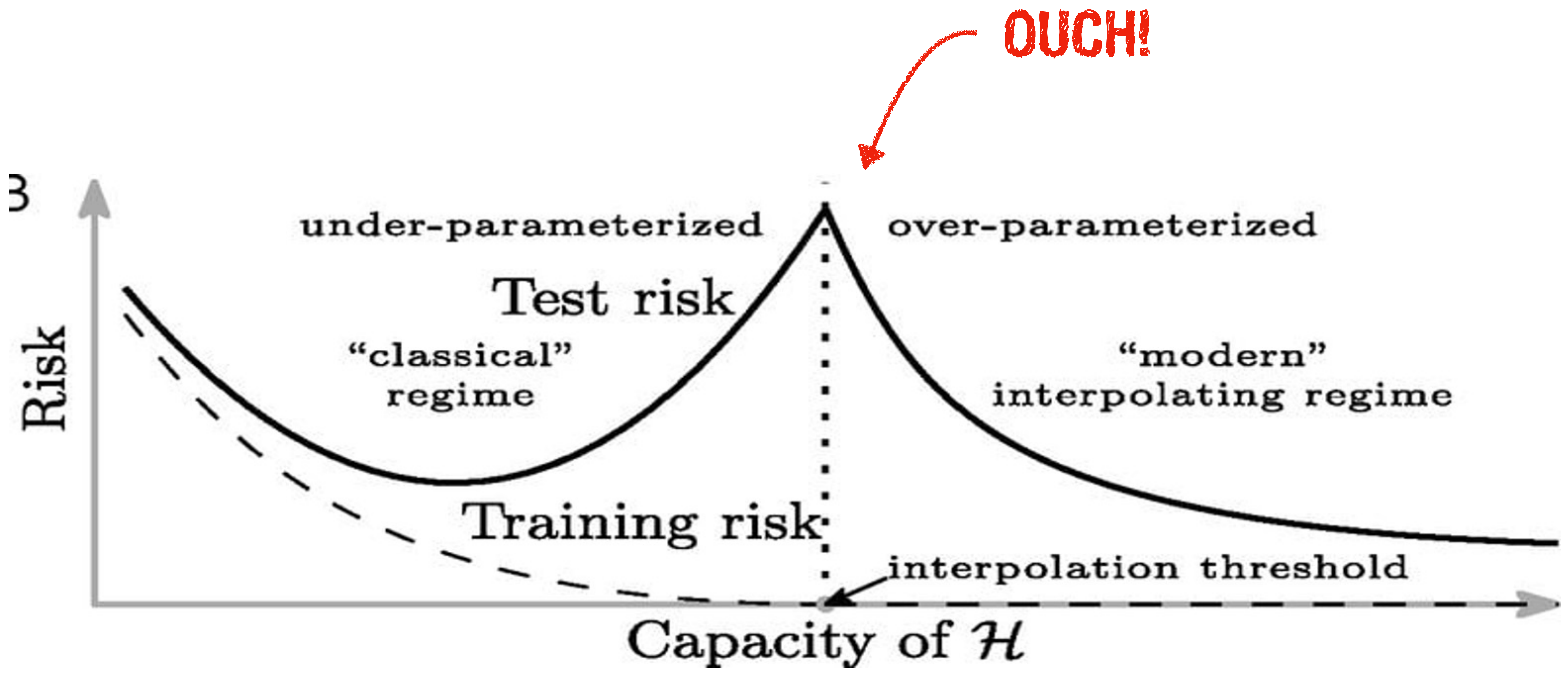
We need to trade-off **model fit** and model **complexity**.

Underfitting vs. Overfitting



the goal is to find the **sweet spot** between underfitting and overfitting

Discovery of Double-Descent and Over-parameterized Models



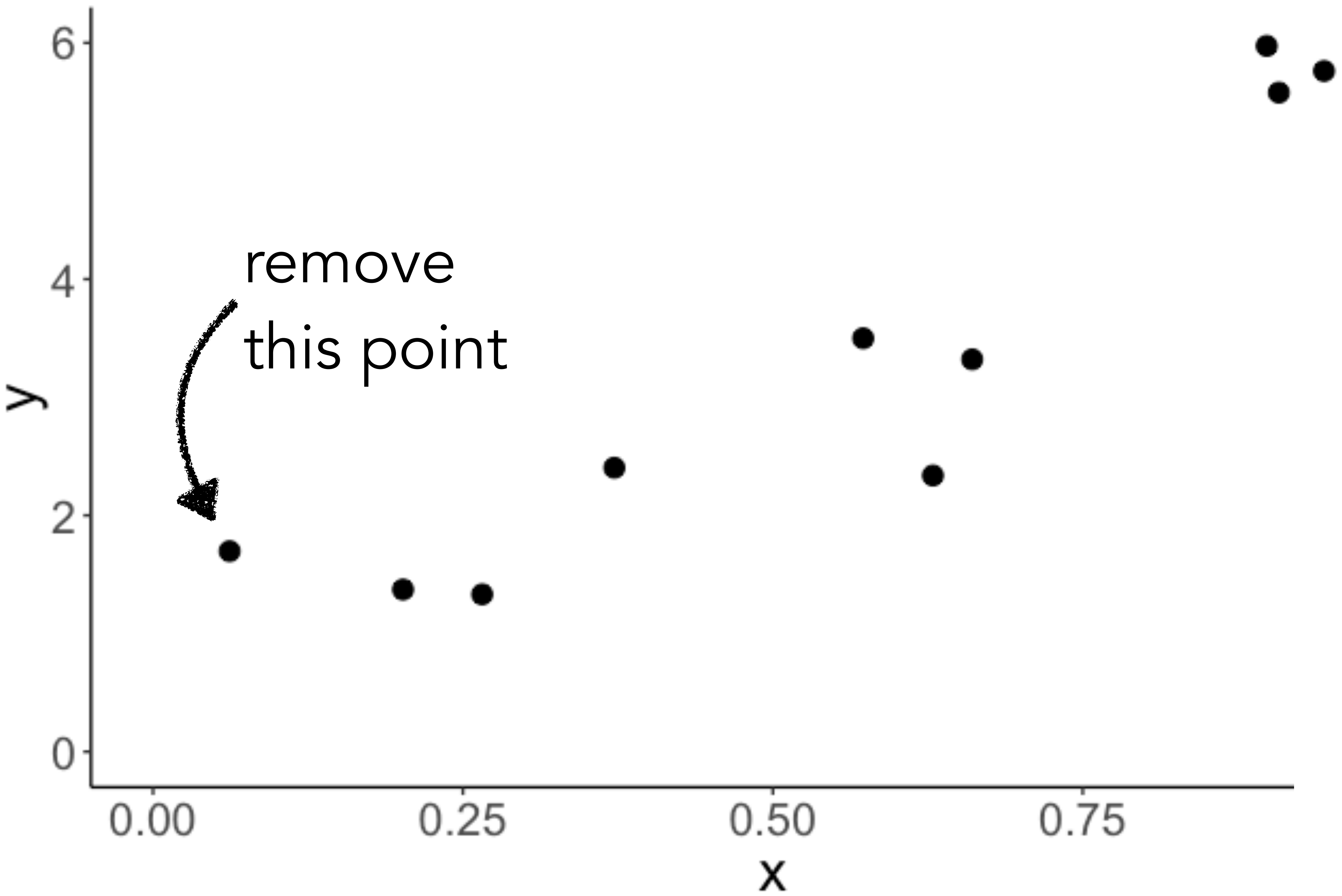
Belkin et al., 2019

Quick recap: Model comparison

1. compare the proportional reduction in error using `anova()`
   - only works for nested models
2. cross-validation
   - works generally, but can take some time …
   - different cross-validation procedures (LOO, k-fold, Monte Carlo)
3. AIC, BIC
   - works as long as we can compute the likelihood of the data
   - some discussion whether number of parameters is a good measure of model complexity
4. Bayesian model comparison

Leave one out cross-validation

LOO



Leave-one out crossvalidation

```
library("modelr")

df.cross = df.data %>%
  crossv_loo() %>%
```

splits the data set into training and test

	train	test	.id
1	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	1
2	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	2
3	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	3
4	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	4
5	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	5
6	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	6
7	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	7
8	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	8
9	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	9
10	list(data = list(participant = 1:10, x = c(0.265508663...	list(data = list(participant = 1:10, x = c(0.265508663...	10

Leave-one out crossvalidation

```r
library("modelr")

df.cross = df.data %>%
  crossv_loo() %>%
  mutate(model_simple = map(train, ~ lm(y ~ 1 + x, data = .)),
         model_correct = map(train, ~ lm(y ~ 1 + x + I(x^2), data = .)),
         model_complex = map(train, ~ lm(y ~ 1 + x + I(x^2) + I(x^3), data = .))) %>%
```

fit three different models to the training data set

Leave-one out crossvalidation

```r
library("modelr")

df.cross = df.data %>%
  crossv_loo() %>%
  mutate(model_simple = map(train, ~ lm(y ~ 1 + x, data = .)),
         model_correct = map(train, ~ lm(y ~ 1 + x + I(x^2), data = .)),
         model_complex = map(train, ~ lm(y ~ 1 + x + I(x^2) + I(x^3), data = .))) %>%
  pivot_longer(cols = contains("model"),
               names_to = "index",
               values_to = "model") %>%
  mutate(rmse = map2_dbl(.x = model, .y = test, .f = ~ rmse(.x, .y)))
```

calculate the root mean squared error for each model on the test data set

```r
df.cross %>%
  group_by(index) %>%
  summarize(mean_rmse = mean(rmse))
```

index	mean_rmse
simple	0.65
correct	0.48
complex	0.70

the correct model has the lowest prediction error

Leave-one out crossvalidation

Any potential problems with the LOO?

can be computationally expensive since it requires fitting the model `n` times (once for each data point) ...

k-fold cross validation

k-fold crossvalidation

Full data set

Iteration 1	Test	Train	Train	Train	Train
Iteration 2	Train	Test	Train	Train	Train
Iteration 3	Train	Train	Test	Train	Train
Iteration 4	Train	Train	Train	Test	Train
Iteration 5	Train	Train	Train	Train	Test

k-fold crossvalidation

k-fold crossvalidation

```
df.cross = df.data %>%
  crossv_kfold(k = 10) %>%
  mutate(model_simple = map(.x = train,
                            .f = ~ lm(y ~ 1 + x, data = .)),
         model_correct = map(.x = train,
                             .f = ~ lm(y ~ 1 + x + I(x^2), data = .)),
         model_complex = map(.x = train,
                             .f = ~ lm(y ~ 1 + x + I(x^2) + I(x^3), data = .))) %>%
  pivot_longer(cols = contains("model"),
               names_to = "model",
               values_to = "fit") %>%
  mutate(rsquare = map2_dbl(.x = fit,
                            .y = test,
                            .f = ~ rsquare(.x, .y)))
```

using R^2 as a measure

why didn't we use R^2 for LOO?

this wouldn't work for LOO since we only have one data point in the test data...

index	median_rsquare
simple	0.839
correct	0.865
complex	0.860

the correct model accounts for the most variance in the test data

k-fold vs. leave-one-out crossvalidation

- **LOO**:
  - trained on **more** data
    - more variance — more overfitting
    - less bias — less underfitting
- **k-fold**:
  - trained on **less** data
    - less variance — less overfitting
    - more bias — more underfitting

- **bias** = error from erroneous model assumptions, miss the relevant relations
- **variance** = error from small fluctuations in the data, fitting the random noise

Monte Carlo crossvalidation

Monte Carlo crossvalidation

random splits into training and test data

```
crossv_mc(n = 50, test = 0.5)
```

number of training-test splits

proportion of test data in each split

- splits can also be done in a **stratified** way (check out the `tidymodels` package)
- for example, generate training data that has the same percentage of cases from each group
- fit data from some participants, test on data from other participants, ...

AIC and BIC

AIC and BIC

- AIC = Akaike Information Criterion
- BIC = Bayesian Information Criterion

not that much Bayesian about it ...

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

?

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L})$$

$\hat{L}$ = maximized value of the likelihood function of the model

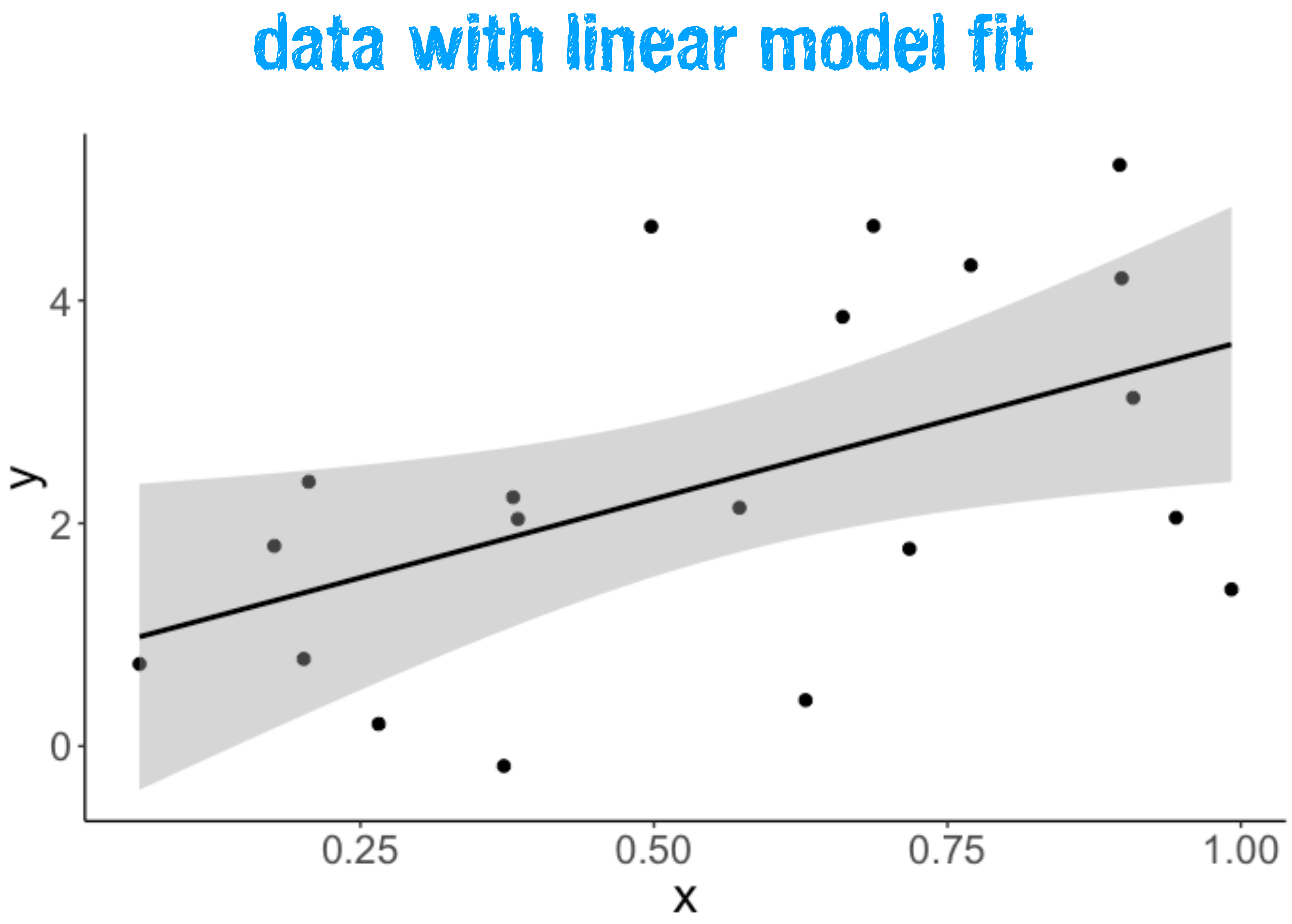
k = number of parameters in the model

n = number of observations

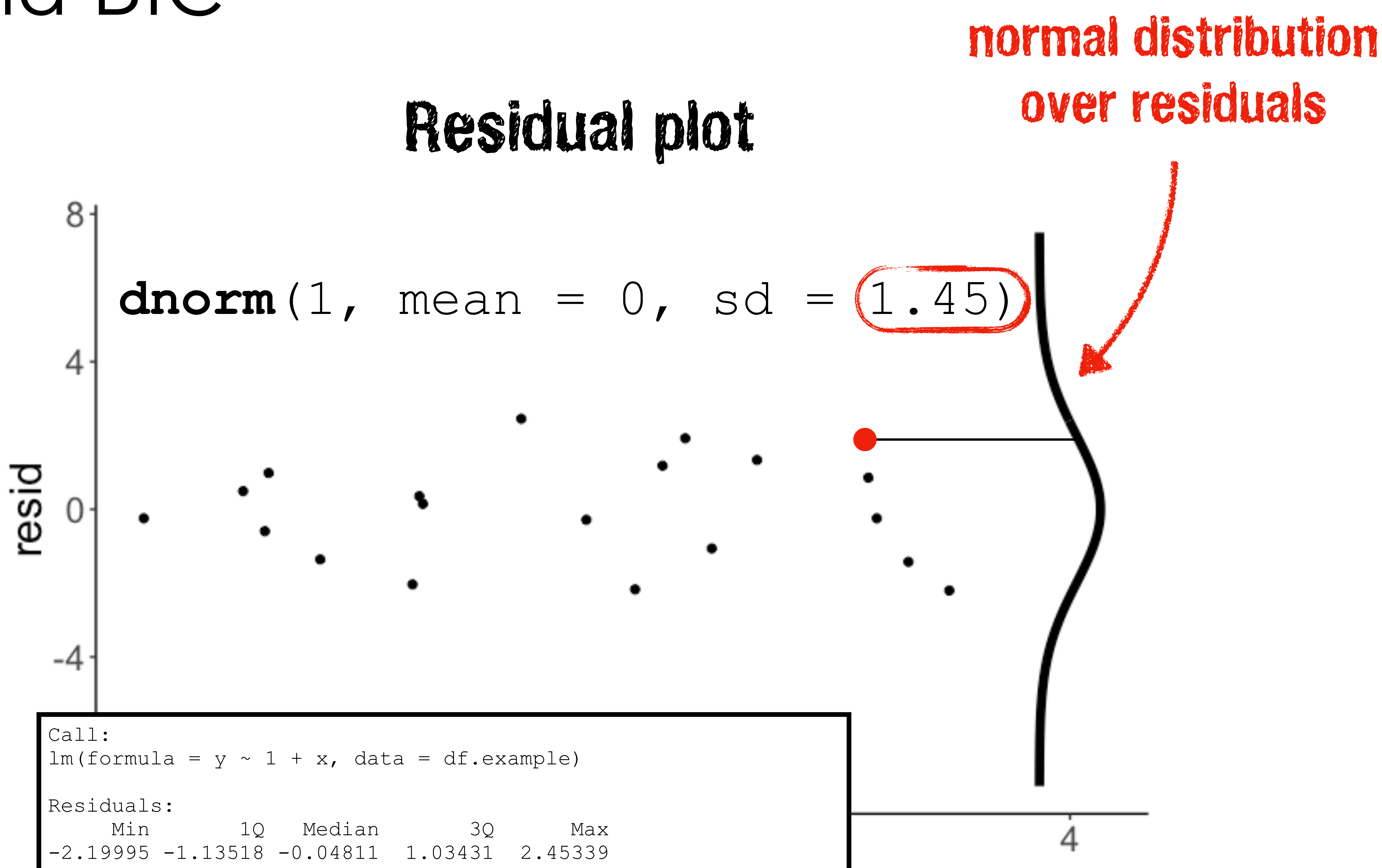
AIC and BIC

- How do we get the likelihood of our model?
  - in a linear regression, minimizing least squares is equivalent to maximizing the likelihood of the data given the model
- Assumptions of the linear model:
  - residuals are normally distributed with:
    - `mean = 0` and `sd = sigma`
  - calculate overall likelihood by computing the likelihood of each residual, and then multiplying

AIC and BIC

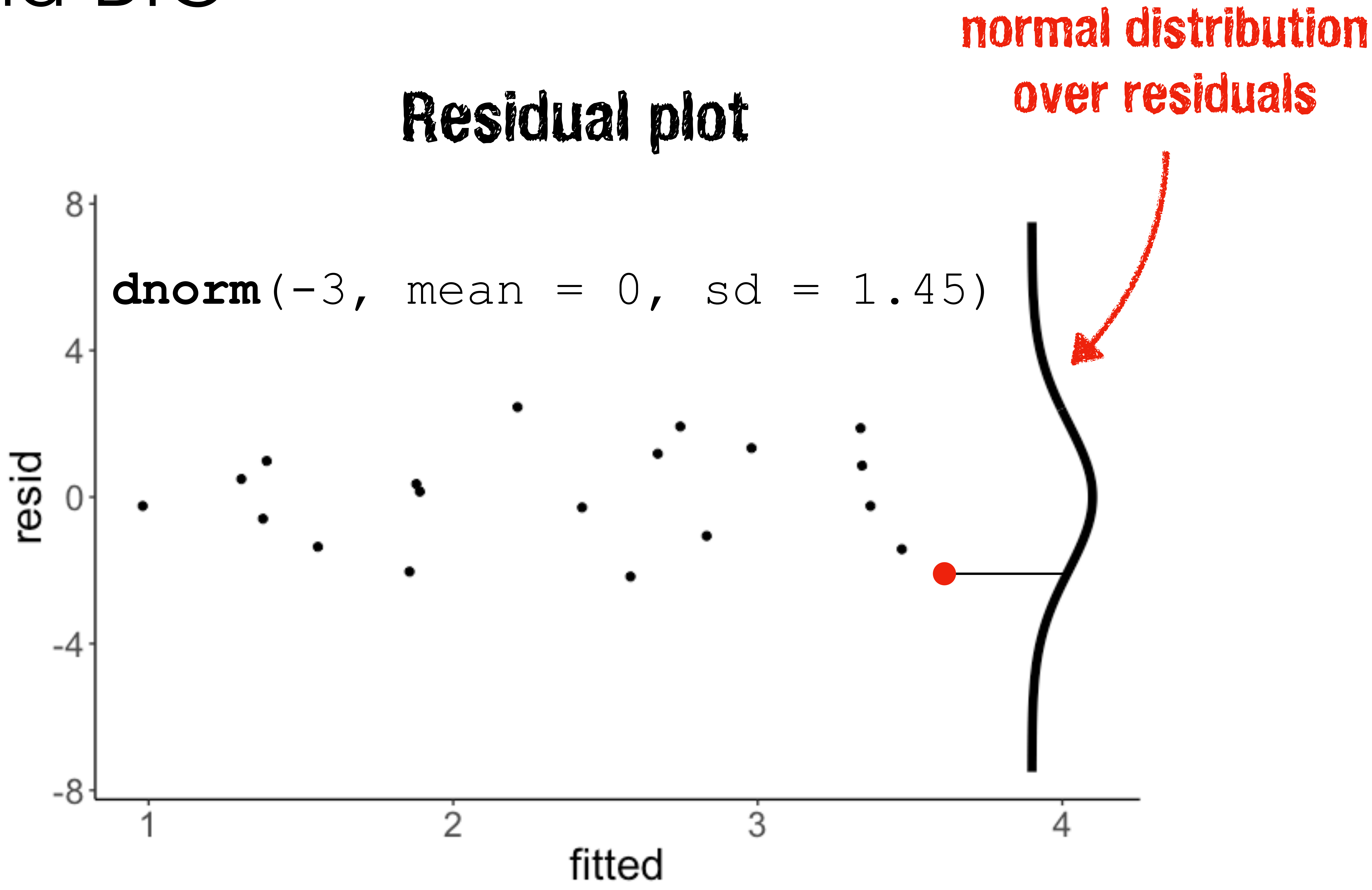


AIC and BIC




```
Call:
lm(formula = y ~ 1 + x, data = df.example)

Residuals:
     Min       1Q   Median       3Q      Max
-2.19995 -1.13518 -0.04811  1.03431  2.45339

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.8062     0.7157   1.126   0.2748
x             2.8232     1.1374   2.482   0.0232 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.449 on 18 degrees of freedom
Multiple R-squared:  0.255,	Adjusted R-squared:  0.2136
F-statistic: 6.161 on 1 and 18 DF,  p-value: 0.02315
```

AIC and BIC



since the data points are independent, we can calculate the overall likelihood by multiplying the likelihood of each observation

AIC and BIC

`dnorm(1.88, mean = 0, sd = 1.45) = 0.12`

```r
# generate some data
df.like = tibble(
  x = runif(20, min = 0, max = 1),
  y = 1 + 3 * x + rnorm(20, sd = 2)
)

# fit the model
fit = lm(formula = y ~ x,
         data = df.like)

# model summary
fit %>%
  glance()
```

x	y	fitted	resid	likelihood
0.90	5.22	3.34	1.88	0.12
0.27	0.20	1.56	-1.36	0.18
0.37	-0.18	1.86	-2.04	0.10
0.57	2.14	2.42	-0.28	0.27
0.91	3.13	3.37	-0.24	0.27
0.20	0.78	1.38	-0.59	0.25
0.90	4.20	3.34	0.86	0.23
0.94	2.05	3.47	-1.42	0.17
0.66	3.85	2.67	1.18	0.20
0.63	0.41	2.58	-2.17	0.09

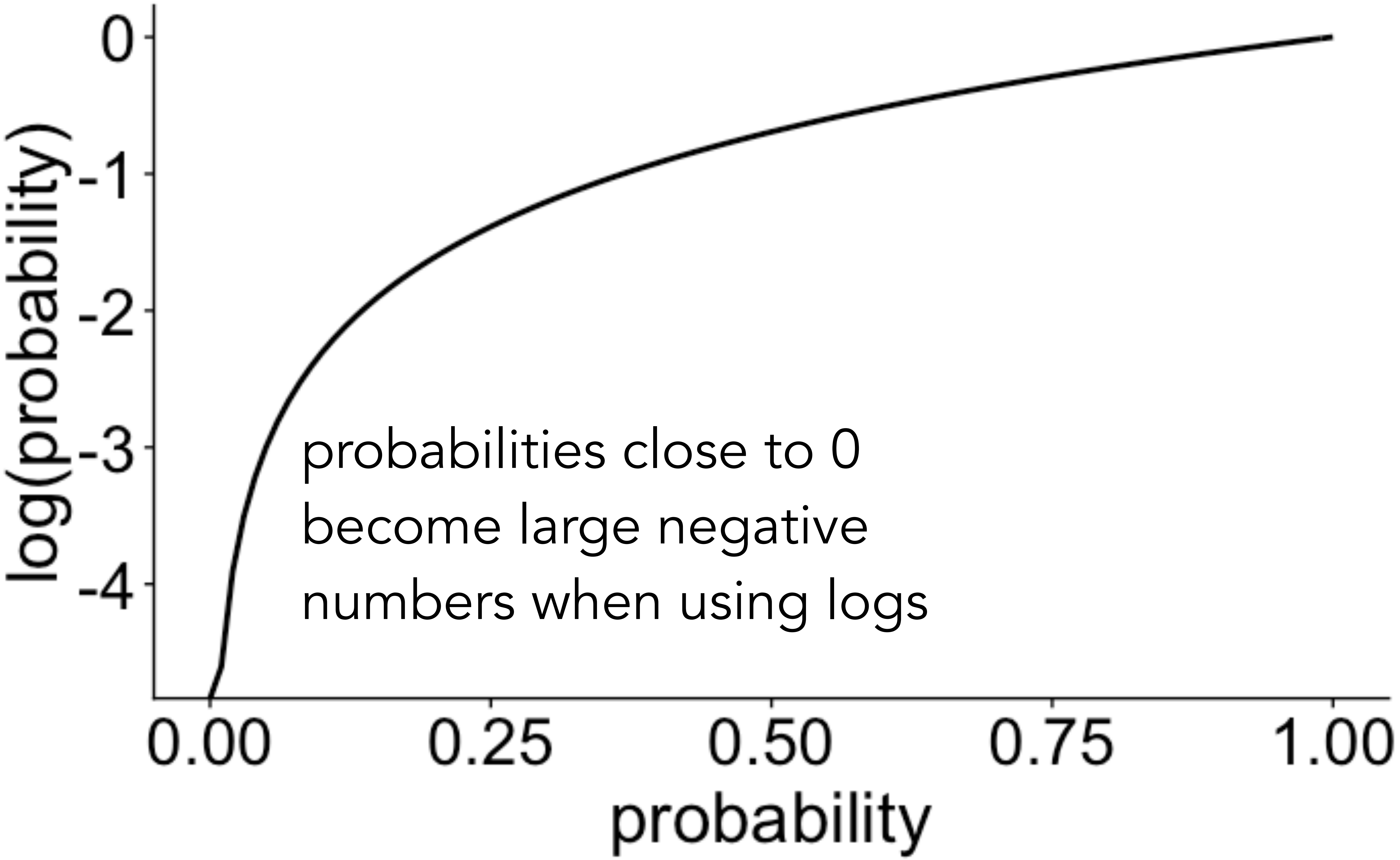
inferred standard deviation of the error

$$\sum_{i=1}^{n} \ln(\text{likelihood})$$

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.25	0.21	1.45	6.16	0.02	2	-34.74	75.47	78.46	37.77	18

$$e \sim \mathcal{N}(\text{mean} = 0,\ \text{sd} = 1.45)$$

`log()` is your friend!



Clue guide to probability

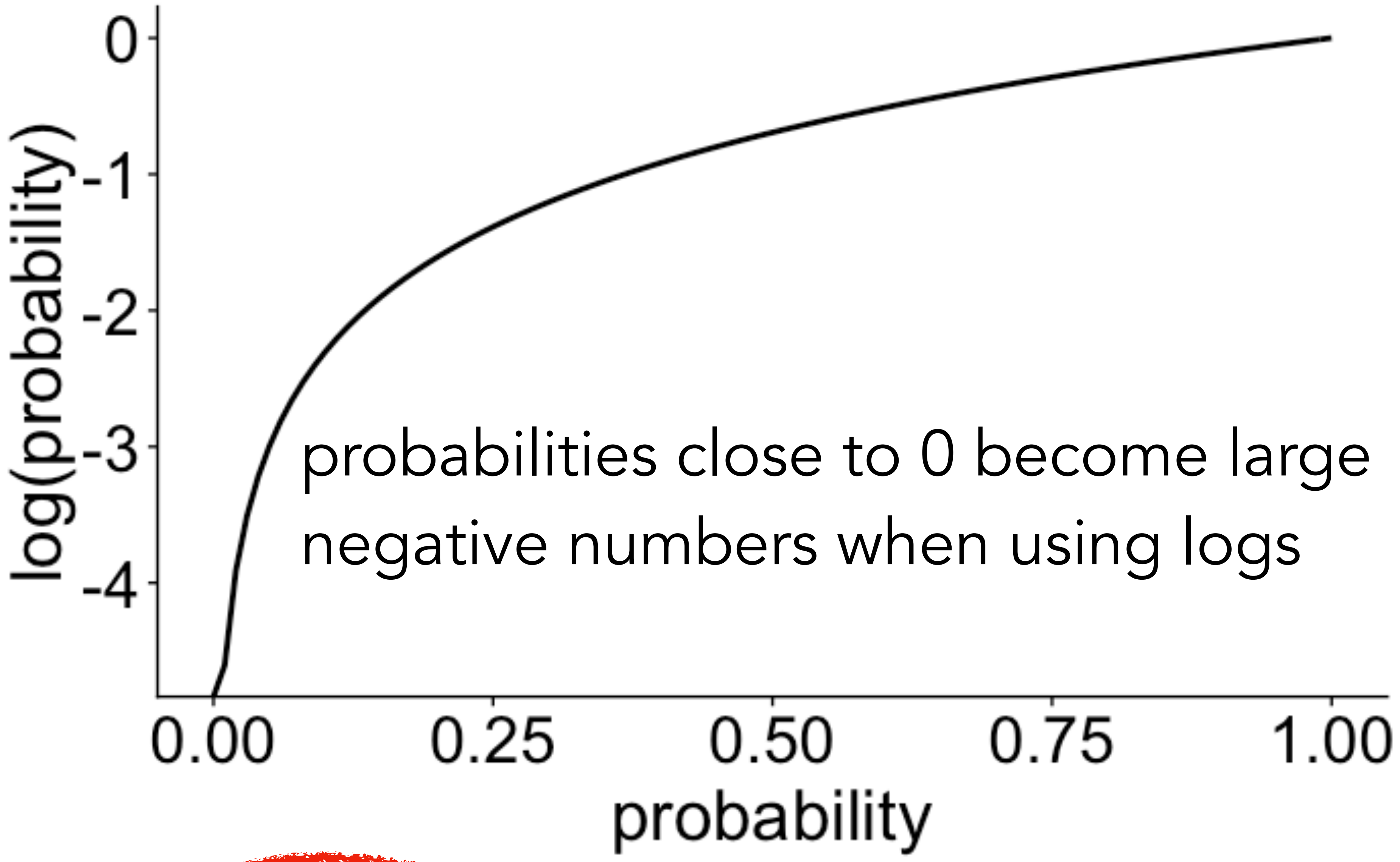
- *joint probability*:
- if A and B are independent then
- **Definition**: $p(A, B) = p(A) \cdot p(B)$

- $p(\text{Prof Plum, candle stick}) =$
  $p(\text{Prof Plum}) \cdot p(\text{candle stick}) =$
  $$\frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

Who?

What?

`log()` is your friend!



multiplying probabilities

$$0.01 \cdot 0.01 \cdot 0.01 \cdot 0.01 = 0.00000001$$

number becomes extremely small quickly

take log()

`log(0.01)= -4.60517`

number becomes large but that's ok

summing logs

$$(-4.60517) + (-4.60517) + (-4.60517) + (-4.60517) = -18.42068$$

transform back into probability

`exp(-18.42068)= 0.00000001`

often not necessary since we just use logLikelihood

AIC and BIC

`dnorm(1.88, mean = 0, sd = 1.45) = 0.12`

```r
# generate some data
df.like = tibble(
  x = runif(20, min = 0, max = 1),
  y = 1 + 3 * x + rnorm(20, sd = 2)
)

# fit the model
fit = lm(formula = y ~ x,
         data = df.like)

# model summary
fit %>%
  glance()
```

x	y	fitted	resid	likelihood
0.90	5.22	3.34	1.88	0.12
0.27	0.20	1.56	-1.36	0.18
0.37	-0.18	1.86	-2.04	0.10
0.57	2.14	2.42	-0.28	0.27
0.91	3.13	3.37	-0.24	0.27
0.20	0.78	1.38	-0.59	0.25
0.90	4.20	3.34	0.86	0.23
0.94	2.05	3.47	-1.42	0.17
0.66	3.85	2.67	1.18	0.20
0.63	0.41	2.58	-2.17	0.09

inferred standard deviation of the error

$$\sum_{i=1}^{n} \ln(\text{likelihood})$$

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.25	0.21	1.45	6.16	0.02	2	-34.74	75.47	78.46	37.77	18

$$e \sim \mathcal{N}(\text{mean} = 0,\ \text{sd} = 1.45)$$

AIC and BIC

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L})$$

$\ln(\hat{L})$ = maximized value of the likelihood function of the model **-34.74**

k = number of parameters in the model **3**

n = number of observations **20**

the `sd` of the normal distribution modeling the residuals counts as a parameter

```
lm(formula = y ~ 1 + x, data = df.example)
```

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.25	0.21	1.45	6.16	0.02	2	-34.74	75.47	78.46	37.77	18

AIC and BIC

$$\text{AIC} = 2k - 2\ln(\hat{L}) = 2 \cdot 3 - 2 \cdot (-34.74) = 75.47$$

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L}) = \ln(20) \cdot 3 - 2 \cdot (-34.74) = 78.46$$

$\ln(\hat{L})$ = maximized value of the likelihood function of the model **-34.74**

k = number of parameters in the model **3**

n = number of observations **20**

the `sd` of the normal distribution modeling the residuals counts as a parameter

```
lm(formula = y ~ 1 + x, data = df.example)
```

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.25	0.21	1.45	6.16	0.02	2	-34.74	75.47	78.46	37.77	18

AIC and BIC

$$\mathrm{AIC} = 2k - 2\ln(\hat{L})$$

$$\mathrm{BIC} = \ln(n)k - 2\ln(\hat{L})$$

ΔBIC	Evidence against higher BIC
0 to 2	Not worth more than a bare mention
2 to 6	Positive
6 to 10	Strong
>10	Very Strong

- for both AIC and BIC, *lower* is better!
- neither provide a test of a model in the sense of testing a null hypothesis
  - AIC or BIC tell us nothing about the absolute quality of a model, only the quality relative to other models
- BIC generally penalizes free parameters more strongly than AIC (though it depends on the size of *n*)

What shall I use when?

- Use it all!
- ideally, the different measures provide converging evidence

Table 2
Summary of the model results. Values for *r* and RMSE indicate means (with 5% and 95% quantiles in parentheses) based on 100 split-half cross-validation runs. BIC scores are based on running the models on the full data set.

Model	*r*	RMSE	BIC
Difference & pivotality	.86 (.66, .95)	10.56 (6.17, 17.21)	158.59
Difference	.70 (.30, .90)	26.92 (16.4, 40.6)	209.74
Pivotality	.63 (.41, .77)	14.23 (11.39, 17.54)	199.53
Optimality	.66 (.42, .84)	14.55 (10.54, 17.91)	199.47

Note: BIC = Bayesian Information Criterion (lower values indicate better model performance).

Quick recap: AIC and BIC

- AIC = Akaike Information Criterion
- BIC = Bayesian Information Criterion

not that much Bayesian about it ...

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

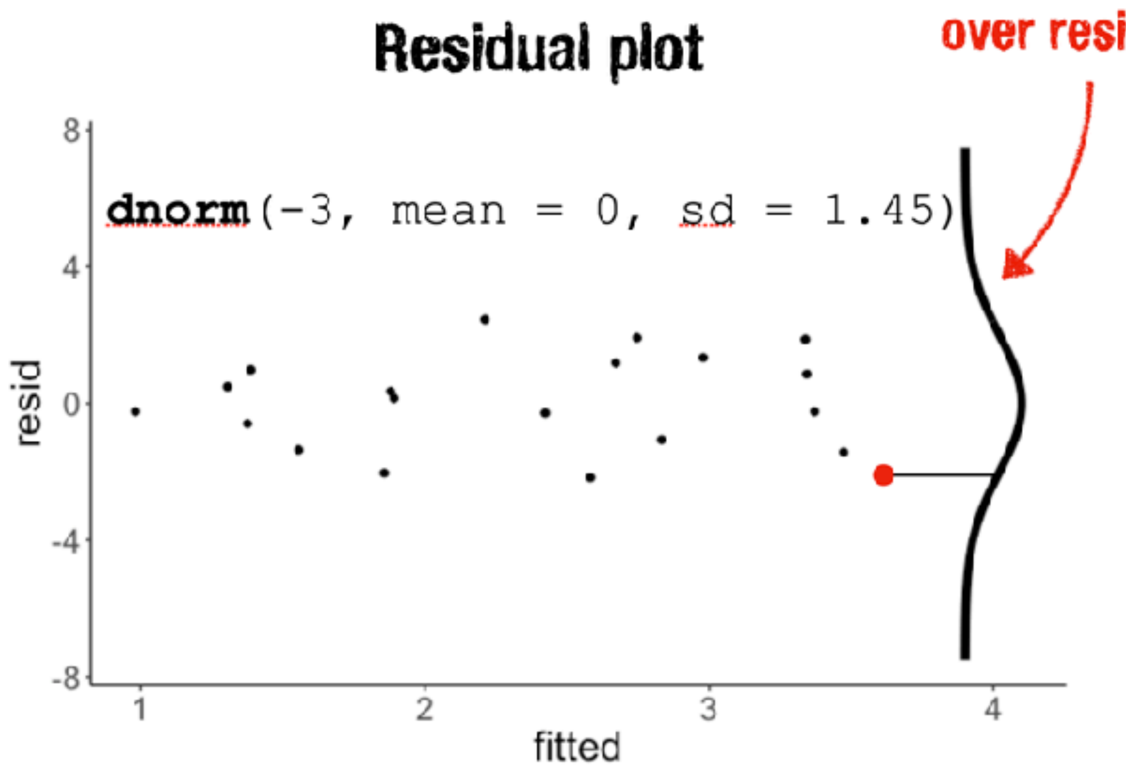
?

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L})$$

$\hat{L}$ = maximized value of the likelihood function of the model
k = number of parameters in the model
n = number of observations

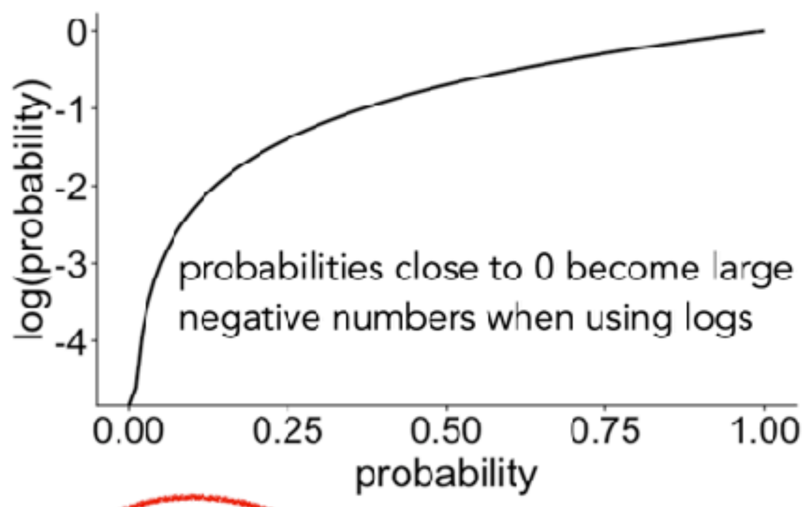
AIC and BIC

normal distribution over residuals



since the data points are independent, we can calculate the overall likelihood by multiplying the likelihood of each observation

`log()` is your friend!



multiplying probabilities

$$0.01 \cdot 0.01 \cdot 0.01 \cdot 0.01 = 0.00000001$$

number becomes extremely small quickly

take log()

`log(0.01)= -4.60517`

summing logs

number becomes large but that's ok

$$(-4.60517) + (-4.60517) + (-4.60517) + (-4.60517) = -18.42068$$

transform back into probability

`exp(-18.42068)= 0.00000001`

often not necessary since we just use logLikelihood

27

AIC and BIC

`dnorm(1.88, mean = 0, sd = 1.45) = 0.12`

```r
# generate some data
df.like = tibble(
  x = runif(20, min = 0, max = 1),
  y = 1 + 3 * x + rnorm(20, sd = 2)
)

# fit the model
fit = lm(formula = y ~ x,
         data = df.like)

# model summary
fit %>%
  glance()
```

x	y	fitted	resid	likelihood
0.90	5.22	3.34	1.88	0.12
0.27	0.20	1.56	-1.36	0.18
0.37	-0.18	1.86	-2.04	0.10
0.57	2.14	2.42	-0.28	0.27
0.91	3.13	3.37	-0.24	0.27
0.20	0.78	1.38	-0.59	0.25
0.90	4.20	3.34	0.86	0.23
0.94	2.05	3.47	-1.42	0.17
0.66	3.85	2.67	1.18	0.20
0.63	0.41	2.58	-2.17	0.09

inferred standard deviation of the error

$$\sum_{i=1}^{n} \ln(\text{likelihood})$$

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.25	0.21	1.45	6.16	0.02	2	-34.74	75.47	78.46	37.77	18

$$e \sim \mathcal{N}(\text{mean} = 0, \text{sd} = 1.45)$$

28

Linear mixed effects models

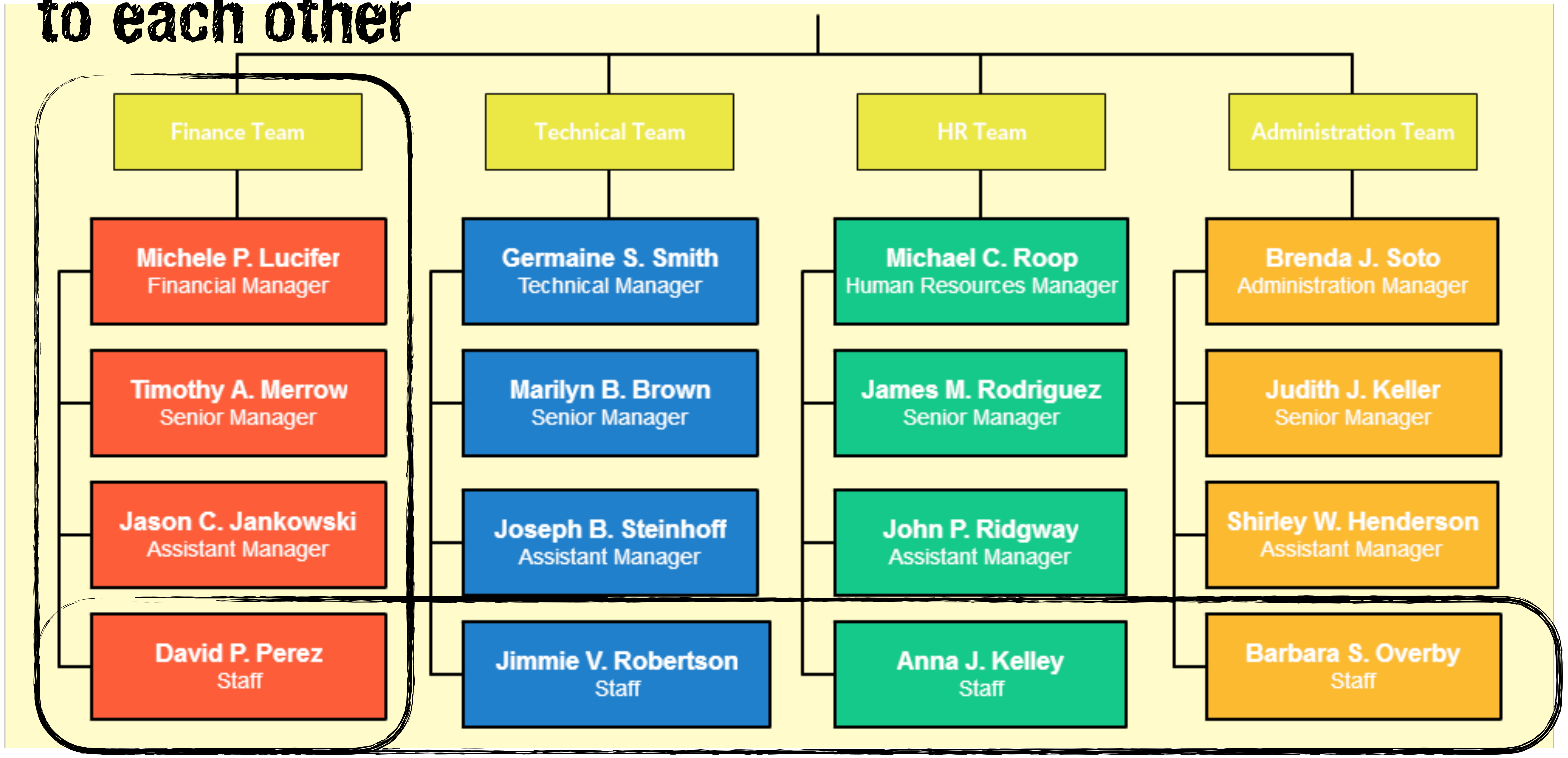
Dependence (vs iid)

- so far, all the models that we've discussed (linear model with different kinds of predictors and contrasts) make the assumption that the data are **iid** (independent, and identically distributed)

- often this assumption is violated
  - **psychology experiments**: many observations from the same participants
  - **survey data**: different populations between different states in the US
  - **time series**: distribution at $t + 1$ depends on t

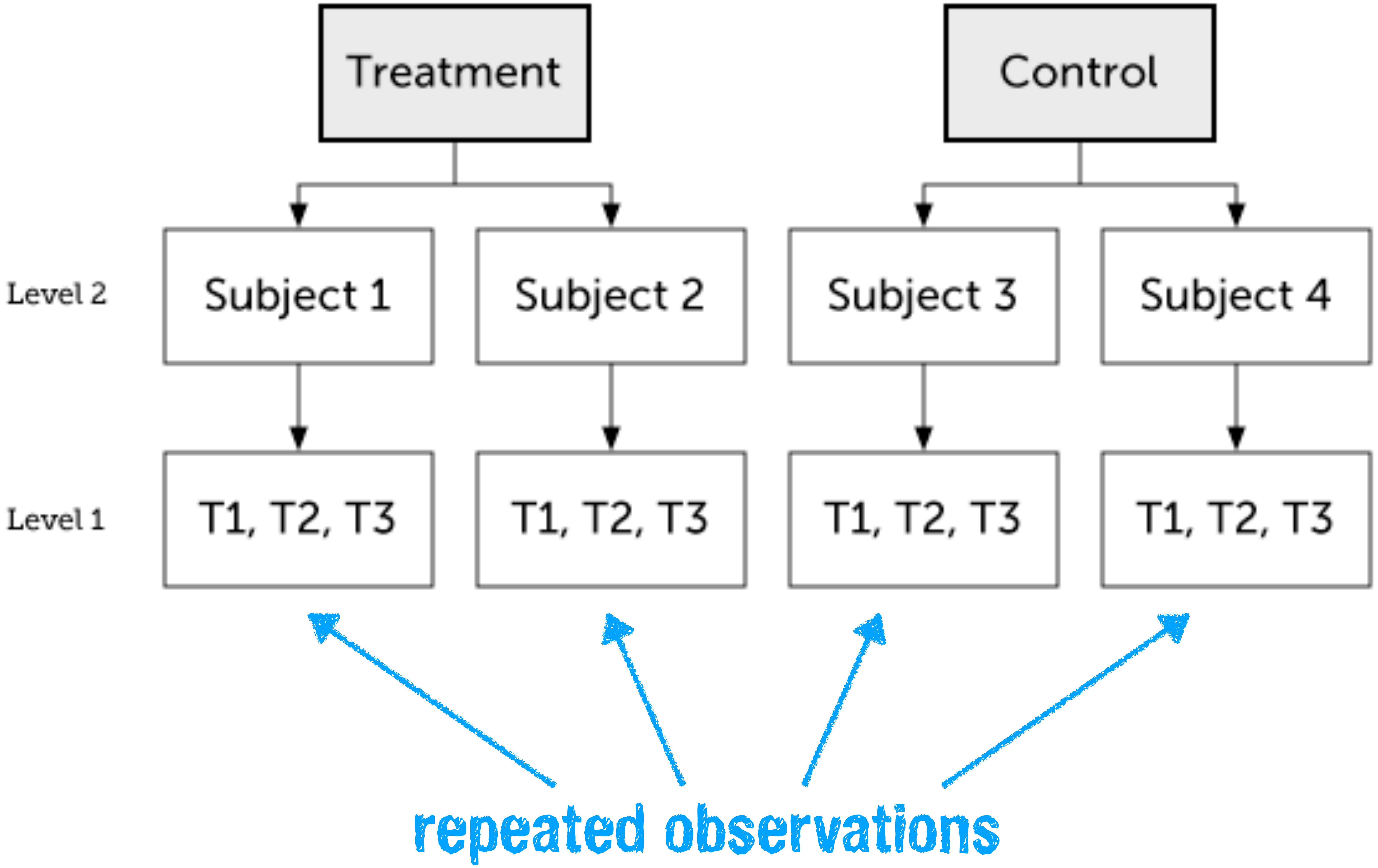
Main use cases: **Hierarchical models**

more similar to each other



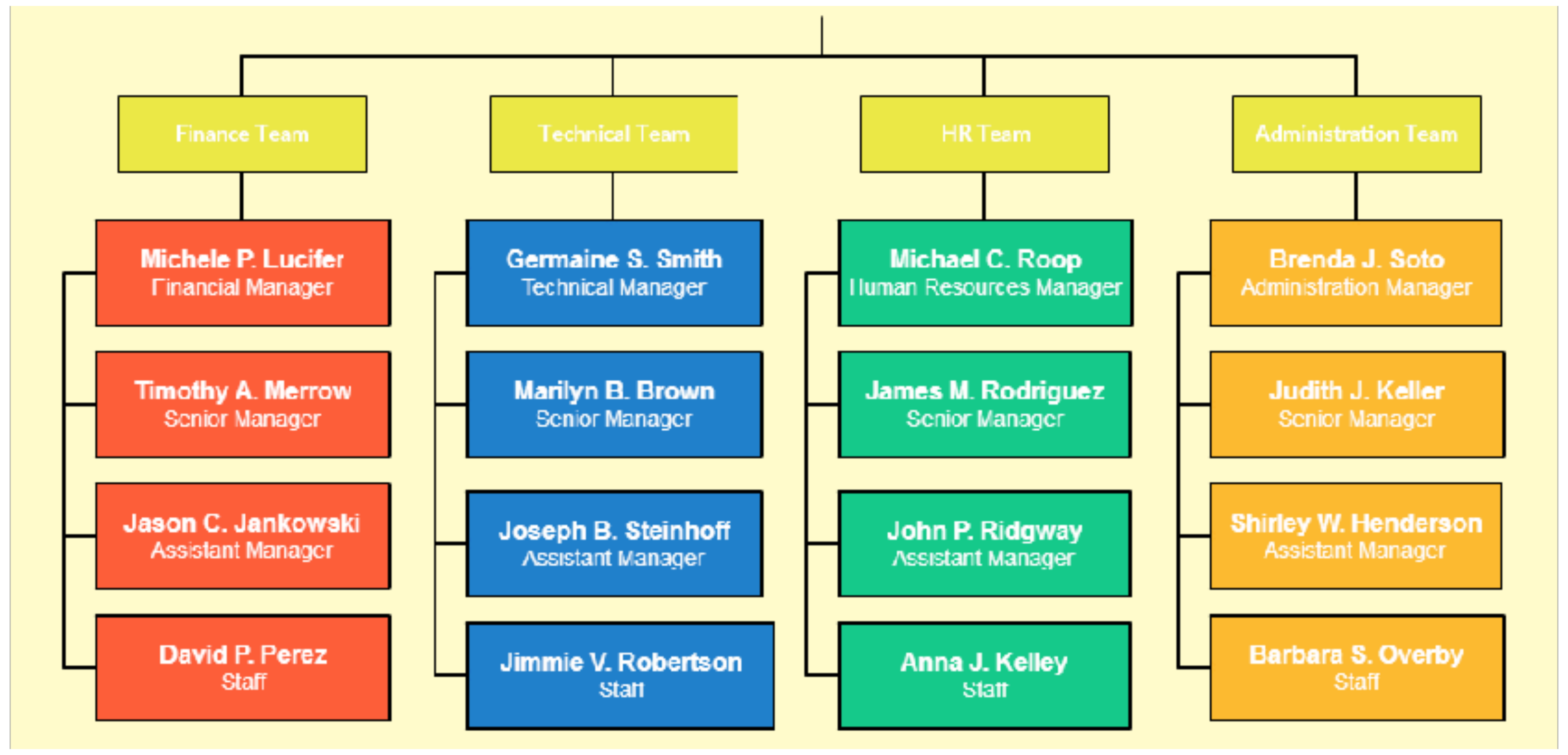
less similar to each other

Main use cases: **Longitudinal models**

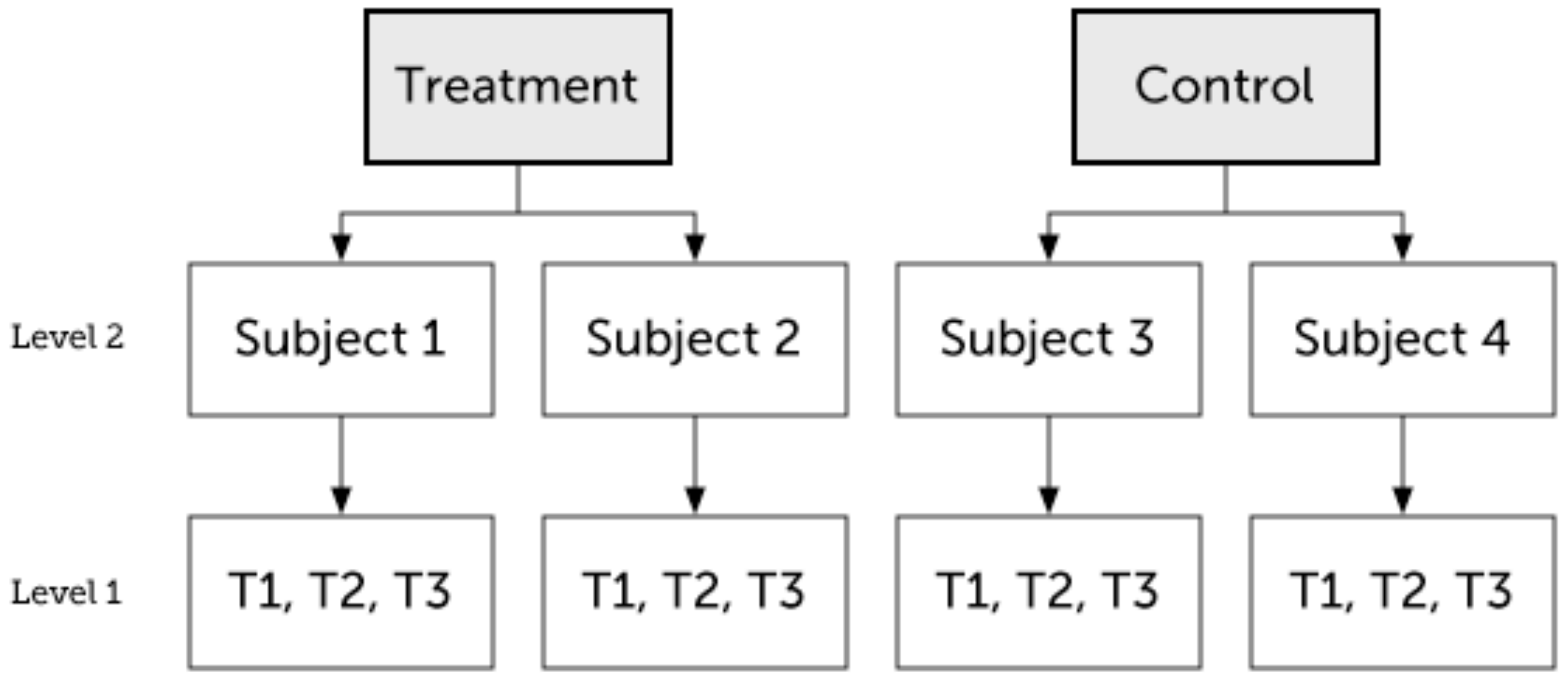


Linear mixed effects models

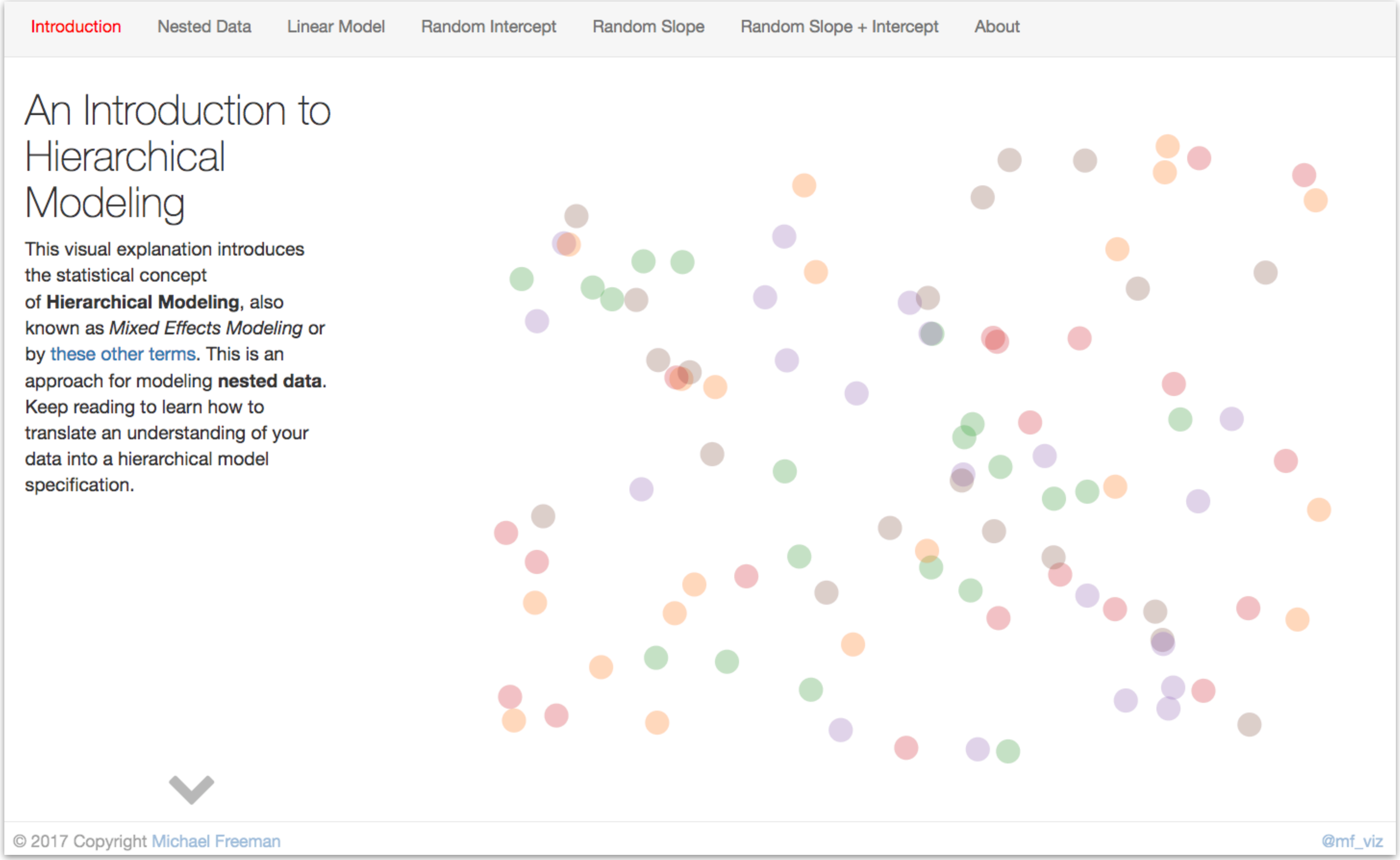
Hierarchical models



Longitudinal models



- allow us to account for dependencies in our data
- **hierarchical models**: schools > teachers > students
- **longitudinal models**: repeated observations from the same people



http://mfviz.com/hierarchical-models/

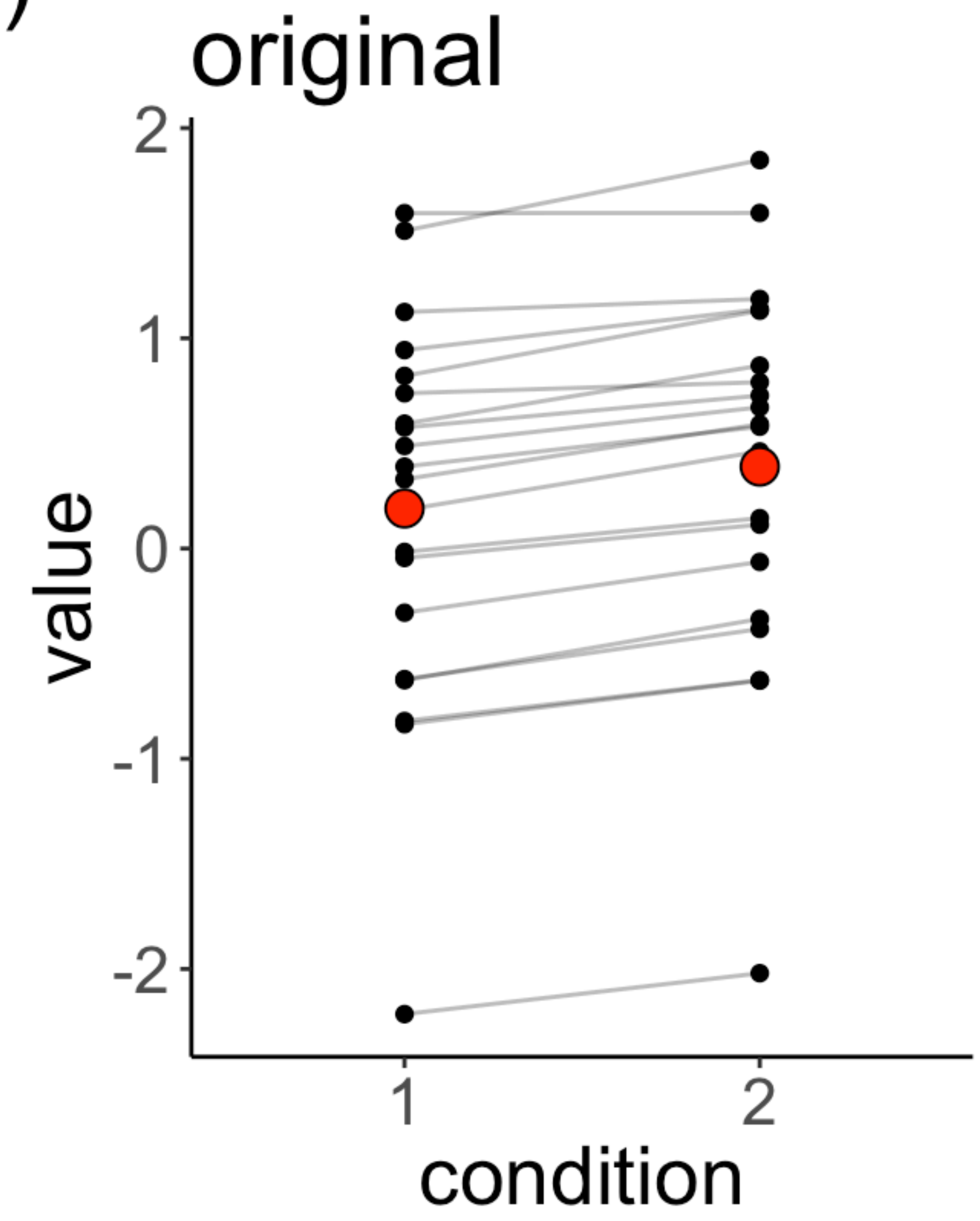
Modeling dependence in data

Dependence

Does it really matter?

Is there a significant difference between conditions 1 and 2?

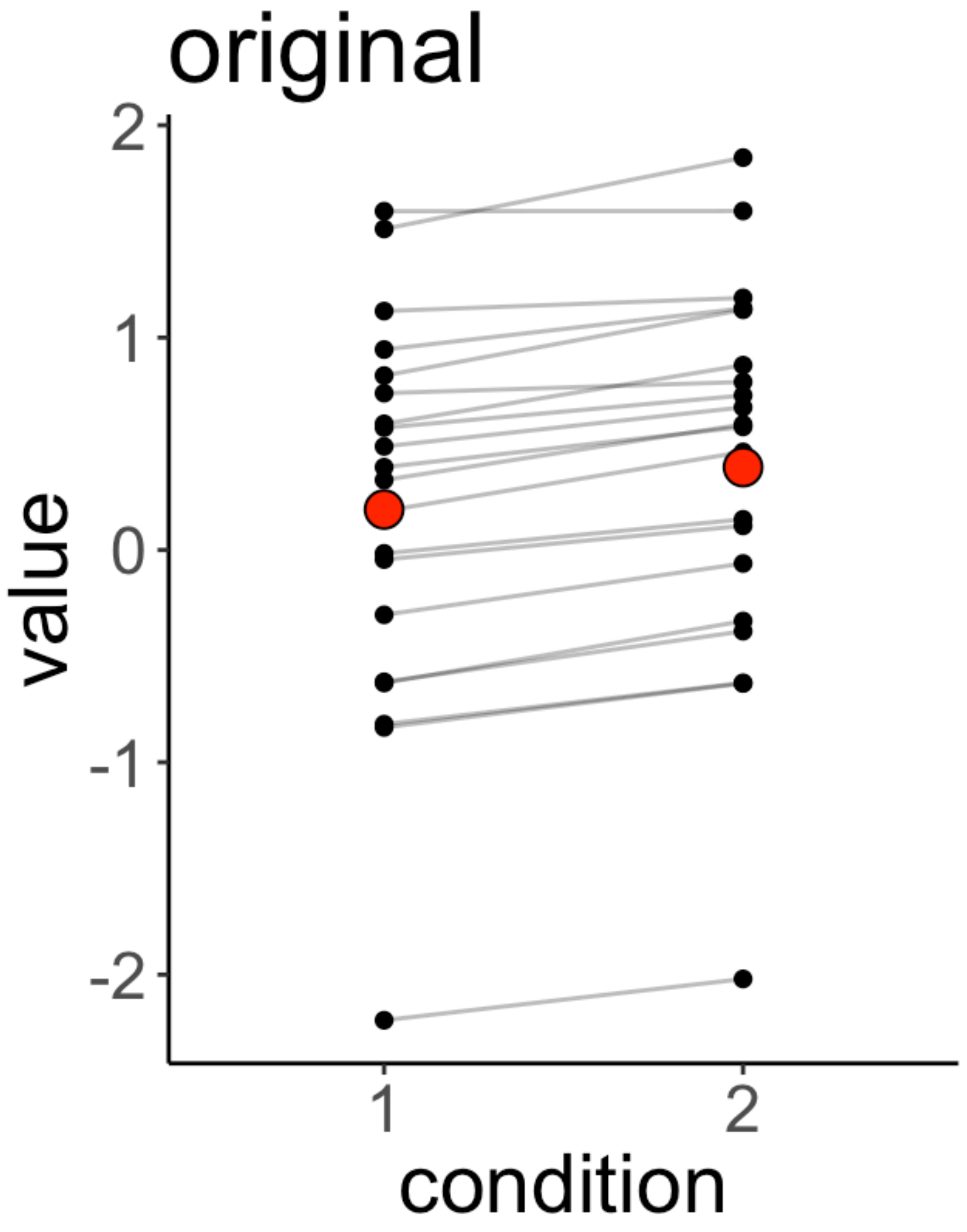
a)



Dependence

assuming independence!

```r
# linear model
lm(formula = value ~ condition,
   data = df.original) %>%
  summary()
```




```
Call:
lm(formula = value ~ condition, data = df.original)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.4100 -0.5530  0.1945  0.5685  1.4578 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.1905     0.2025   0.941    0.353
condition2    0.1994     0.2864   0.696    0.491

Residual standard error: 0.9058 on 38 degrees of freedom
Multiple R-squared:  0.01259, Adjusted R-squared:  -0.0134 
F-statistic: 0.4843 on 1 and 38 DF,  p-value: 0.4907
```

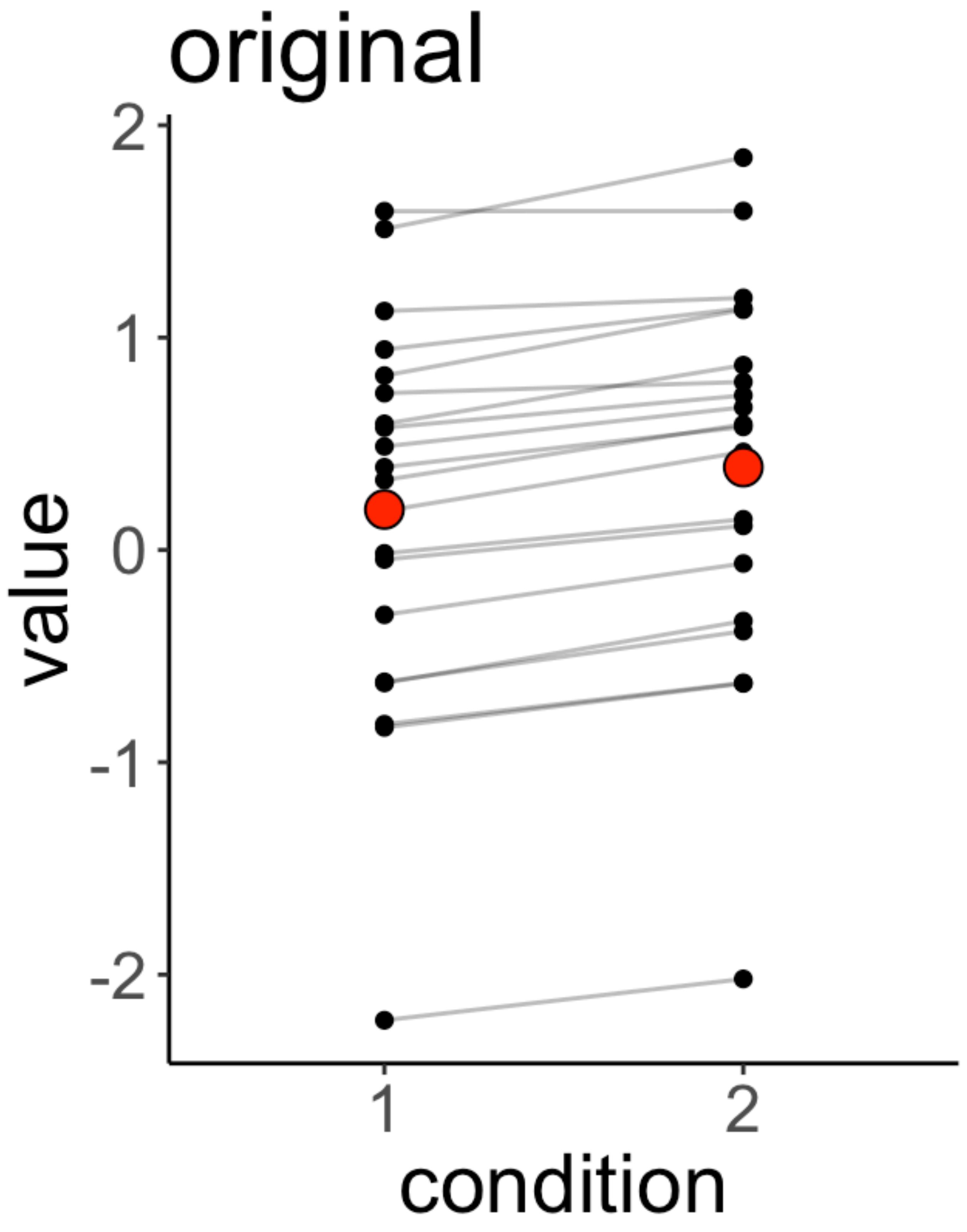
- we ignore the fact that we have repeated observations from the same participants
- in the data it looks like there is a small but consistent effect of condition

meet **`lmer`**()

Dependence

new syntax

```r
# fit a linear mixed effects model
lmer(formula = value ~ condition + (1 | participant),
                      data = df.original) %>%
  summary()
```




```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ condition + (1 | participant)
   Data: df.original

REML criterion at convergence: 17.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.55996 -0.36399 -0.03341  0.34400  1.65823

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 0.816722 0.90373
 Residual                0.003796 0.06161
Number of obs: 40, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.19052    0.20255   0.941
condition2   0.19935    0.01948  10.231

Correlation of Fixed Effects:
           (Intr)
condition2 -0.048
```

no p-value!

No P-value

Dependence

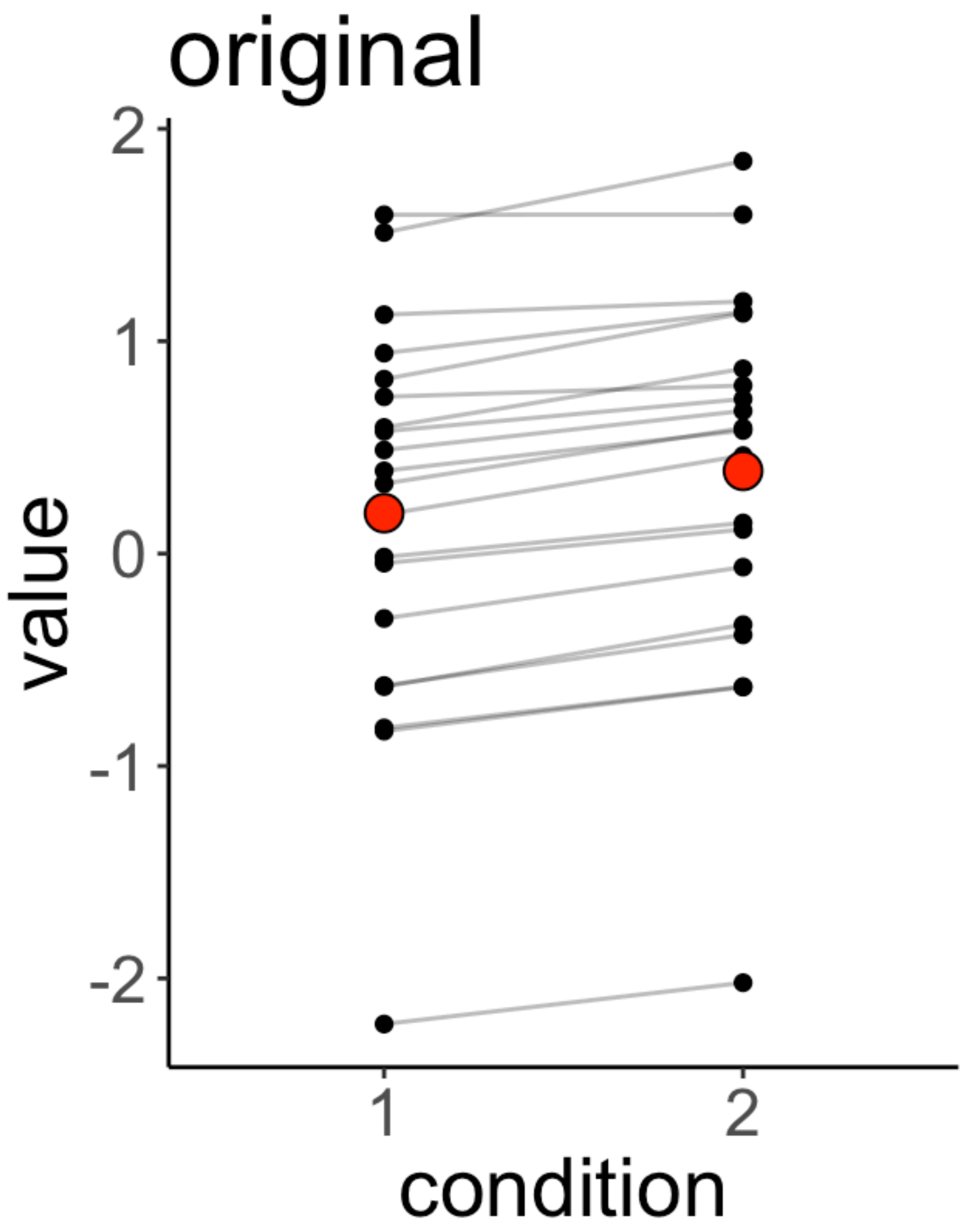
we can still do our good ol model comparison trick

```
# fit models
fit.compact = lmer(formula = value ~ 1 + (1 | participant),
                   data = df.original)

fit.augmented = lmer(formula = value ~ 1 + condition + (1 | participant),
                     data = df.original)

# compare via Chisq-test
anova(fit.compact, fit.augmented)
```

```
refitting model(s) with ML (instead of REML)
Data: df.original
Models:
fit.compact: value ~ 1 + (1 | participant)
fit.augmented: value ~ 1 + condition + (1 | participant)
              Df    AIC    BIC   logLik deviance  Chisq Chi Df Pr(>Chisq)
fit.compact    3 53.315 58.382 -23.6575   47.315
fit.augmented  4 17.849 24.605  -4.9247    9.849 37.466      1  9.304e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Dependence

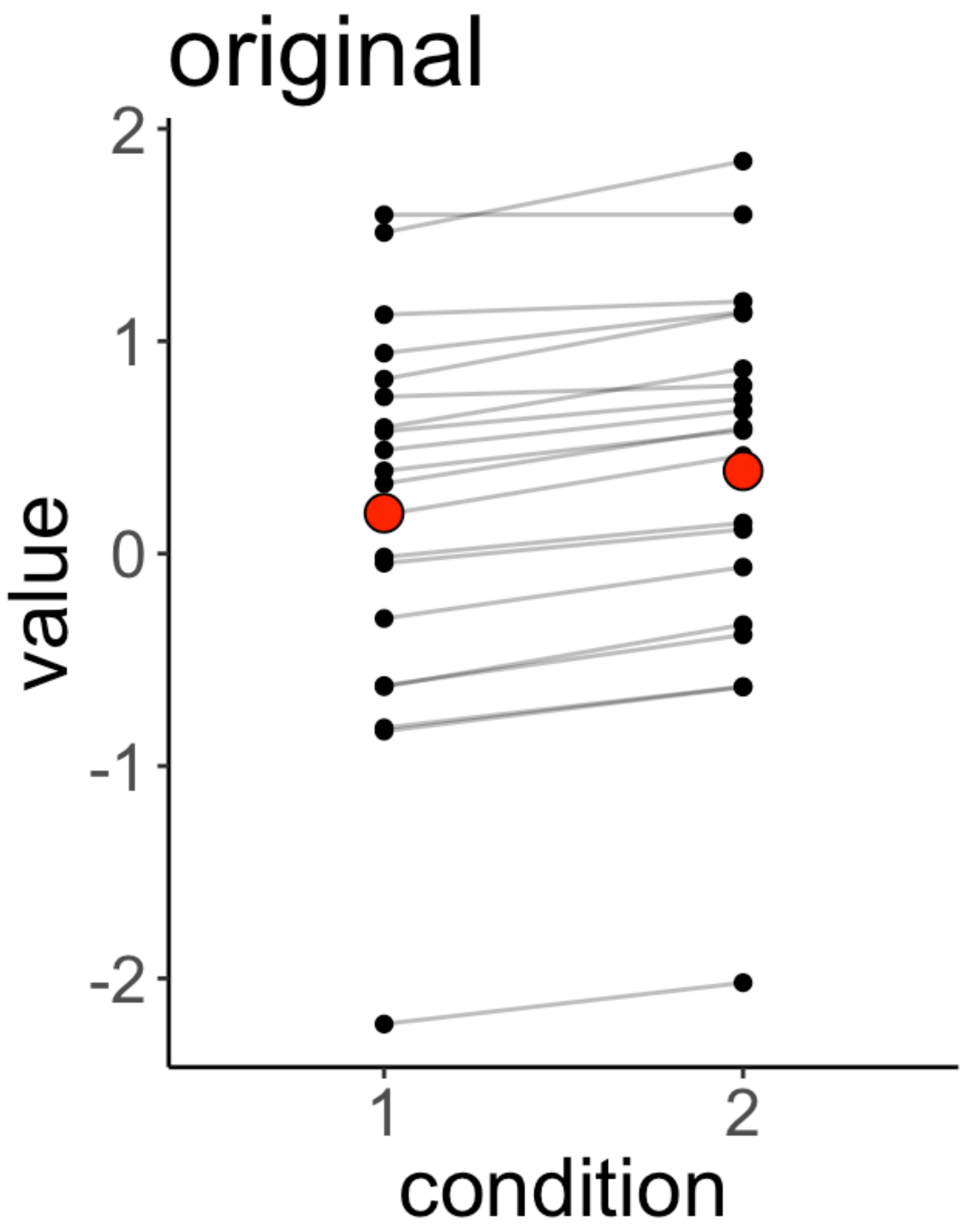
Why is the effect of condition significant when we account for the dependence in the data?



- there are large interindividual differences in the baseline
- the variance explained by the effect of condition is (much) smaller than the interindividual variance
- **but**: the effect of condition is highly consistent

Dependence

Why is the effect of condition significant when we account for the dependence in the data?



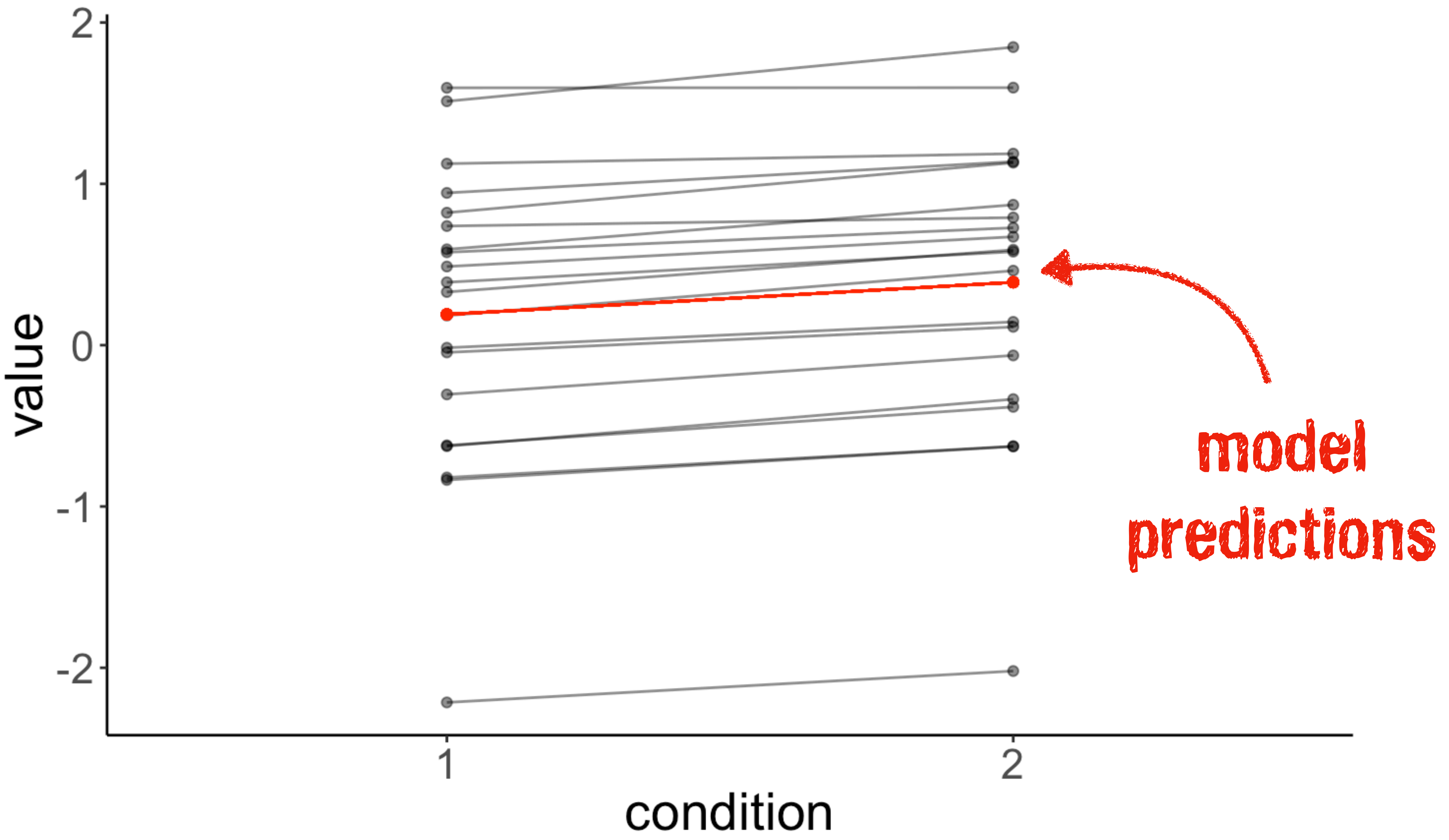
- by explicitly modeling the dependence in the data, we account for the interindividual differences

let's visualize the model predictions!

Linear model (assuming independence)

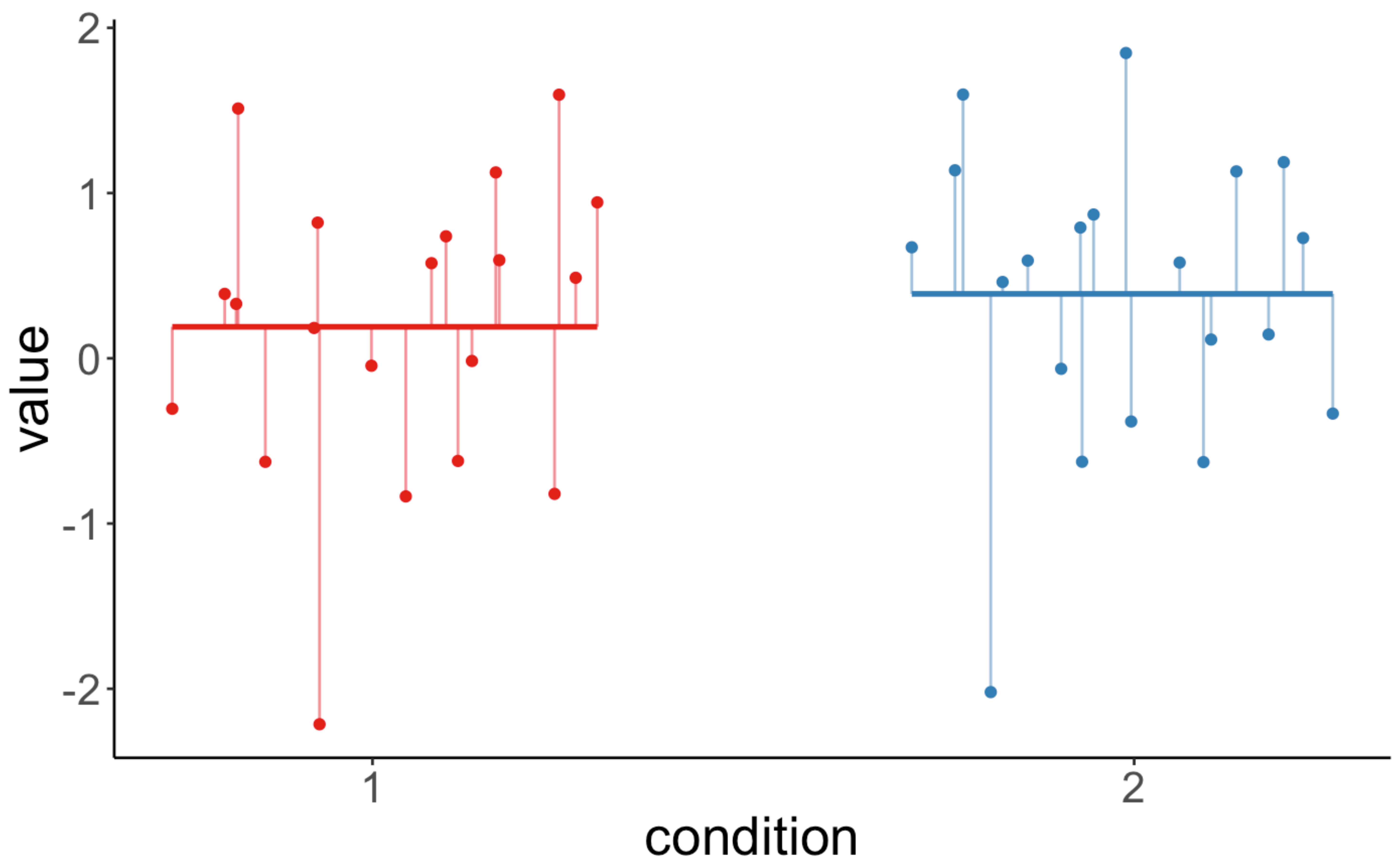
Predictions by the linear model which assumes independence

```
lm(formula = value ~ 1 + condition,
   data = df.original)
```



Linear model (assuming independence)

Residuals of the model

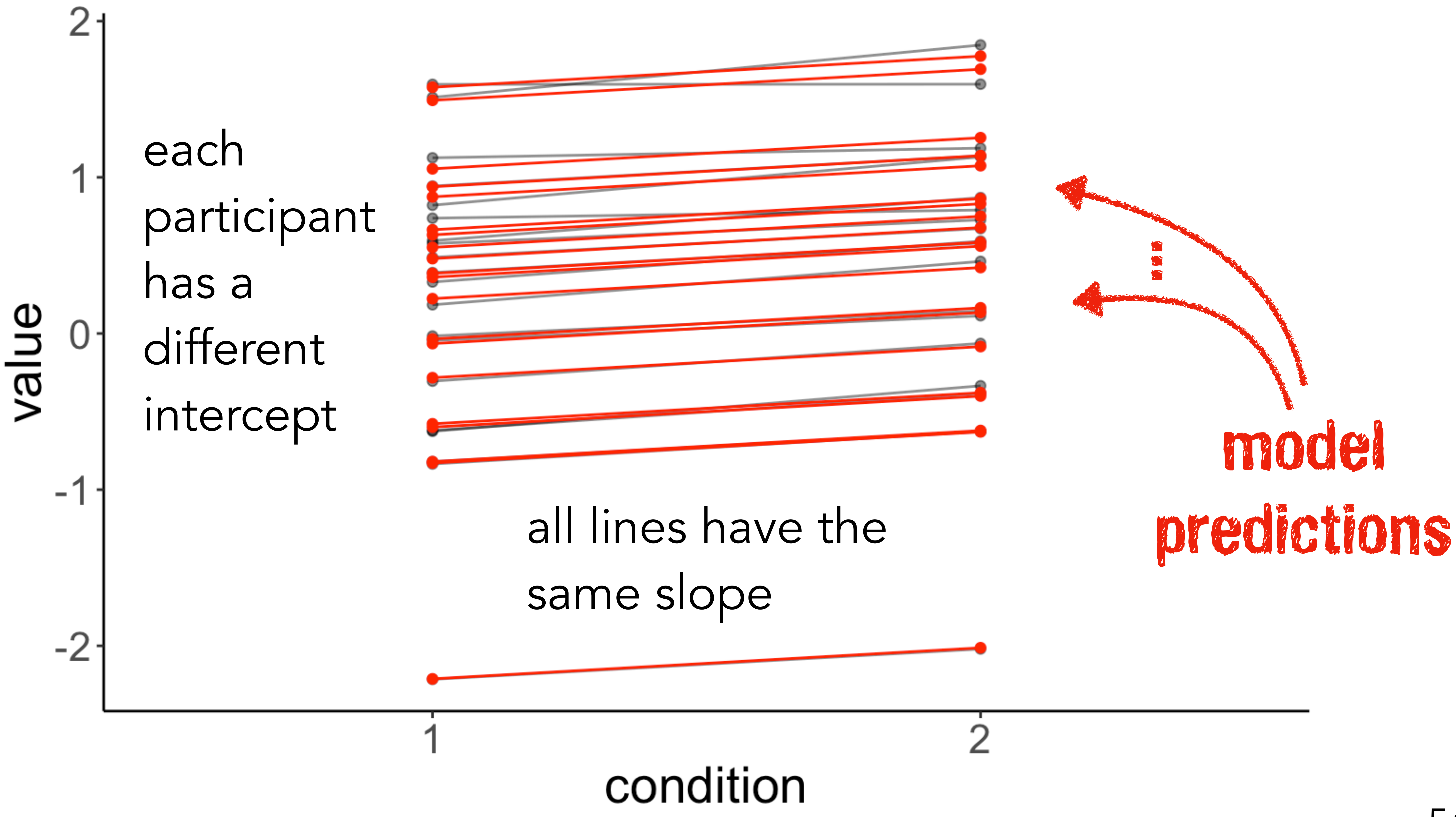


This is not much better than fitting a single line (point).

Linear mixed effects model (accounting for dependence)

Predictions by the linear mixed effects model

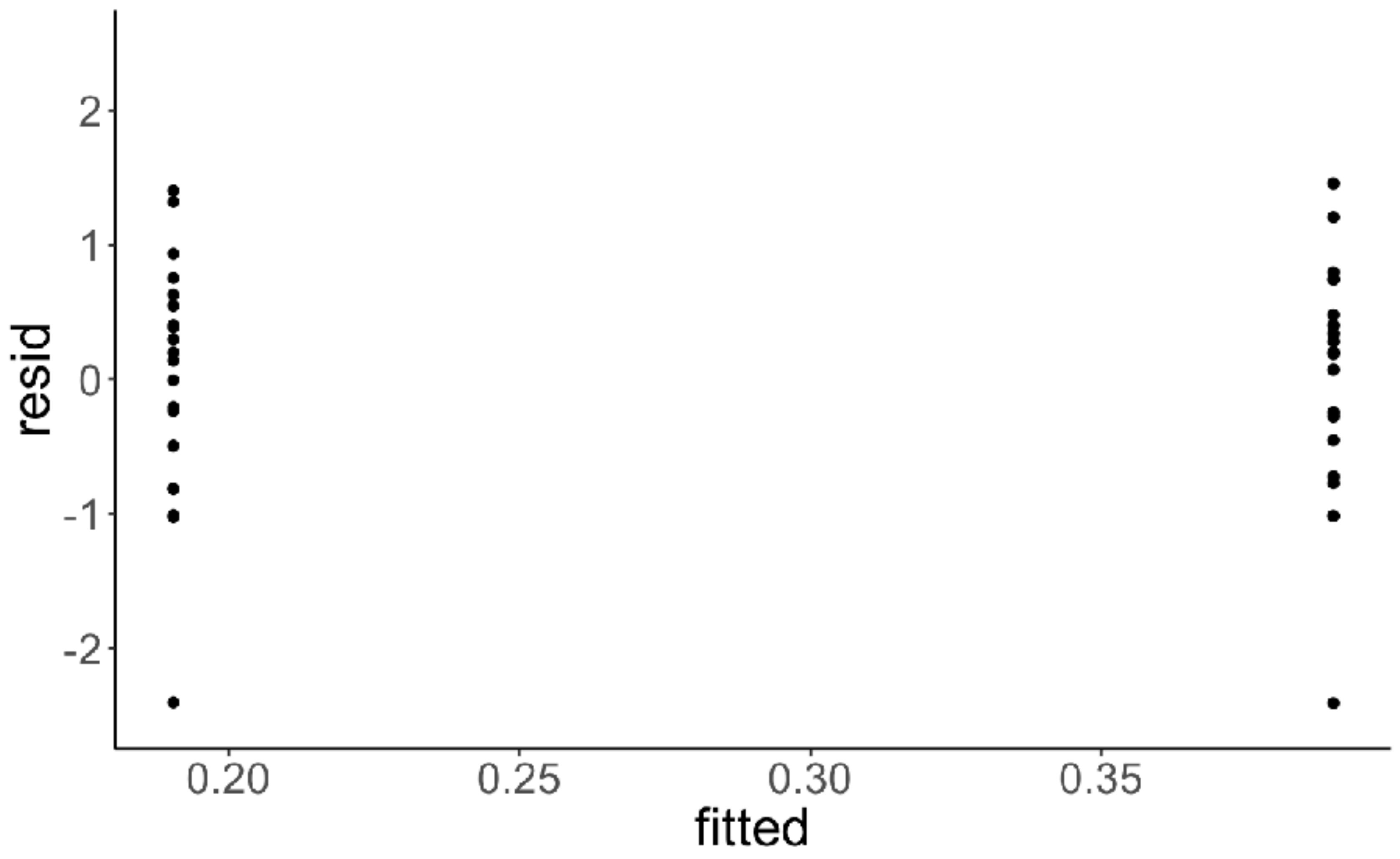
```
lmer(formula = value ~ 1 + condition + (1 | participant),
     data = df.original)
```



Model comparison

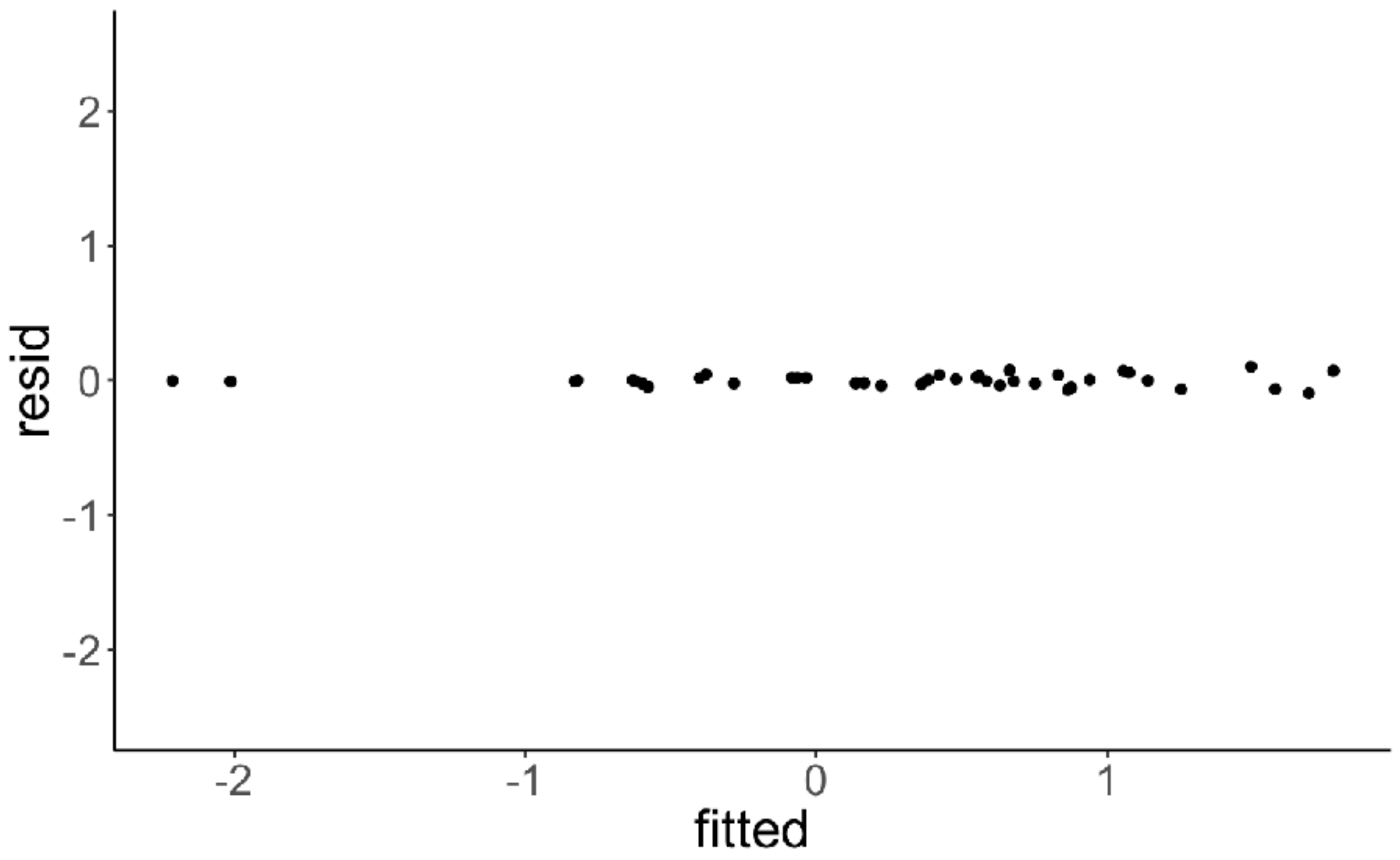
Residual plots

```
lm(formula = value ~ 1 + condition,
   data = df.original)
```



much variance left to be explained

```
lmer(formula = value ~ 1 + condition +
                (1 | participant),
     data = df.original)
```



almost all variance explained

Model comparison

Hypothesis test

Is taking into account individual differences worth it?

```r
# fit models (without and with dependence)
fit.compact = lm(formula = value ~ 1 + condition,
                 data = df.original)

fit.augmented = lmer(formula = value ~ 1 + condition + (1 | participant),
                     data = df.original)

# compare models
# note: the lmer model has to be supplied first
anova(fit.augmented, fit.compact)
```

```
refitting model(s) with ML (instead of REML)
Data: df.original
Models:
fit.compact: value ~ 1 + condition
fit.augmented: value ~ 1 + condition + (1 | participant)
              Df     AIC     BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
fit.compact    3 109.551 114.617 -51.775  103.551
fit.augmented  4  17.849  24.605  -4.925    9.849 93.701      1  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Linear model

```
lm(formula = value ~ 1 + condition,
   data = df.original)
```

$$\text{value}_i = b_0 + b_1 \cdot \text{condition}_i + e_i$$

i = observation

$$e_i \sim \mathcal{N}(\text{mean} = 0, \text{sd} = s_{\text{error}})$$

3 parameters: b_0, b_1, s_{error}

Linear mixed effects model

```
lmer(formula = value ~ 1 + condition +
                 (1 | participant),
     data = df.original)
```

$$\text{value}_{i,j} = b_0 + b_1 \cdot \text{condition}_{i,j} + U_i + e_i$$

i = participant,

j = time point

$$e_i \sim \mathcal{N}(\text{mean} = 0, \text{sd} = s_{\text{error}})$$

$$U_i \sim \mathcal{N}(\text{mean} = 0, \text{sd} = s_{\text{U}})$$

b_0, b_1 = fixed effects

U_i = random effect

here: random intercept

4 parameters: b_0, b_1, s_{error}, s_{U}

Model coefficients

Linear model

```
fit = lm(formula = value ~ 1 + condition,
         data = df.original)
coef(fit)
```

```
(Intercept)   condition2
  0.1905239    0.1993528
```

- one intercept
- one slope for condition

Linear mixed effects model

```
fit = lmer(formula = value ~ 1 + condition +
                       (1 | participant),
           data = df.original)
coef(fit)
```

```
$participant
   (Intercept)  condition2
1  -0.57839428  0.1993528
2   0.22299824  0.1993528
3  -0.82920677  0.1993528
4   1.49310938  0.1993528
5   0.36042775  0.1993528
6  -0.82060123  0.1993528
7   0.47929171  0.1993528
8   0.66401020  0.1993528
9   0.55135879  0.1993528
10 -0.28306703  0.1993528
11  1.57681676  0.1993528
12  0.38457642  0.1993528
13 -0.59969682  0.1993528
14 -2.21148391  0.1993528
15  1.05439374  0.1993528
16 -0.06476643  0.1993528
17 -0.03505690  0.1993528
18  0.93945348  0.1993528
19  0.87495531  0.1993528
20  0.63135911  0.1993528

attr(,"class")
[1] "coef.mer"
```

- different intercept for each participant
- one slope for condition

Understanding the `lmer()` summary

Understanding the **lmer**() summary

```r
# fit a linear mixed effects model
lmer(formula = value ~ 1 + condition + (1 | participant),
                         data = df.original) %>%
  summary()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
   Data: df.original

REML criterion at convergence: 17.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.55996 -0.36399 -0.03341  0.34400  1.65823

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 0.816722 0.90373
 Residual                0.003796 0.06161
Number of obs: 40, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.19052    0.20255   0.941
condition2   0.19935    0.01948  10.231

Correlation of Fixed Effects:
           (Intr)
condition2 -0.048
```

REML = restricted maximum likelihood

method for fitting models with **random effects**

Understanding the `lmer()` summary

```r
# fit a linear mixed effects model
lmer(formula = value ~ 1 + condition + (1 | participant),
                          data = df.original) %>%
  summary()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
   Data: df.original

REML criterion at convergence: 17.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.55996 -0.36399 -0.03341  0.34400  1.65823

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 0.816722 0.90373
 Residual                0.003796 0.06161
Number of obs: 40, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.19052    0.20255   0.941
condition2   0.19935    0.01948  10.231

Correlation of Fixed Effects:
           (Intr)
condition2 -0.048
```

fitting **`lmer()`** doesn't always work ...

`lmer()` complains when it didn't work

Understanding the **lmer()** summary

```
# fit a linear mixed effects model
lmer(formula = value ~ 1 + condition + (1 | participant),
                         data = df.original) %>%
  summary()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
   Data: df.original

REML criterion at convergence: 17.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.55996 -0.36399 -0.03341  0.34400  1.65823

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 0.816722 0.90373
 Residual                0.003796 0.06161
Number of obs: 40, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.19052    0.20255   0.941
condition2   0.19935    0.01948  10.231

Correlation of Fixed Effects:
           (Intr)
condition2 -0.048
```

summary information about residuals

Understanding the `lmer()` summary

```r
# fit a linear mixed effects model
lmer(formula = value ~ 1 + condition + (1 | participant),
                          data = df.original) %>%
  summary()
```

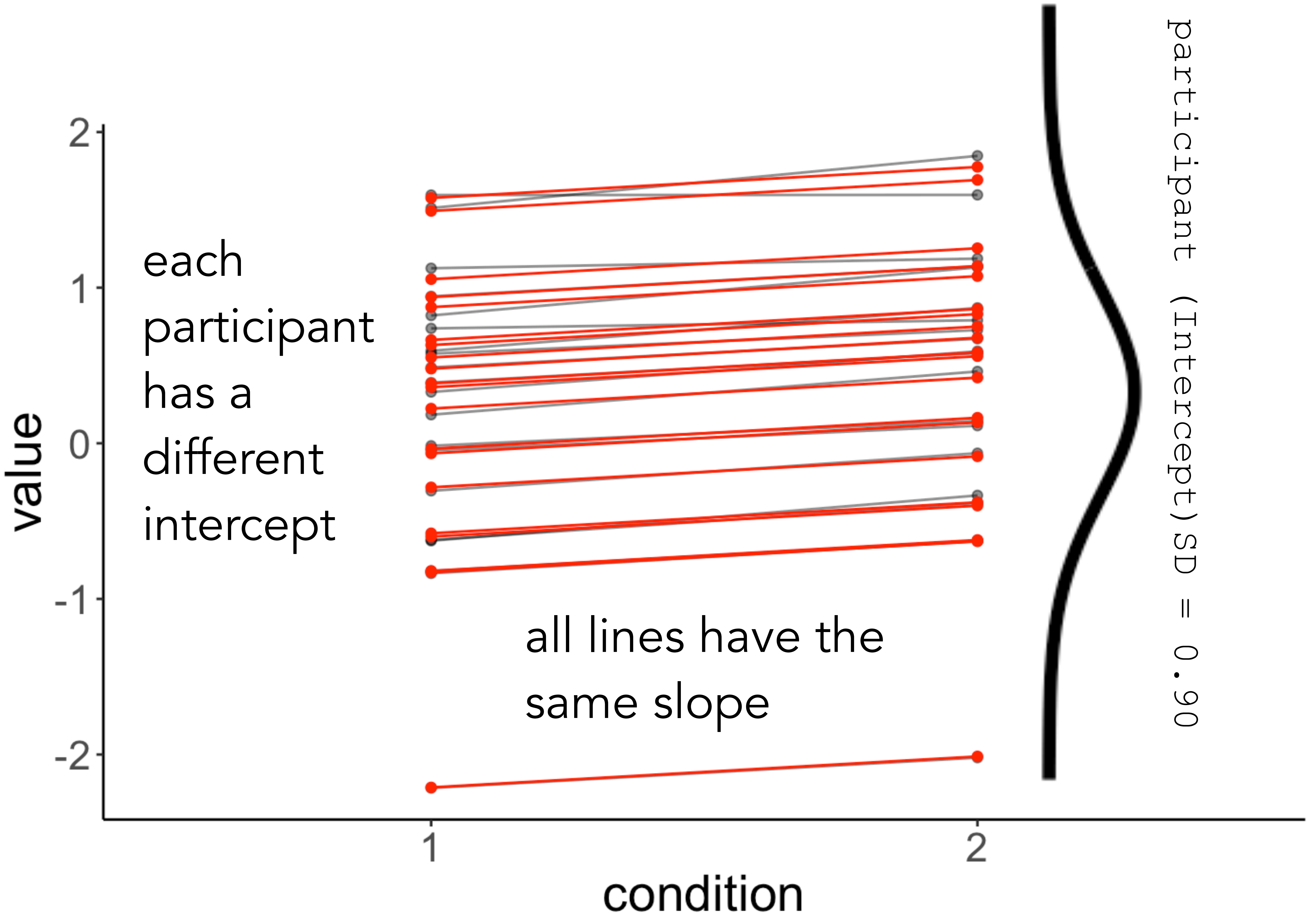
```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
   Data: df.original

REML criterion at convergence: 17.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.55996 -0.36399 -0.03341  0.34400  1.65823

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 0.816722 0.90373
 Residual                0.003796 0.06161
Number of obs: 40, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.19052    0.20255   0.941
condition2   0.19935    0.01948  10.231

Correlation of Fixed Effects:
           (Intr)
condition2 -0.048
```

Random effects

one parameter to capture the variance between participants (gives us a sense for whether there are interindividual differences)

one parameter to capture the residual variance (just like sigma in an `lm()`)

Understanding the `lmer()` summary



Understanding the `lmer()` summary

```r
# fit a linear mixed effects model
lmer(formula = value ~ 1 + condition + (1 | participant),
                     data = df.original) %>%
  summary()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
   Data: df.original

REML criterion at convergence: 17.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.55996 -0.36399 -0.03341  0.34400  1.65823

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 0.816722 0.90373
 Residual                0.003796 0.06161
Number of obs: 40, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.19052    0.20255   0.941
condition2   0.19935    0.01948  10.231

Correlation of Fixed Effects:
           (Intr)
condition2 -0.048
```

Fixed effects

one parameter for the global intercept (value for the baseline condition)

one parameter for the condition effect (difference between the two conditions)

interpretation the same as for **`lm()`**, also: we can use contrasts!

Understanding the **lmer()** summary

```r
# fit a linear mixed effects model
lmer(formula = value ~ 1 + condition + (1 | participant),
                          data = df.original) %>%
  summary()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
   Data: df.original

REML criterion at convergence: 17.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.55996 -0.36399 -0.03341  0.34400  1.65823

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 0.816722 0.90373
 Residual                0.003796 0.06161
Number of obs: 40, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.19052    0.20255   0.941
condition2   0.19935    0.01948  10.231

Correlation of Fixed Effects:
           (Intr)
condition2 -0.048
```

correlation between intercept and condition2

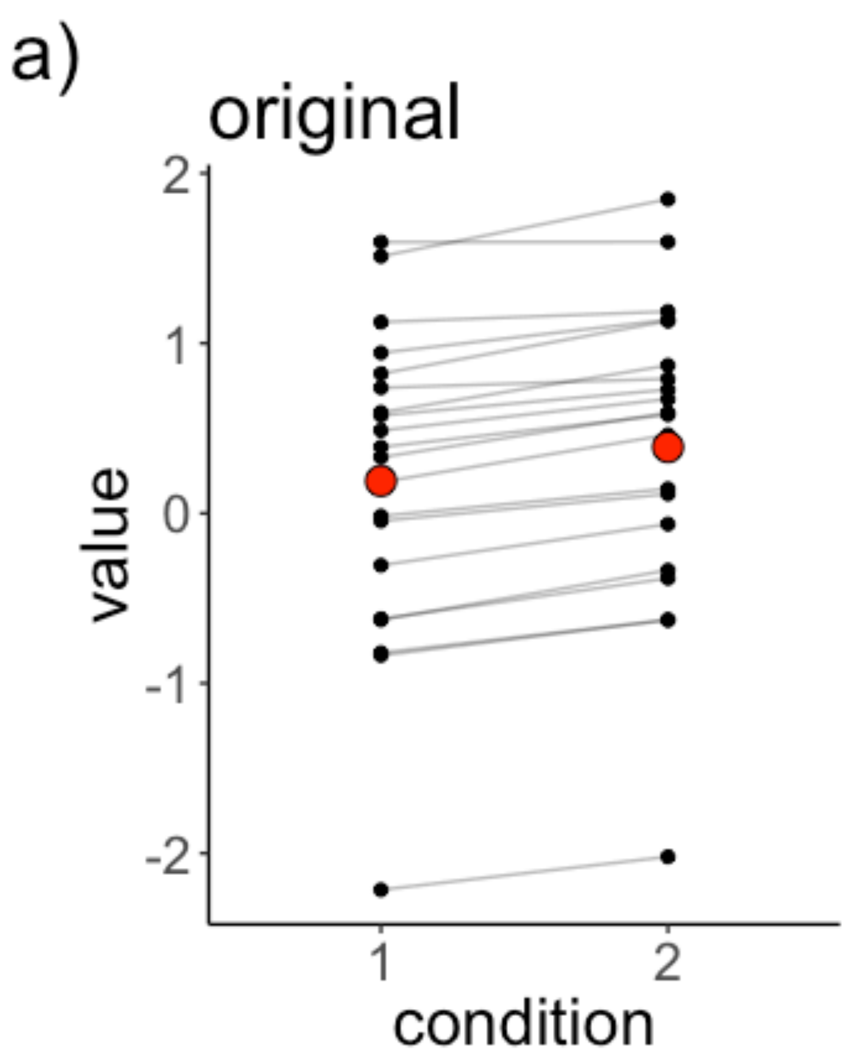
The "correlation of fixed effects" output doesn't have the intuitive meaning that most would ascribe to it. Specifically, is not about the correlation of the variables. It is in fact about the expected correlation of the regression coefficients.

we just performed a paired t-test ...

```
t.test(df.original$value[df.original$condition == "1"],
       df.original$value[df.original$condition == "2"],
       alternative = "two.sided",
       paired = T)
```

```
Paired t-test

data:  df.original$value[df.original$condition == "1"] and
df.original$value[df.original$condition == "2"]
t = -10.231, df = 19, p-value = 3.636e-09
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.2401340 -0.1585717
sample estimates:
mean of the differences
             -0.1993528
```

If we take the differences for each participant between condition 1 and condition 2, are these difference scores significantly different from 0?



we just performed a paired t-test …

```r
lmer(formula = value ~ 1 + condition + (1 | participant),
     data = df.original)
```

- explicitly models the interindividual variation
- much more flexible …

```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
   Data: df.original

REML criterion at convergence: 17.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.55996 -0.36399 -0.03341  0.34400  1.65823

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 0.816722 0.90373
 Residual                0.003796 0.06161
Number of obs: 40, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.19052    0.20255   0.941
condition2   0.19935    0.01948  10.231

Correlation of Fixed Effects:
           (Intr)
condition2 -0.048
```

```r
t.test(df.original$value[df.original$condition == "1"],
       df.original$value[df.original$condition == "2"],
       alternative = "two.sided",
       paired = T)
```

```
Paired t-test

data:  df.original$value[df.original$condition == "1"] and
df.original$value[df.original$condition == "2"]
t = -10.231, df = 19, p-value = 3.636e-09
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.2401340 -0.1585717
sample estimates:
mean of the differences
             -0.1993528
```

general points about **`lmer`**`()`

- **fixed effects**:
  - parameters are estimated
  - often: factors that we manipulate experimentally
- **random effects**:
  - variation we want to take into account
  - often: differences between participants (or items) in our experiment
  - sampling viewpoint: we explicitly model the variation in participants

general points about `lmer()`

- Why don't we just run individual regressions?
  - overfitting ...
  - inflating type 1 error
  - larger uncertainty in parameter estimates because only few data points are used for each model
  - unclear how to aggregate the results to make an overall statement
- Why don't we just run a regression on the means?
  - we throw away a lot of information
  - what to do when the design is unbalanced?
- Mixed effects model:
  - makes use of all available information
  - addresses the main problems of the other two approaches

let's take a look at an example

A worked example

Higher Order Functions   Posts   About Me   CV

Tristan Mahr

Language and data scientist

- Madison, WI
- Email
- Twitter
- GitHub
- Stackoverflow
- R Bloggers

Plotting partial pooling in mixed-effects models

In this post, I demonstrate a few techniques for plotting information from a relatively simple mixed-effects model fit in R. These plots can help us develop intuitions about what these models are doing and what “partial pooling” means.

The `sleepstudy` dataset

For these examples, I’m going to use the `sleepstudy` dataset from the lme4 package. The outcome measure is reaction time, the predictor measure is days of sleep deprivation, and these measurements are nested within participants—we have 10 observations per participant. I am also going to add two fake participants with incomplete data to illustrate partial pooling.

https://www.tjmahr.com/plotting-partial-pooling-in-mixed-effects-models/

Data set

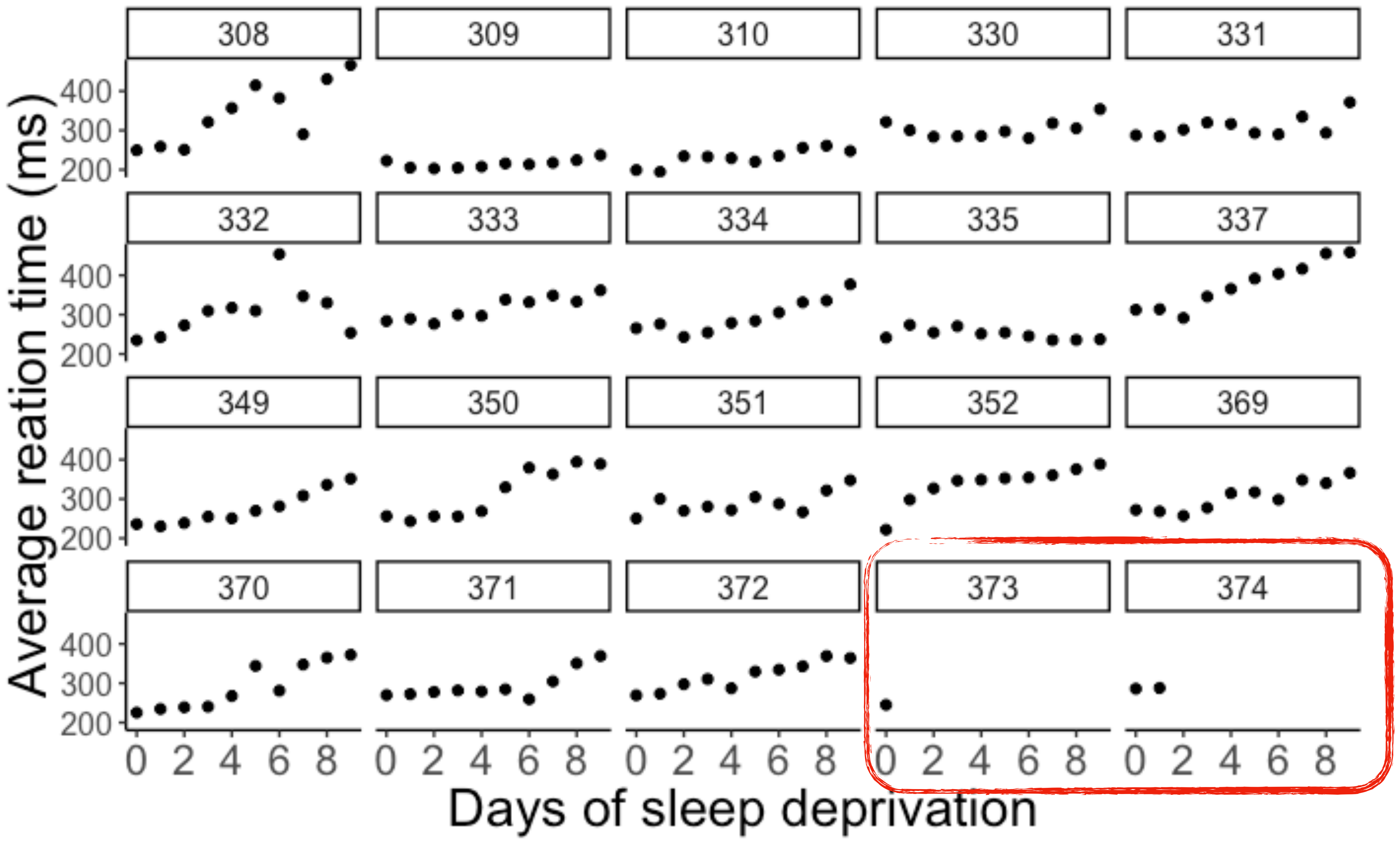
How does sleep deprivation affect reaction time?

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71
309	4	207.72

Data set

How does sleep deprivation affect reaction time?

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71
309	4	207.72



20 participants

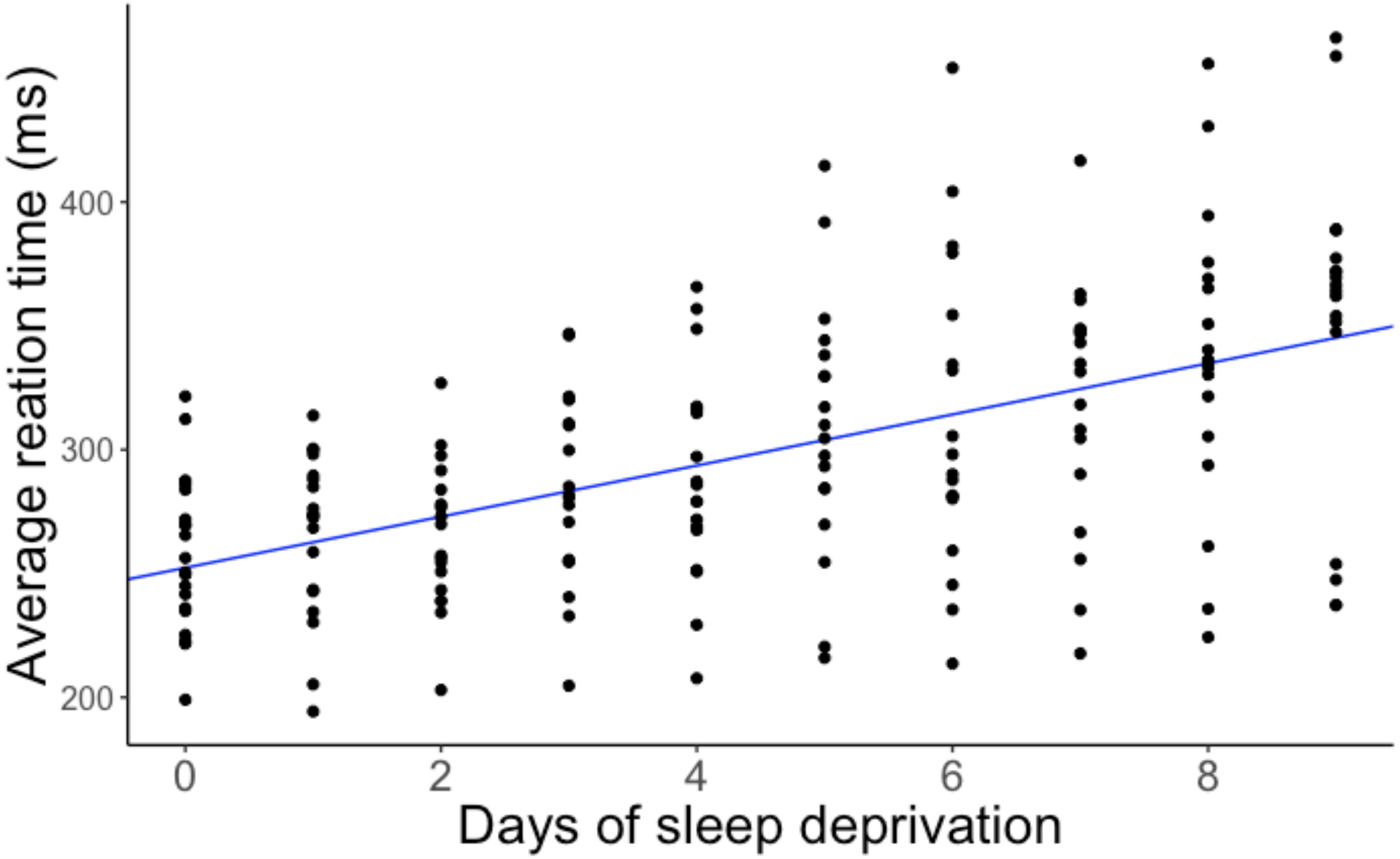
2 with incomplete information

Pooling information

- **complete pooling**
  - combine data from all participants and fit one global regression
- **no pooling**
  - don't combine any of the data and fit a separate regression to each individual participant
- **partial pooling**
  - take into account all information by explicitly modeling the variation between participants

Complete pooling: Fit one global regression

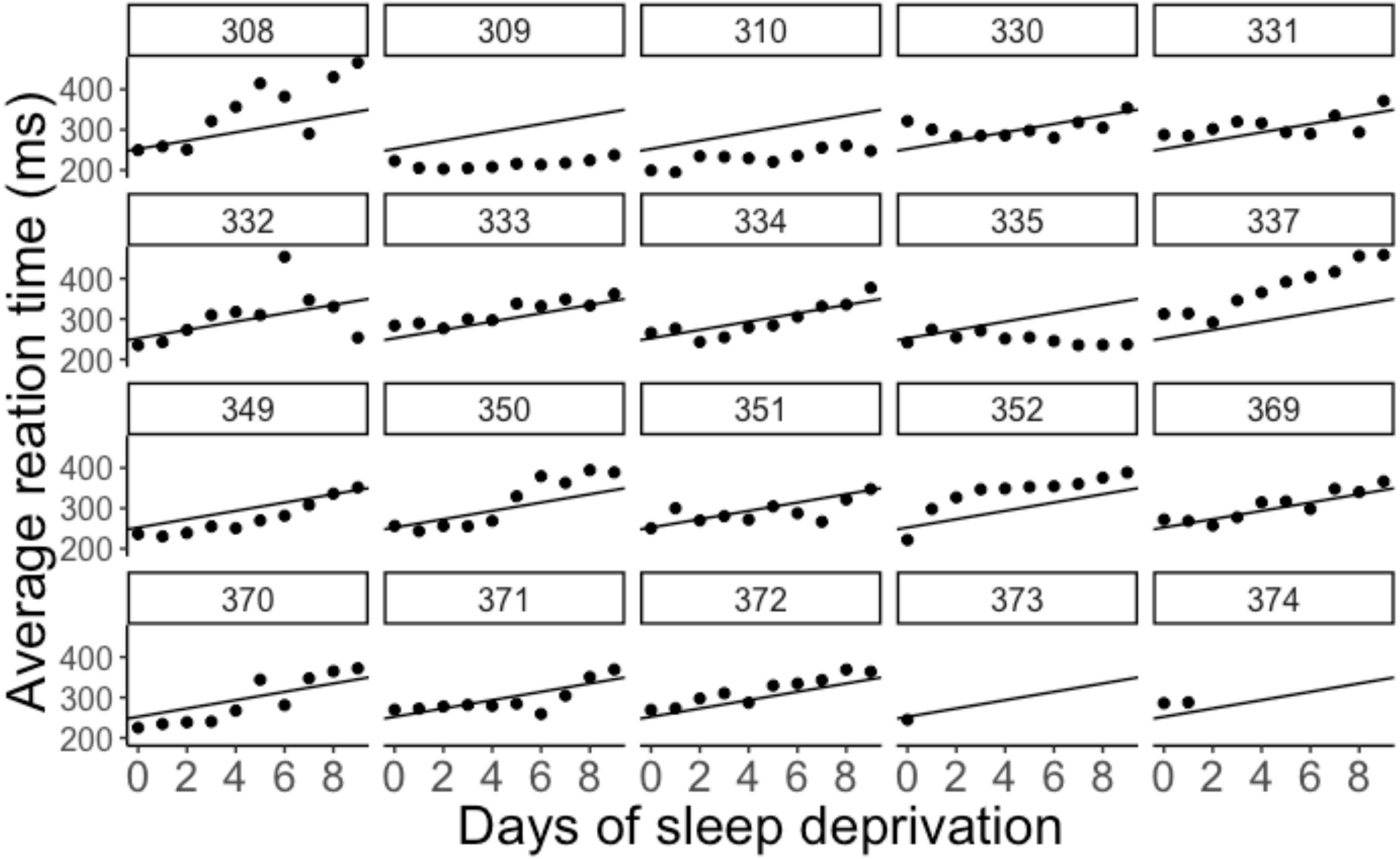
```
lm(formula = reaction ~ 1 + days,
   data = df.sleep)
```



Complete pooling: Fit one global regression

```
lm(formula = reaction ~ 1 + days,
   data = df.sleep)
```

same line for
each participant



No pooling: Fit separate regressions

```r
df.no_pooling = df.sleep %>%
  group_by(subject) %>%
  nest(data = c(days, reaction)) %>%
  mutate(fit = map(data, ~ lm(reaction ~ 1 + days, data = .)),
         params = map(fit, tidy)) %>%
  unnest(c(params)) %>%
  select(subject, term, estimate) %>%
  complete(subject, term, fill = list(estimate = 0)) %>%
  pivot_wider(names_from = term, values_from = estimate) %>%
  clean_names()
```

data — regression fit — extracted parameters

	subject	data	fit	params
1	308	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(2...	list(coefficients = c(`(Intercept)` = 244.1926690909...	list(term = c("(Intercept)", "days"), estimate = c(244.1...
2	309	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(2...	list(coefficients = c(`(Intercept)` = 205.0549454545...	list(term = c("(Intercept)", "days"), estimate = c(205.0...
3	310	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(1...	list(coefficients = c(`(Intercept)` = 203.4842254545...	list(term = c("(Intercept)", "days"), estimate = c(203.4...
4	330	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(3...	list(coefficients = c(`(Intercept)` = 289.6850927272...	list(term = c("(Intercept)", "days"), estimate = c(289.6...
5	331	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(2...	list(coefficients = c(`(Intercept)` = 285.7389654545...	list(term = c("(Intercept)", "days"), estimate = c(285.7...
6	332	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(2...	list(coefficients = c(`(Intercept)` = 264.2516145454...	list(term = c("(Intercept)", "days"), estimate = c(264.2...
7	333	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(2...	list(coefficients = c(`(Intercept)` = 275.0191054545...	list(term = c("(Intercept)", "days"), estimate = c(275.0...
8	334	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(2...	list(coefficients = c(`(Intercept)` = 240.1629145454...	list(term = c("(Intercept)", "days"), estimate = c(240.1...
9	335	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(2...	list(coefficients = c(`(Intercept)` = 263.0346927272...	list(term = c("(Intercept)", "days"), estimate = c(263.0...
10	337	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(3...	list(coefficients = c(`(Intercept)` = 290.1041272727...	list(term = c("(Intercept)", "days"), estimate = c(290.1...
11	349	list(days = c(0, 1, 2, 3, 4, 5, 6, 7, 8, 9), reaction = c(2...	list(coefficients = c(`(Intercept)` = 215.1117727272...	list(term = c("(Intercept)", "days"), estimate = c(215.1...
			⋮	
19	374	list(days = c(0, 1), reaction = c(286, 288))	list(coefficients = c(`(Intercept)` = 286, days = 2.000...	list(term = c("(Intercept)", "days"), estimate = c(286, 2...
20	373	list(days = 0, reaction = 245)	list(coefficients = c(`(Intercept)` = 245, days = NA), r...	list(term = "(Intercept)", estimate = 245, std.error = ...

No pooling: Fit separate regressions

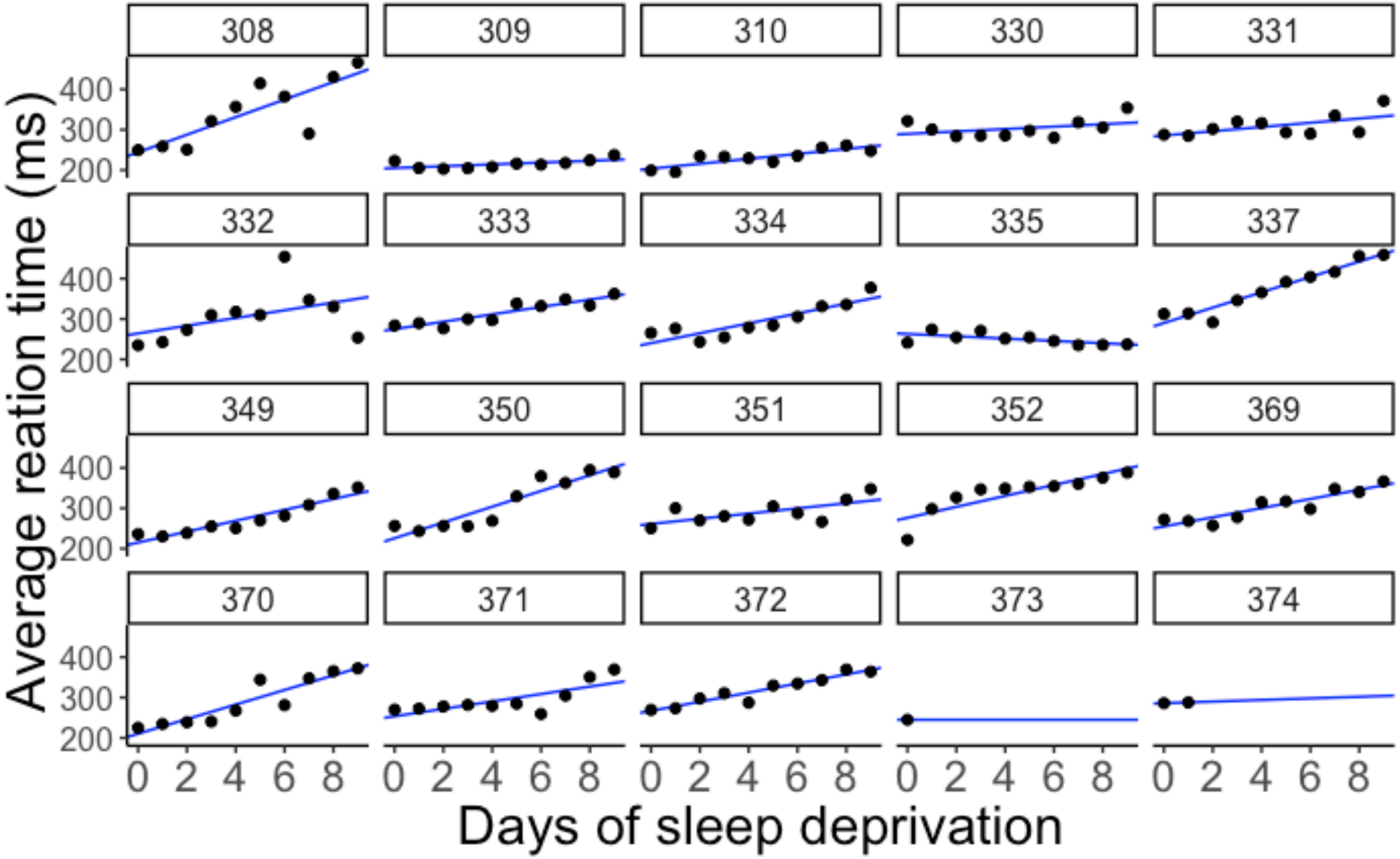
```r
df.no_pooling = df.sleep %>%
  group_by(subject) %>%
  nest(data = c(days, reaction)) %>%
  mutate(fit = map(data, ~ lm(reaction ~ 1 + days, data = .)),
         params = map(fit, tidy)) %>%
  unnest(c(params)) %>%
  select(subject, term, estimate) %>%
  complete(subject, term, fill = list(estimate = 0)) %>%
  pivot_wider(names_from = term, values_from = estimate) %>%
  clean_names()
```

separate intercept and
slope for each participant

	subject	intercept	days
1	308	244.1927	21.764702
2	309	205.0549	2.261785
3	310	203.4842	6.114899
4	330	289.6851	3.008073
5	331	285.7390	5.266019
6	332	264.2516	9.566768
7	333	275.0191	9.142045
8	334	240.1629	12.253141
9	335	263.0347	-2.881034
10	337	290.1041	19.025974
11	349	215.1118	13.493933
12	350	225.8346	19.504017
13	351	261.1470	6.433498
14	352	276.3721	13.566549
15	369	254.9681	11.348109
16	370	210.4491	18.056151
17	371	253.6360	9.188445
18	372	267.0448	11.298073
19	373	245.0000	0.000000
20	374	286.0000	2.000000

No pooling: Fit separate regressions

different line for each participant



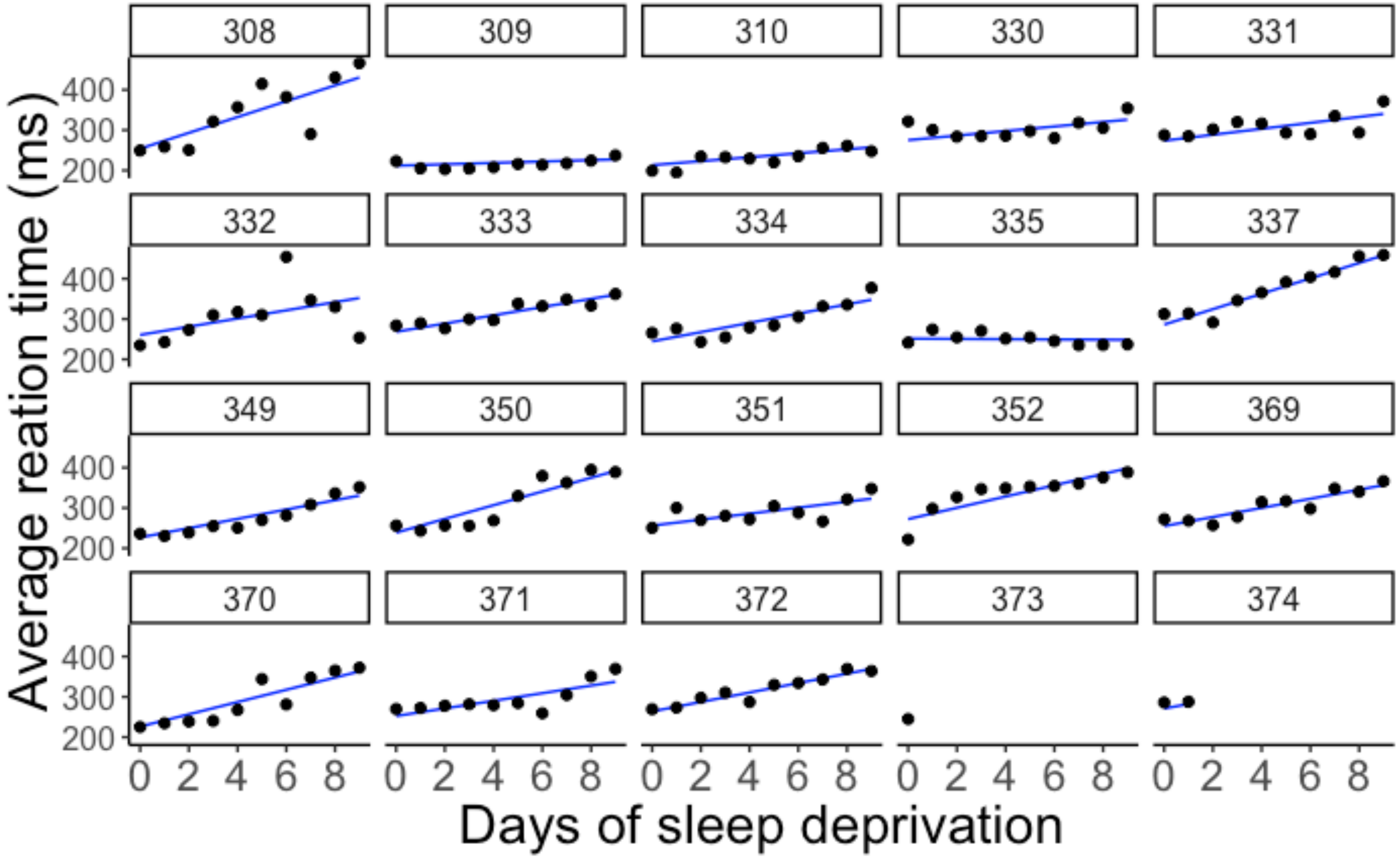
Partial pooling: Fit mixed effects model

intercepts and slopes differ between participants

random intercept

random slope

```
lmer(formula = reaction ~ 1 + days + (1 + days | subject),
     data = df.sleep)
```

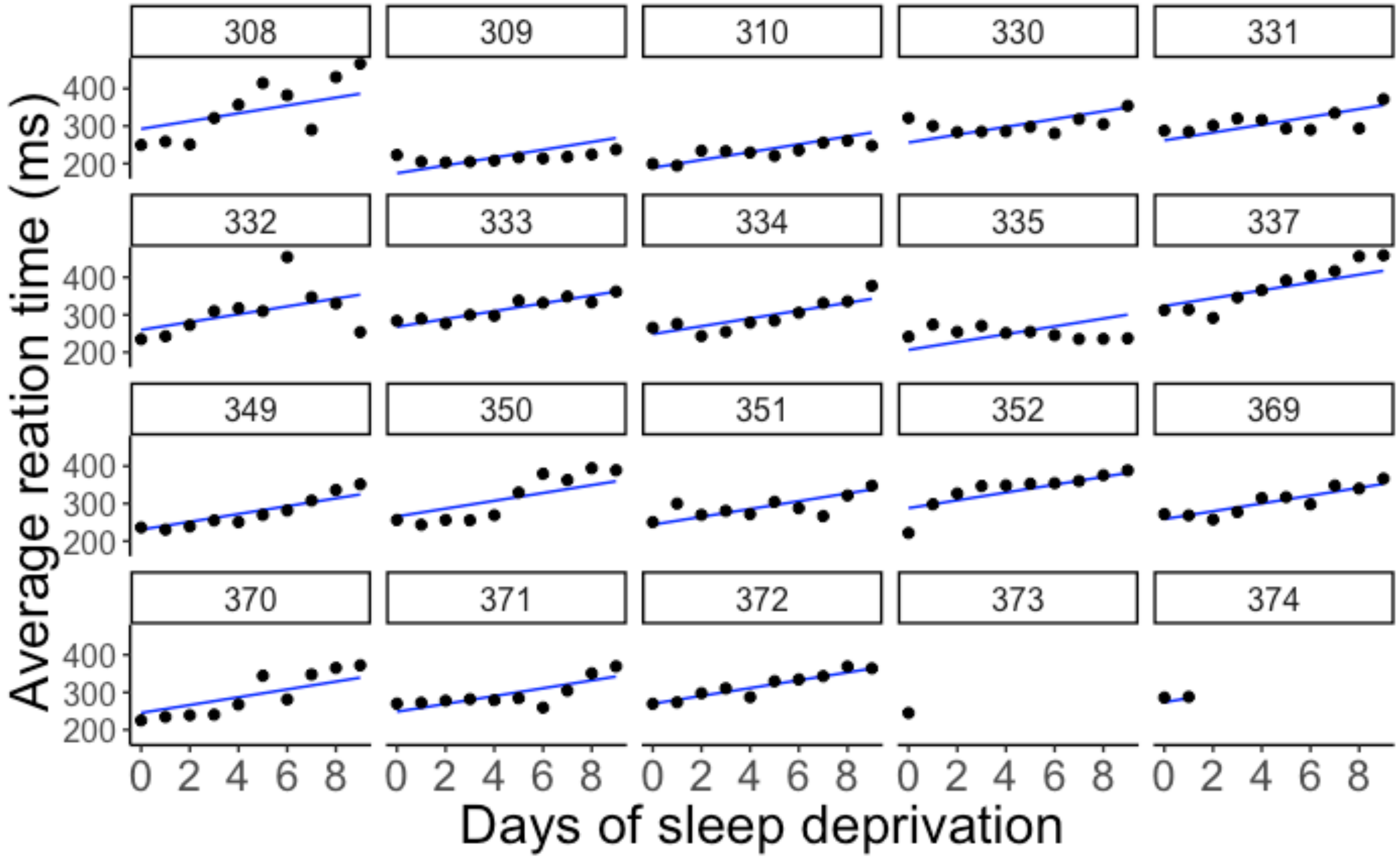


Partial pooling: Fit mixed effects model

only intercepts differ between participants

random intercept

```
lmer(formula = reaction ~ 1 + days + (1 | subject),
     data = df.sleep)
```

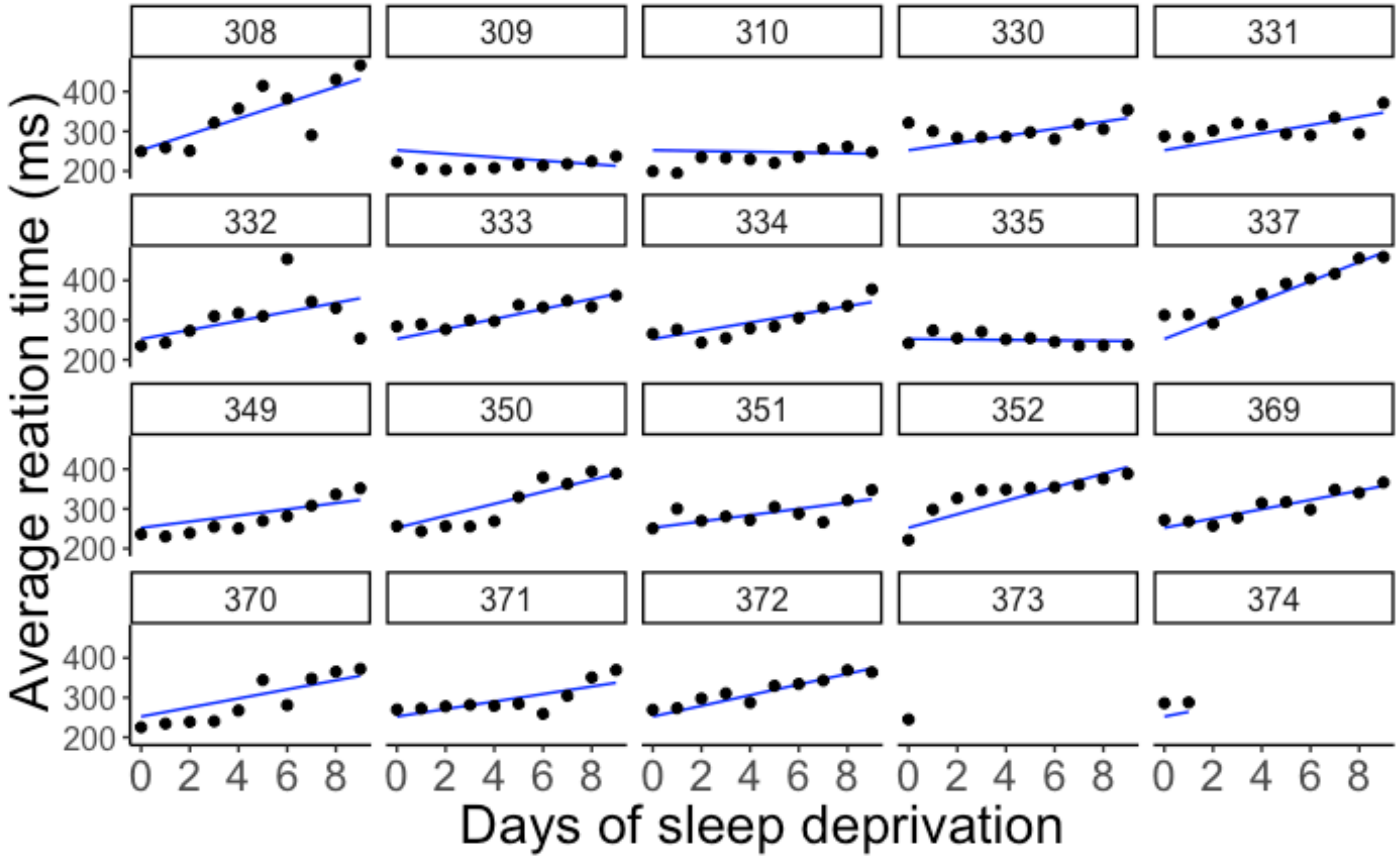


Partial pooling: Fit mixed effects model

only slopes differ between participants

random slope

```
lmer(formula = reaction ~ 1 + days + (0 + days | subject),
     data = df.sleep)
```



Coefficients

```
lmer(formula = reaction ~ 1 + days + ...,
     data = df.sleep)
```

`(1 | subject)` `(0 + days | subject)` `(1 + days | subject)`

random intercepts — random slopes — random intercepts and slopes (+ correlation)

random intercepts `(1 | subject)`

```
$subject
    (Intercept)     days
308    292.2749 10.43191
309    174.0559 10.43191
310    188.7454 10.43191
330    256.0247 10.43191
331    261.8141 10.43191
332    259.8262 10.43191
333    268.0765 10.43191
334    248.6471 10.43191
335    206.5096 10.43191
337    323.5643 10.43191
349    230.5114 10.43191
350    265.6957 10.43191
351    243.7988 10.43191
352    287.8850 10.43191
369    258.6454 10.43191
370    245.2931 10.43191
371    248.3508 10.43191
372    269.6861 10.43191
373    248.2086 10.43191
374    273.9400 10.43191
```

random slopes `(0 + days | subject)`

```
$subject
    (Intercept)       days
308    252.2965 19.9526801
309    252.2965 -4.3719650
310    252.2965 -0.9574726
330    252.2965  8.9909957
331    252.2965 10.5394285
332    252.2965 11.3994289
333    252.2965 12.6074020
334    252.2965 10.3413879
335    252.2965 -0.5722073
337    252.2965 24.2246485
349    252.2965  7.7702676
350    252.2965 15.0661415
351    252.2965  7.9675415
352    252.2965 17.0002999
369    252.2965 11.6982767
370    252.2965 11.3939807
371    252.2965  9.4535879
372    252.2965 13.4569059
373    252.2965 10.4142695
374    252.2965 11.9097917
```

random intercepts and slopes (+ correlation) `(1 + days | subject)`

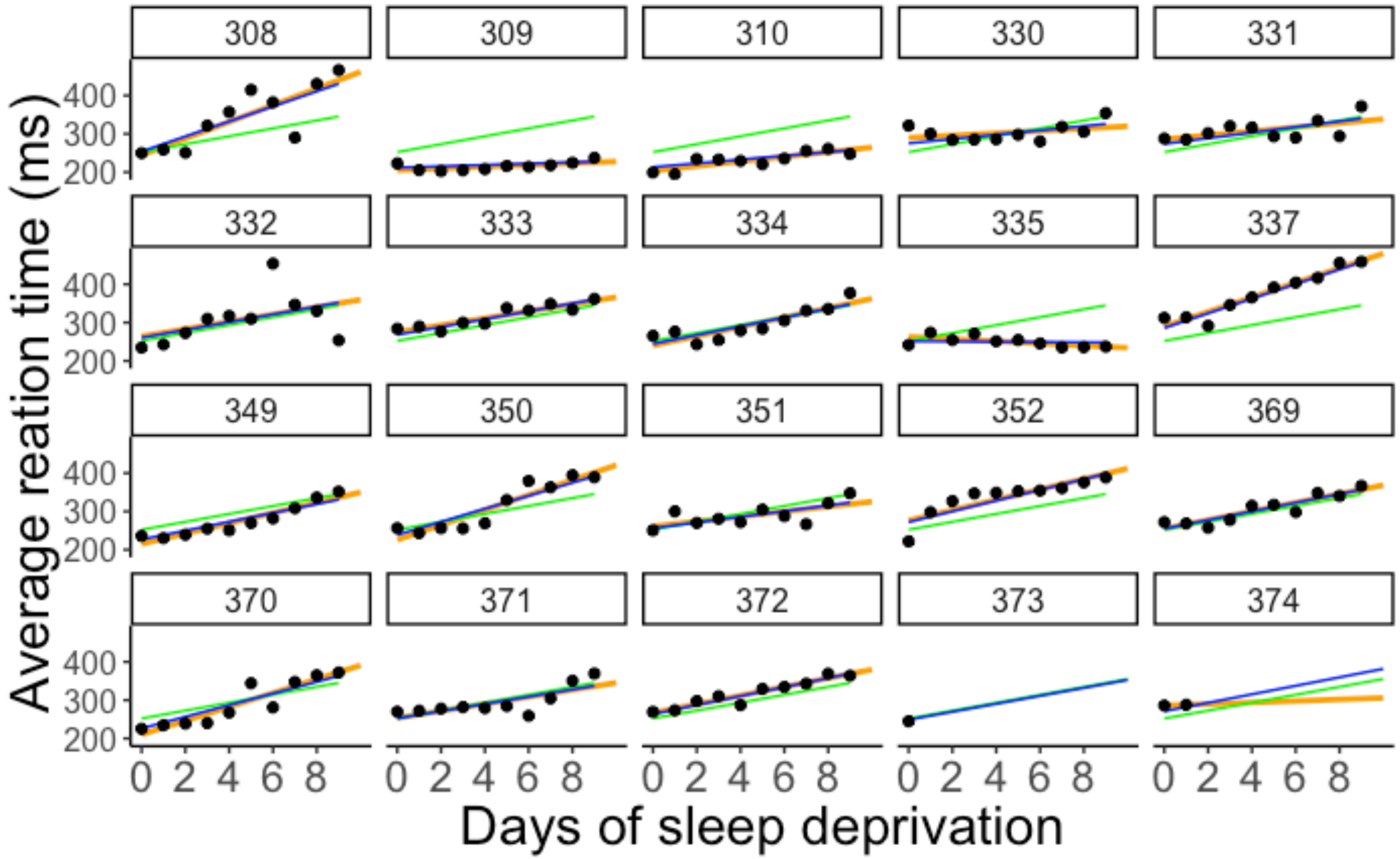
```
$subject
    (Intercept)       days
308    253.9479 19.6264139
309    211.7328  1.7319567
310    213.1579  4.9061843
330    275.1425  5.6435987
331    273.7286  7.3862680
332    260.6504 10.1632535
333    268.3684 10.2245979
334    244.5523 11.4837825
335    251.3700 -0.3355554
337    286.2321 19.1090061
349    226.7662 11.5531963
350    238.7807 17.0156766
351    256.2344  7.4119501
352    272.3512 13.9920698
369    254.9484 11.2985741
370    226.3701 15.2027922
371    252.5051  9.4335432
372    263.8916 11.7253342
373    248.9752 10.3915245
374    271.1451 11.0782697
```

Comparison

complete pooling

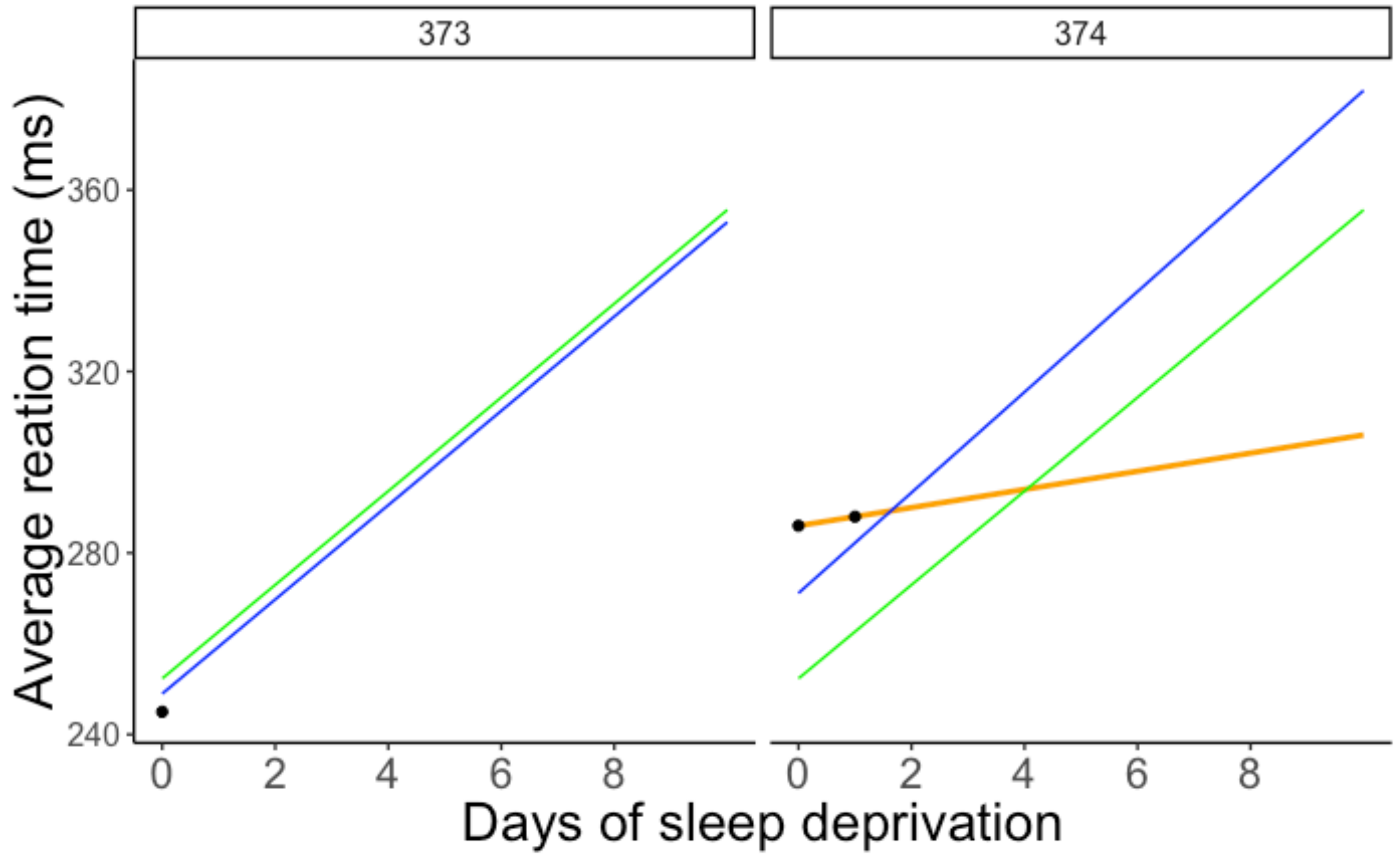
no pooling

partial pooling



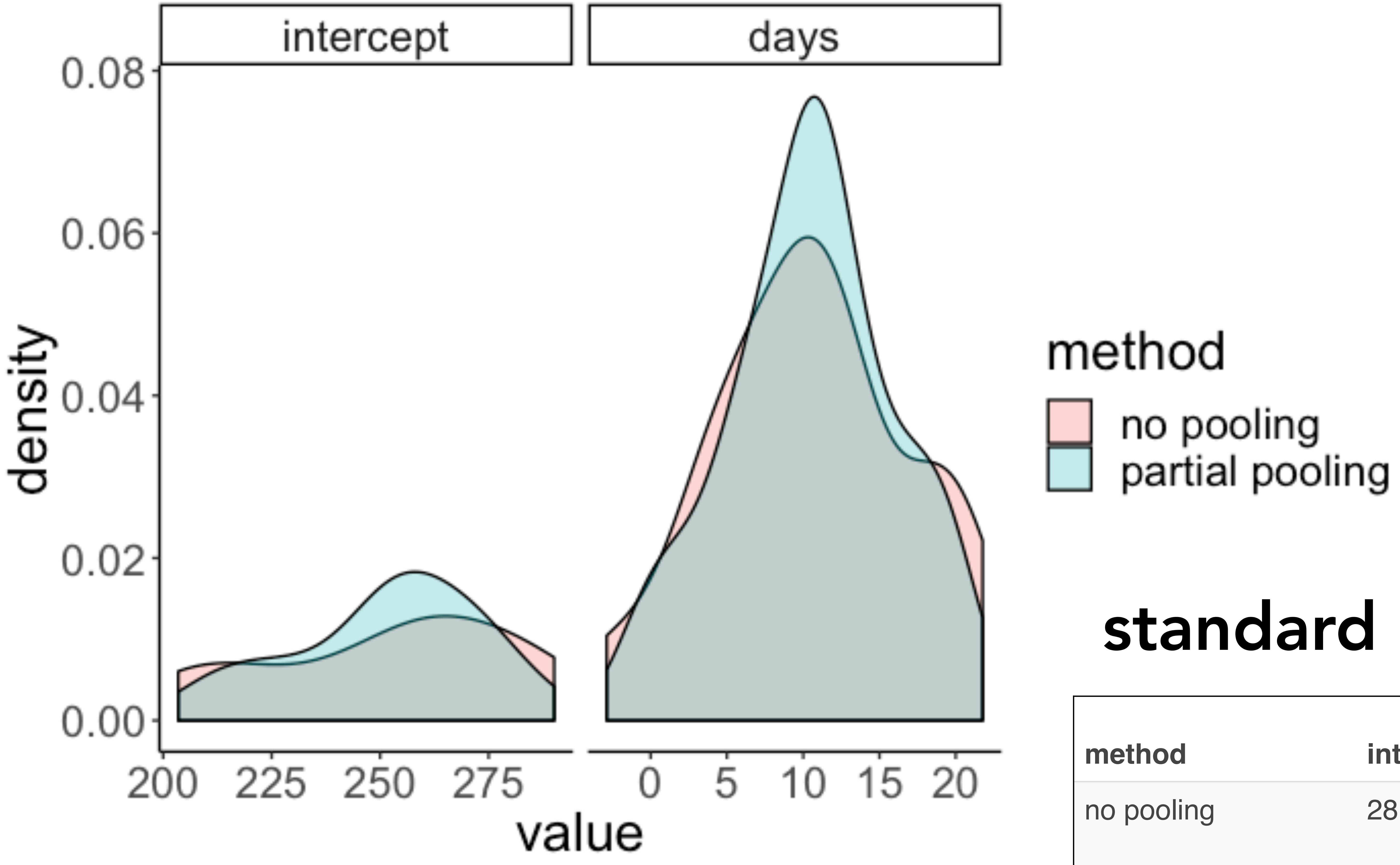
Comparison

complete pooling
no pooling
partial pooling



- **complete pooling**:
  - doesn't account for any individual variation
- **no pooling**:
  - doesn't yield predictions when we only have one observation
  - doesn't consider the general effect of sleep deprivation when making predictions
- **partial pooling**:
  - draws on all the information in the data
  - extrapolates based on information about the individual participants, as well as information based on the whole sample

Shrinkage

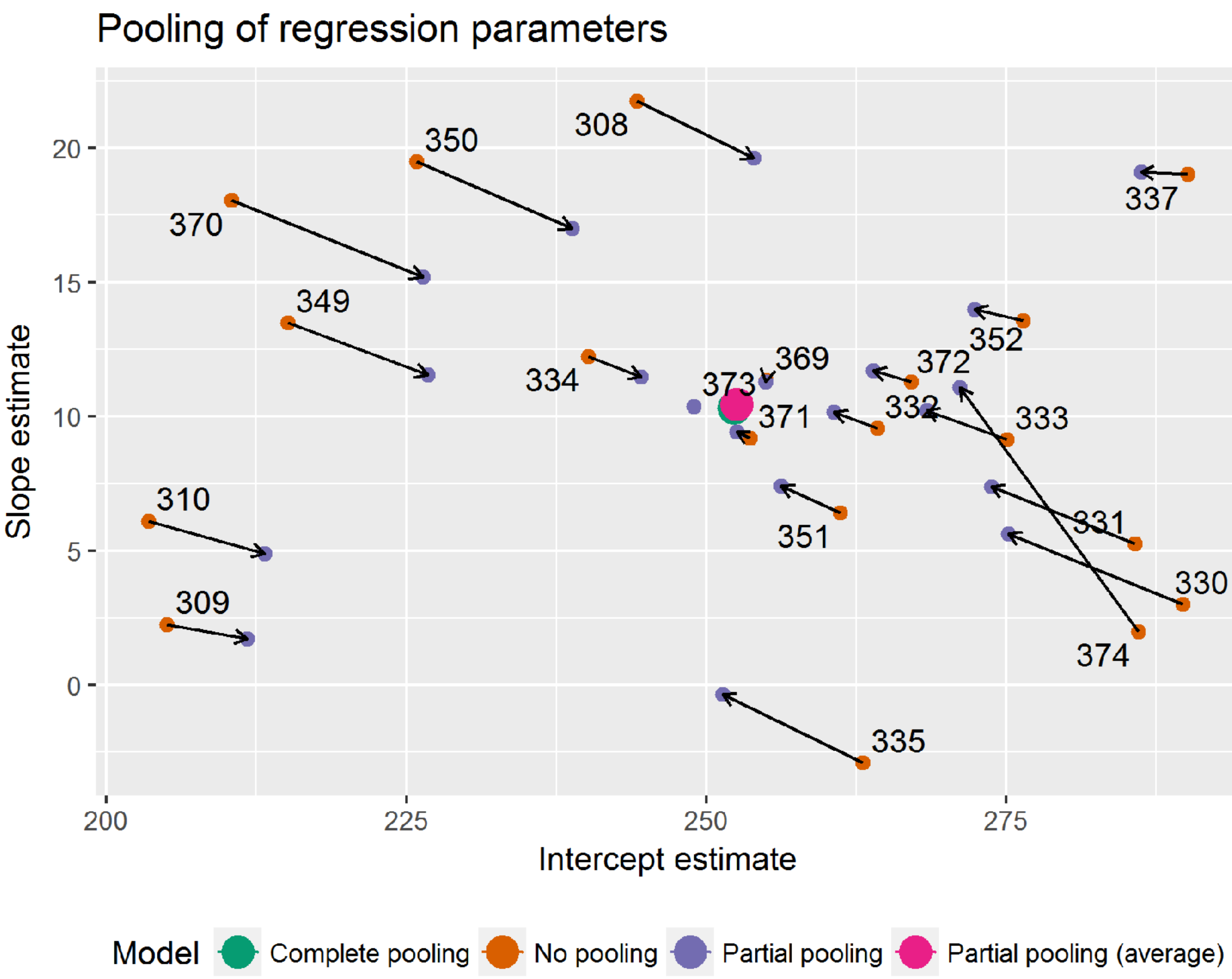


standard deviation

method	intercept	days
no pooling	28.95	6.56
partial pooling	21.59	5.46

variance "shrinks"

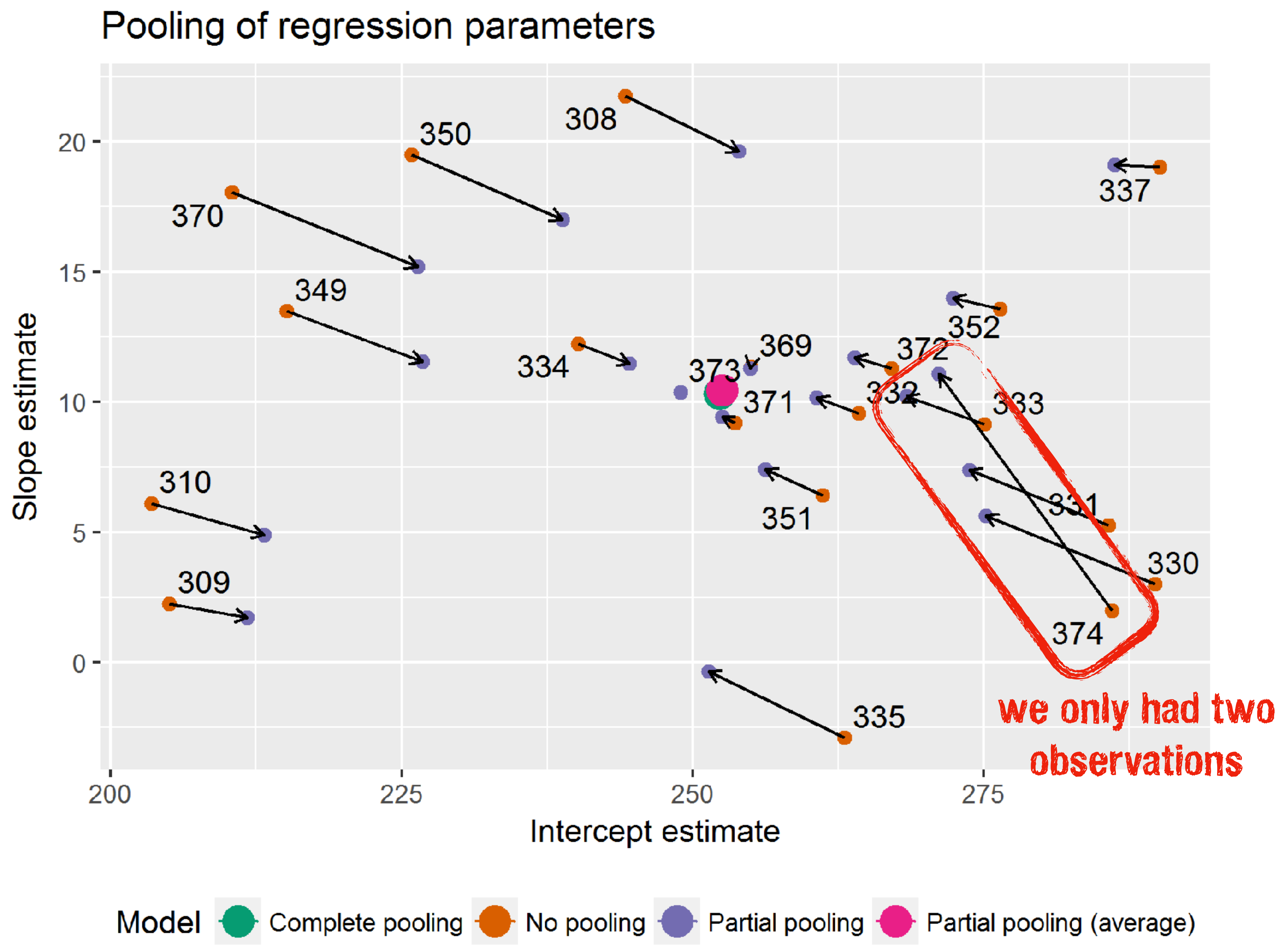
Shrinkage



Pooling of regression parameters
Slope estimate
Intercept estimate
20
15
10
5
0
200
225
250
275
308
350
337
370
349
334
369
373
372
352
371
332
333
351
331
310
330
309
374
335
Model
Complete pooling
No pooling
Partial pooling
Partial pooling (average)

Shrinkage

- more shrinkage when estimate is further from the average
- more shrinkage when estimate is more uncertain (based on fewer observations); more information "borrowed" from other clusters

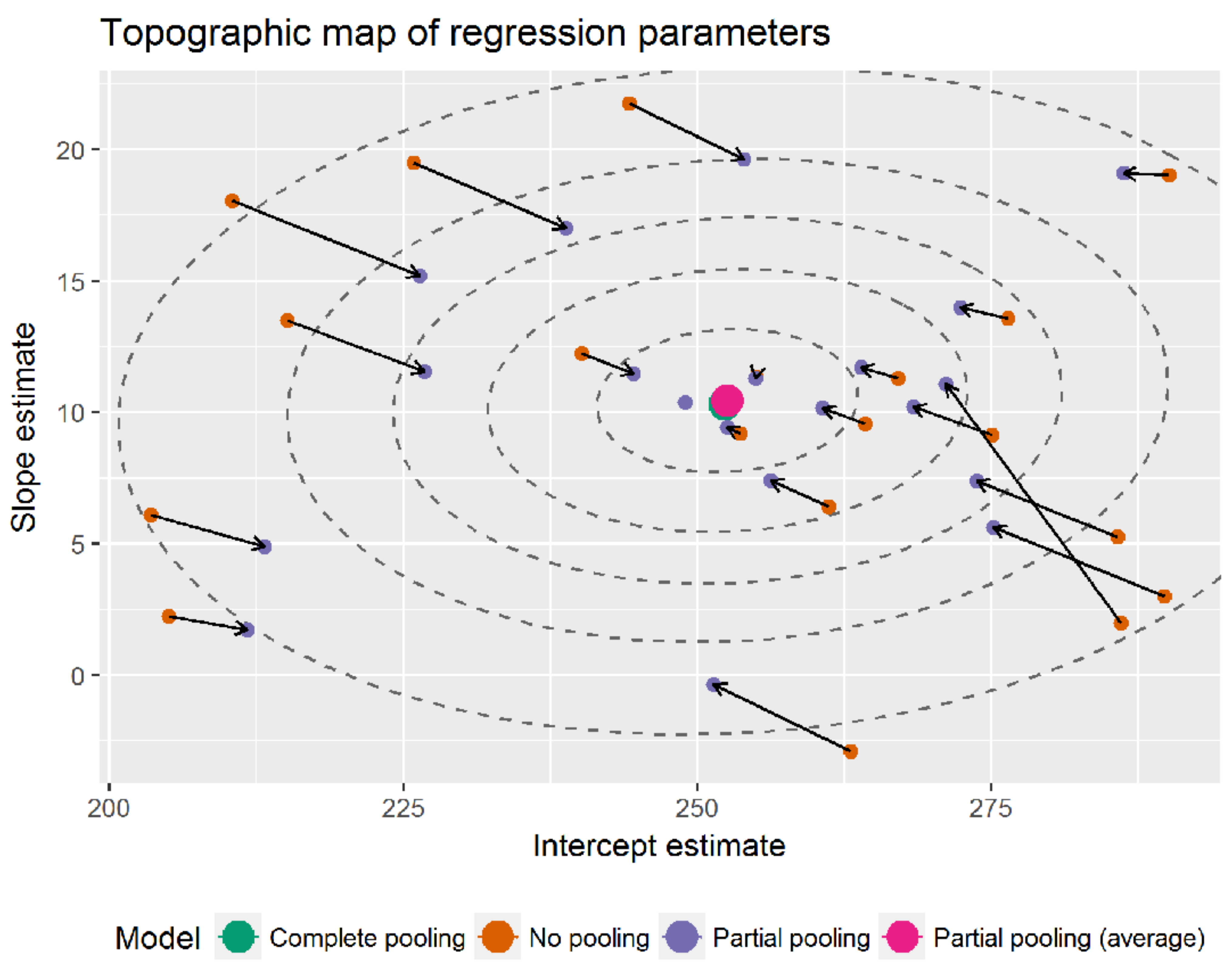


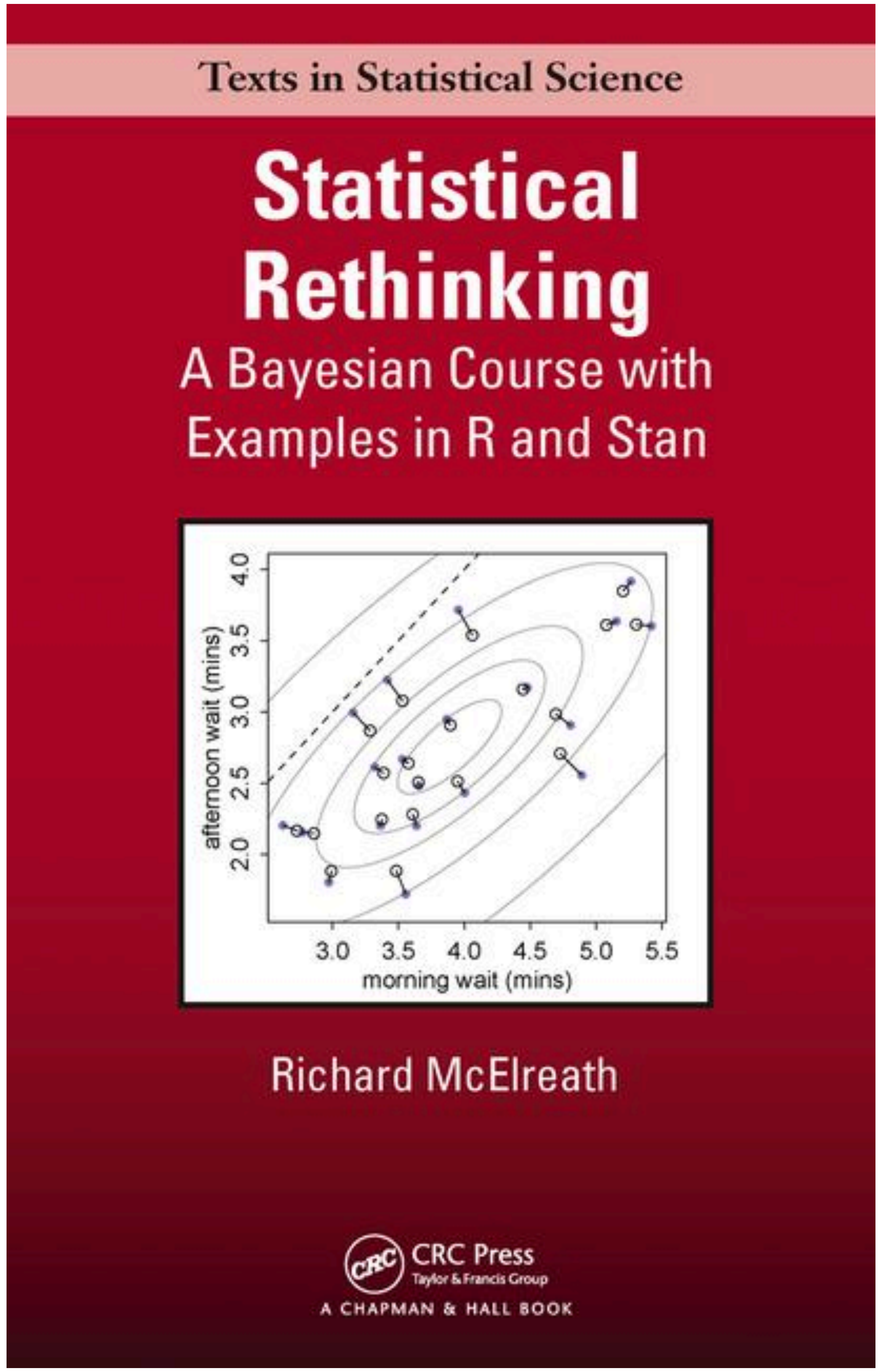
In the lme4 book, Douglas Bates provides an alternative to *shrinkage*:

The term “shrinkage” may have negative connotations. John Tukey preferred to refer to the process as the estimates for individual subjects **“borrowing strength” from each other**.

This is a fundamental difference in the models underlying mixed-effects models versus strictly fixed effects models. In a mixed-effects model we assume that the levels of a grouping factor are a selection from a population and, as a result, can be expected to share characteristics to some degree. Consequently, the predictions from a mixed-effects model are attenuated relative to those from strictly fixed-effects models.

Shrinkage

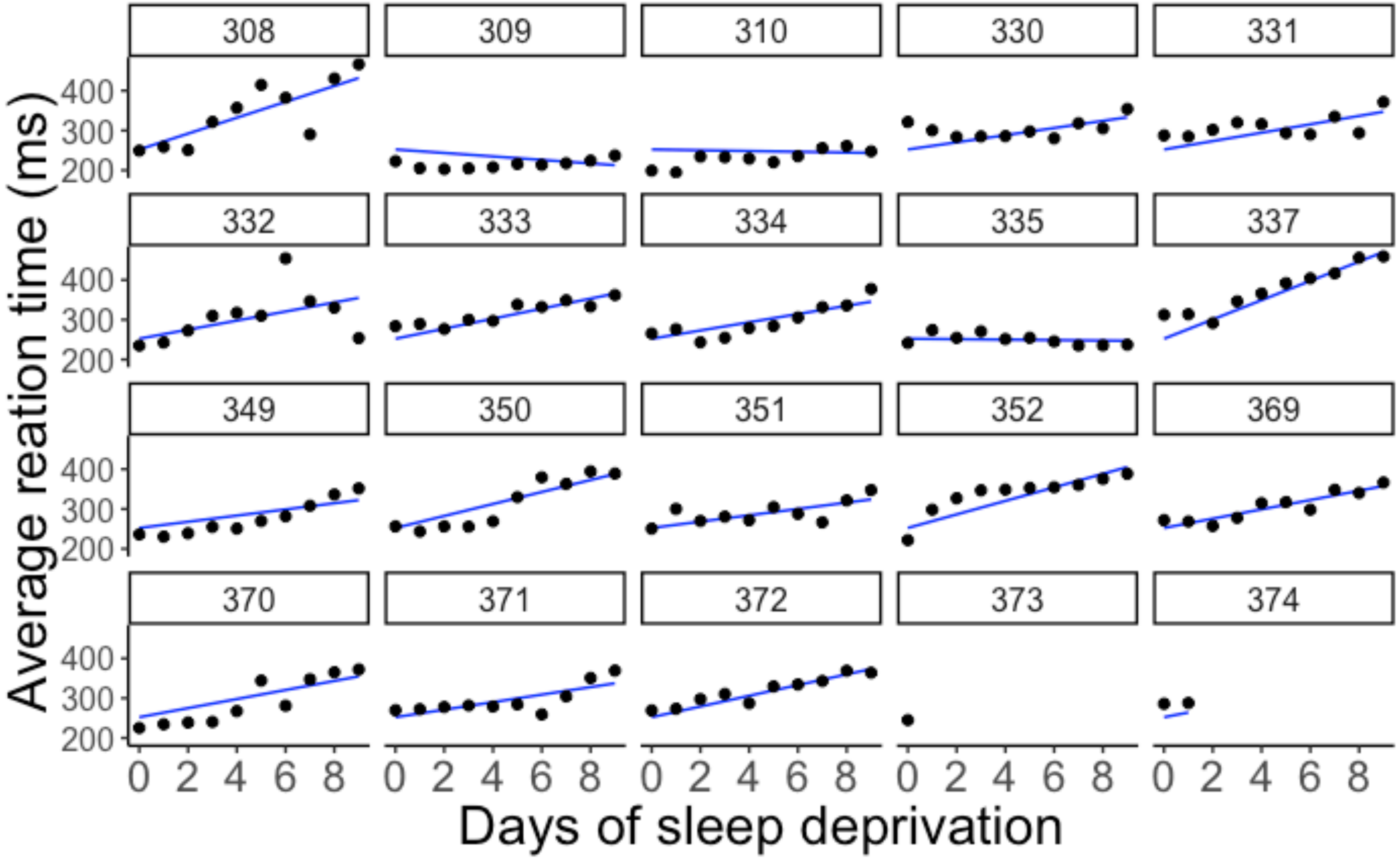




mixed effects model estimates a multi-variate Gaussian to account for (possible) correlations between intercepts and slopes

Reporting results

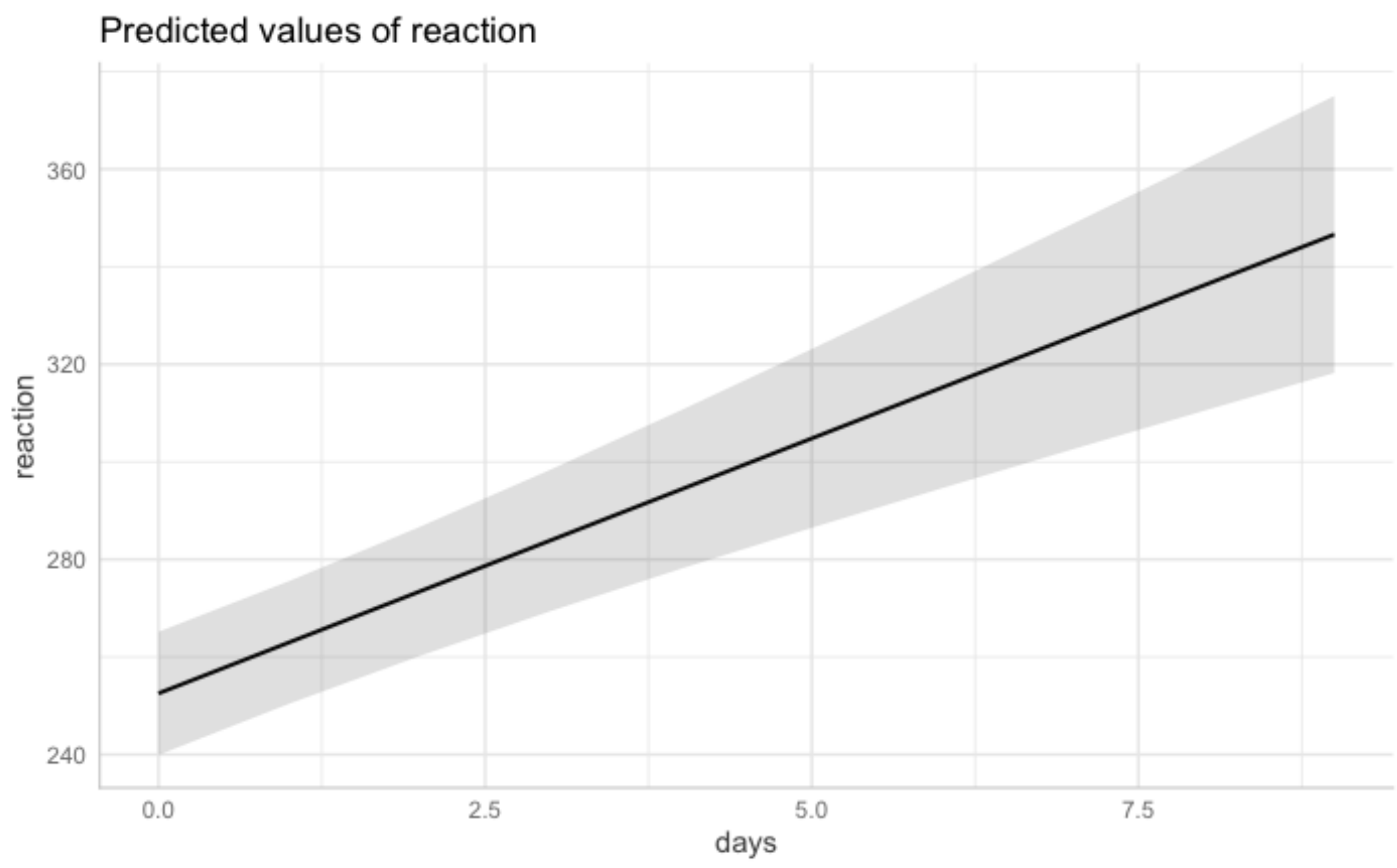
Visualization



show the data together with the model predictions

Visualization

```r
library("ggeffects")

ggpredict(model = fit.random_intercept_slope,
          terms = "days",
          type = "fe") %>%
  plot()
```



- the relationship between the variables of interest (marginalizing over other variables)

show the (marginalized) model prediction

Reporting results

7.1. In Writing

Our reports include a description of the following parts (also see Meteyard & Davies, 2019; Barr et al., 2013):

- Model specification, including:
    - Dependent variable, and all fixed and random effects (intercepts, slopes, correlations), both in words and possibly also by providing the model equation/ R-pseudo code (so-called Wilkinson notation)
    - Transformation of variables, e.g., standardizing or centering variables
    - Contrast coding (typically sum-to-zero coding)
- Inference:
    - Description of how *p*-values were obtained (in case of a frequentist approach) or what other (Bayesian) decision rule was used for inference.
    - Description of what post-hoc or follow-up tests were performed
    - Any convergence issues that may arise while running the model (in particular if they require adjustments in the model specification) and how they were dealt with should be described, as well as the subsequent adjustments that were made.
- Model output, at minimum the following:
    - Model results: (un)standardized regression coefficients, standard errors and/or confidence / credible intervals, test statistics, degrees of freedom, *p*-values

Meteyard, L. & Davies, R. A. I. (2020). Best practice guidance for linear mixed-effects models in psychological science. *Journal of Memory and Language*.

Reporting results

Table 2

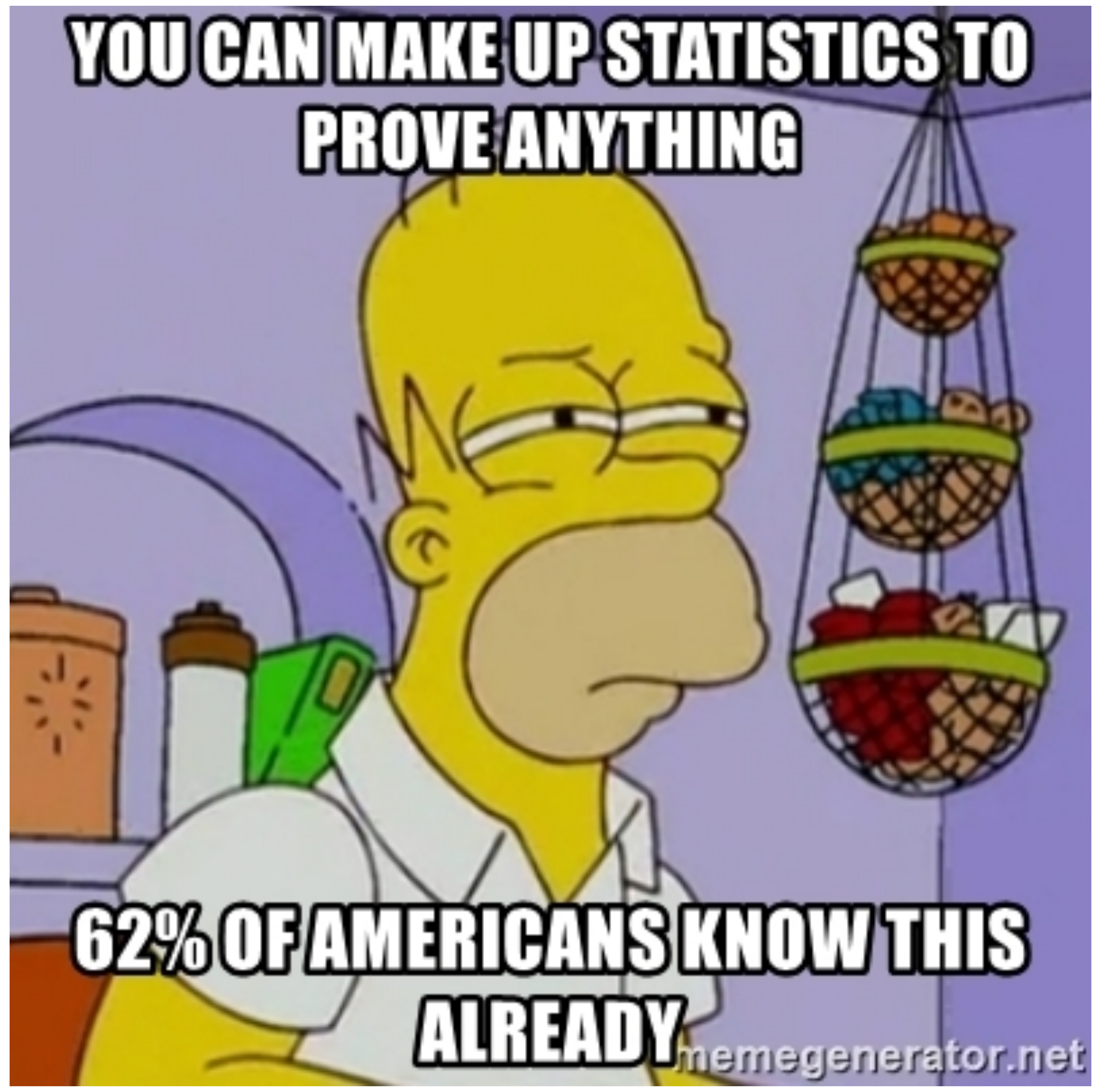
***Experiment 1 – Normality inference**: Estimates of the posterior mean and 95% highest density intervals (HDIs) for the different predictors in the Bayesian regression model. Note: For the dependent variable (normality rating), 100 = abnormal and 0 = normal.*

model specification: `normality rating ~ 1 + structure * norm`

term	estimate	lower 95% CI	upper 95% CI
intercept	62.83	57.57	68.11
structure	21.22	16.27	26.35
norm	−1.47	−6.37	3.83
structure:norm	−1.46	−6.41	3.53

[7] All categorical predictors were coded using sum contrasts. We adopt the convention of calling something an effect if the 95% highest density interval (HDI) of the estimated parameter in the Bayesian model excludes 0.

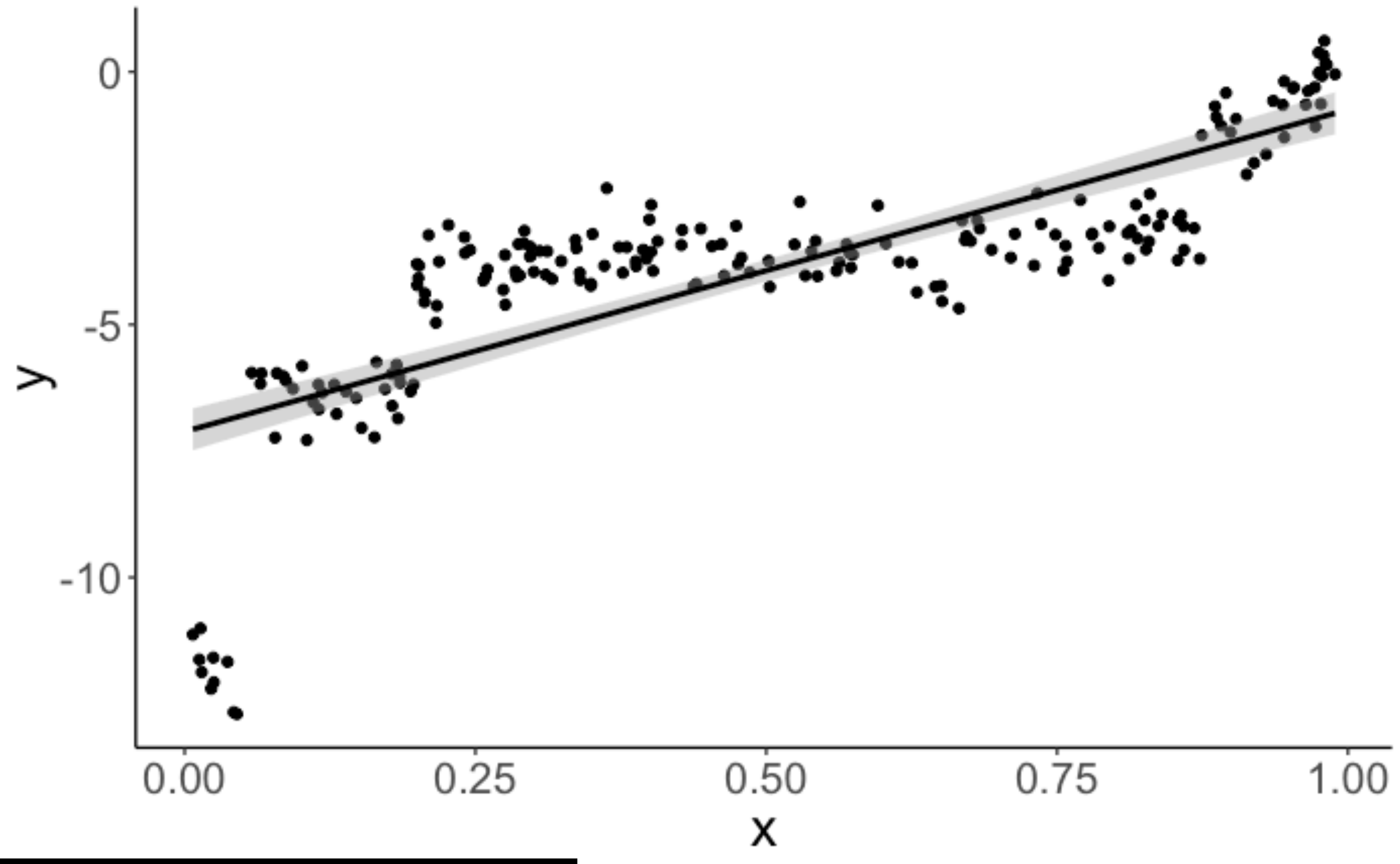
Where are the Random Effects??



Simpsons Paradox

Simpson's paradox

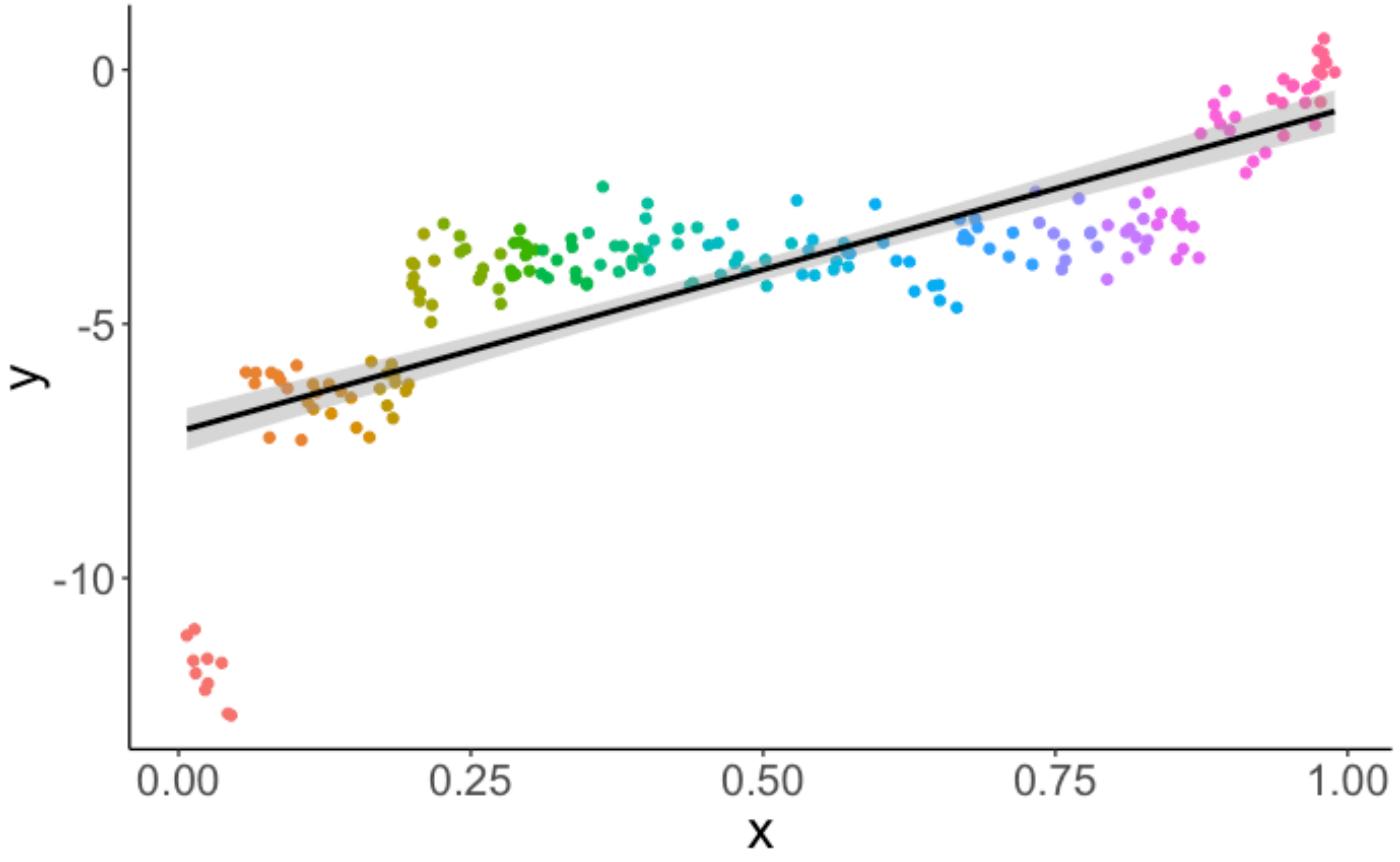
```
1 lm(formula = y ~ 1 + x,
2    data = df.simpson) %>%
3   summary()
```




```
Call:
lm(formula = y ~ x, data = df.simpson)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.8731 -0.6362  0.2272  1.0051  2.6410 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -7.1151     0.2107  -33.76   <2e-16 ***
x             6.3671     0.3631   17.54   <2e-16 ***

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.55 on 198 degrees of freedom
Multiple R-squared:  0.6083,    Adjusted R-squared:  0.6064 
F-statistic: 307.5 on 1 and 198 DF,  p-value: < 2.2e-16
```

positive relationship
between x and y

Simpson's paradox




```r
lmer(formula = y ~ 1 + x + (1 | participant),
     data = df.simpson) %>%
  summary()
```

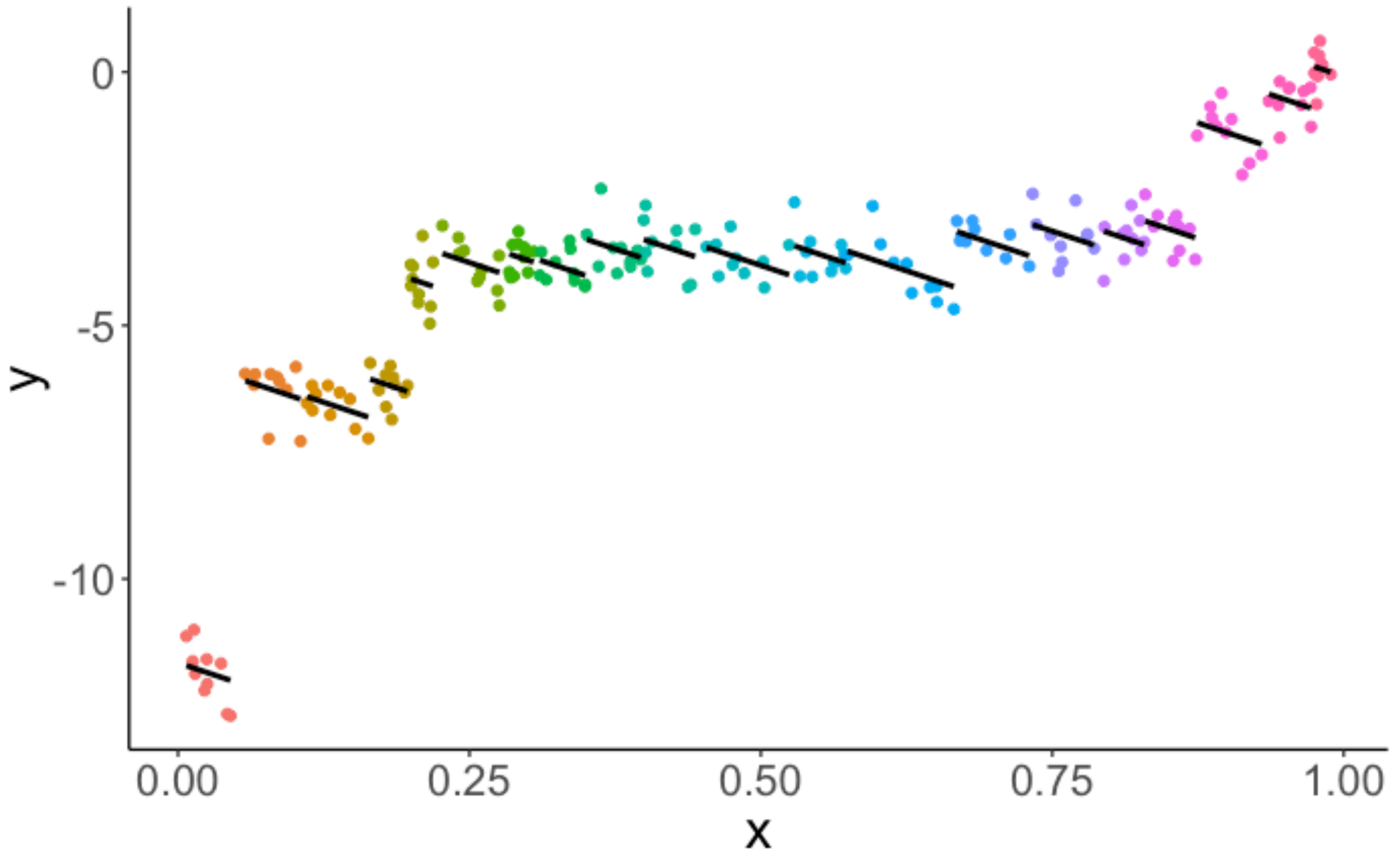
```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ 1 + x + (1 | participant)
   Data: df.simpson

REML criterion at convergence: 345.1

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.43394 -0.59687  0.04493  0.62694  2.68828

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 21.4898  4.6357
 Residual                 0.1661  0.4075
Number of obs: 200, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  -0.1577     1.3230  -0.119
x            -7.6678     1.6572  -4.627

Correlation of Fixed Effects:
  (Intr)
x -0.621
```

negative (!) relationship between x and y

Simpson's paradox

```r
lmer(formula = y ~ 1 + x + (1 | participant),
     data = df.simpson) %>%
  summary()
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ 1 + x + (1 | participant)
   Data: df.simpson

REML criterion at convergence: 345.1

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.43394 -0.59687  0.04493  0.62694  2.68828

Random effects:
 Groups      Name        Variance Std.Dev.
 participant (Intercept) 21.4898  4.6357
 Residual                 0.1661  0.4075
Number of obs: 200, groups:  participant, 20

Fixed effects:
            Estimate Std. Error t value
(Intercept)  -0.1577     1.3230  -0.119
x            -7.6678     1.6572  -4.627

Correlation of Fixed Effects:
  (Intr)
x -0.621
```



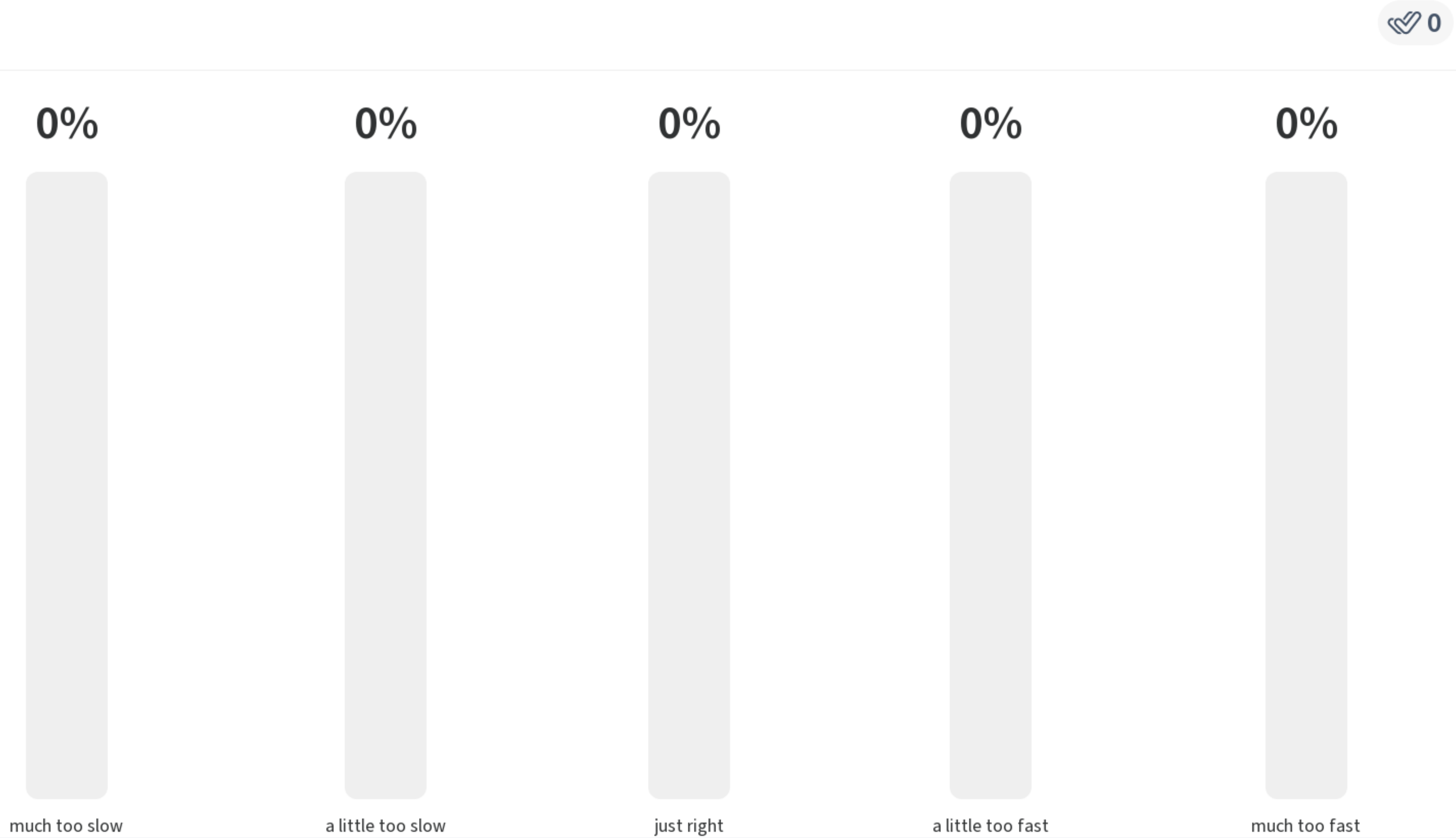
negative (!) relationship between x and y

(once we take into account individual differences)

Plan for today

- Quick recap
- Linear Mixed Model
  - Accommodating non-idependence in data
  - Understanding lmer() syntax
  - A worked example
  - Reporting results
  - Simpsons Paradox

0



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Thank you!